Chemical Thermodynamics

Question1

For the reaction, $N_2O_4 \rightleftharpoons 2NO_2$, graph is plotted as shown below. Identify correct statements.

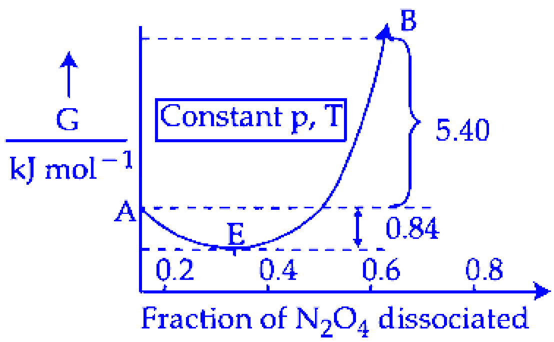
A. Standard free energy change for the reaction is $-5.40 \text{ kJ mol}^{-1}$.

B. As $\Delta G^{\ominus}$ in graph is positive, N_2O_4 will not dissociate into NO_2 at all.

C. Reverse reaction will go to completion.

D. When 1 mole of N_2O_4 changes into equilibrium mixture, value of $\Delta G^{\ominus} = -0.84 \text{ kJ mol}^{-1}$

E. When 2 mole of NO_2 changes into equilibrium mixture, $\Delta G^{\ominus}$ for equilibrium mixture is $-6.24 \text{ kJ mol}^{-1}$.



Choose the correct answer from the options given below :

JEE Main 2026 (Online) 21st January Morning Shift

Options:

A.

C and E only

B.

D and E only

C.

A and D only

D.

B and C only

Answer: B

Solution:

(A) The standard free energy change ($\Delta_r G^\circ$) is found by subtracting the G° value of reactant (A) from that of product (B).

(A) Calculation:
$\Delta_r G^\circ = G_B^\circ - G_A^\circ$

In the graph, this value is positive (i.e., +ve).

(B) A positive $\Delta_r G^\circ$ means the reaction is not spontaneous in the forward direction under standard conditions. But it does not mean that N_2O_4 will not change into NO_2 at all. Some amount will still convert, and equilibrium will be set up with both N_2O_4 and NO_2 present.

(C) The reverse reaction does not go to completion. Because at equilibrium (point E in the graph), both NO_2 and N_2O_4 are present. This means the reverse reaction is also only partially complete.

(D) When 1 mole of N_2O_4 is converted to the equilibrium mixture, the free energy change is $\Delta_r G^\circ = -0.84 \text{ kJ mol}^{-1}$ (as shown on the graph).

(E) When 2 moles of NO_2 change to the equilibrium mixture, the standard free energy change is:

$-5.40 \text{ kJ mol}^{-1} + (-0.84 \text{ kJ mol}^{-1}) = -6.24 \text{ kJ mol}^{-1}$

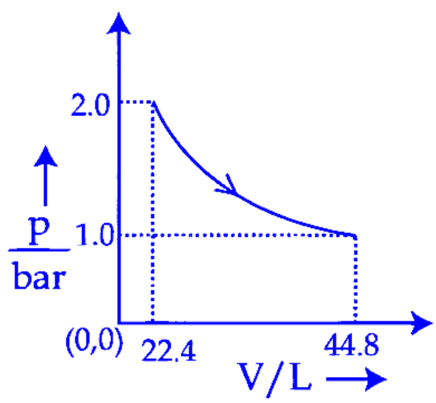
Question2

Which of the following graphs between pressure ' p ' versus volume ' V ' represents the maximum work done?

JEE Main 2026 (Online) 21st January Morning Shift

Options:

A.



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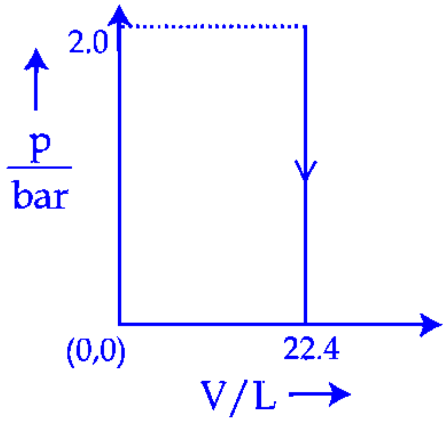
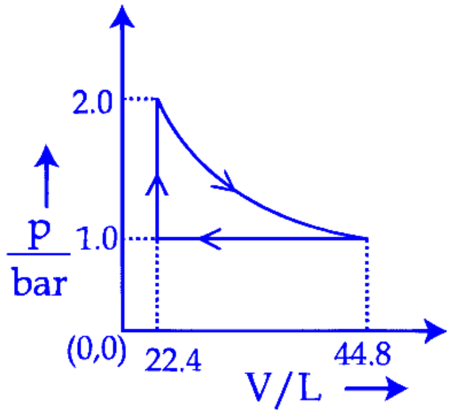
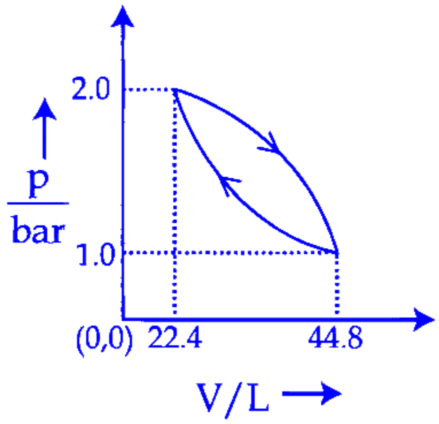


C.



D.



Answer: A

Solution:

Work Done is Area Under the Curve

In a graph where pressure (P) is plotted against volume (V), the area under the curve shows the amount of work done during the process.

Option B: Zero Work

In option (B), the area under the curve is zero. This means no work is done in this case.

Other Options: Work is Negative (Expansion)

For the other options, the area is below the axis, so the work done is negative. This happens because the gas is expanding.

Magnitude of Work

The answer given by NTA looks only at the amount (magnitude) of work, not if it is negative or positive.

$W_1 = (44.8) \times \ln 2 = 31.04$
$W_3 < 31.04$
$W_4 < 31.04$

Question3

Consider the following data :

$\Delta_f H^\circ$ (methane, g) = $-X$ kJ mol $^{-1}$

Enthalpy of sublimation of graphite = Y kJ mol $^{-1}$

Dissociation enthalpy of $H_2 = Z$ kJ mol $^{-1}$

The bond enthalpy of C–H bond is given by :

JEE Main 2026 (Online) 21st January Evening Shift

Options:

A.

$\frac{X+Y+2Z}{4}$

B.

$\frac{X+Y+4Z}{2}$

C.

$\frac{-X+Y+Z}{4}$

D.

$X+Y+Z$

Answer: A

Solution:

$\mathrm{C(s)} + 2\mathrm{H_2(g)} \rightarrow \mathrm{CH_4(g)}$

$-x = (\Delta H_{\text{sub}}\ \text{ of carbon}) + 2 \times (\text{B.E. of H-H}) - 4 \times (\text{B.E. of } C - H)$

$-x = y + 2z - 4(\text{B.E. of C-H})$

$\text{B.E of C} - \text{H} = \frac{\text{y+2z+x}}{4}$

Question4

Match the LIST-I with LIST-II

	List-I Thermodynamic Process	List-II Magnitude in kJ	
A.	Work done in reversible, isothermal expansion of 2 mol of ideal gas from $2\mathrm{dm}^3$ to $20\mathrm{dm}^3$ at 300 K .	I.	4
B.	Work done in irreversible isothermal expansion of 1 mol ideal gas from $1\ \mathrm{m}^3$ to $3\ \mathrm{m}^3$ at 300 K against a constant pressure of 3 kPa .	II.	11.5
C.	Change in internal energy for adiabatic expansion of a 1 mol ideal gas with change of temperature $= 320$ K and $\overline{\mathrm{C}}_{\mathrm{V}} = \frac{3}{2}\mathrm{R}$.	III.	6
D.	Change in enthalpy at constant pressure of 1 mol ideal gas with change of temperature $= 337$ K and $\overline{\mathrm{C}}_{\mathrm{p}} = \frac{5}{2}\mathrm{R}$.	IV.	7

JEE Main 2026 (Online) 22nd January Morning Shift

Options:

A.

A-I, B-II, C-III, D-IV

B.

A-III, B-II, C-IV, D-I

C.

A-II, B-I, C-III, D-IV

D.

A-II, B-III, C-I, D-IV

Answer: D

Solution:

Option (A)

This is a **reversible isothermal expansion** of an ideal gas, so we use the NCERT formula for isothermal reversible work:

$$\begin{aligned}&\mathrm{W} = -\mathrm{nRT}\ln\frac{\mathrm{V}_2}{\mathrm{V}_1}\\&= \frac{-2\times 8.314\times 300}{1000}\times\ln\left(\frac{20}{2}\right)\mathrm{kJ}\\&= -11.5\ \mathrm{kJ}\end{aligned}$$

The negative sign shows work is done **by the gas** during expansion. The **magnitude** of work is 11.5 kJ.

Option (B)

This is an **irreversible isothermal expansion** against a constant external pressure, so work is calculated by:

$$\begin{aligned}&\mathrm{W} = -\mathrm{P}_{\mathrm{ext}}\ \ [\mathrm{V}_2 - \mathrm{V}_1]\\&= -3[3-1]\\&= -6\ \mathrm{kJ}\end{aligned}$$

The negative sign again means work is done **by the gas**. The **magnitude** of work is 6 kJ.

Option (C)

For an ideal gas, internal energy depends only on temperature. For any process (including adiabatic), we use:

$$\begin{aligned}&\Delta\mathrm{U} = \mathrm{nC_v}\Delta\ \mathrm{T}\\&= 1\times\frac{3}{2}\times\frac{8.314\times 320}{1000}100\ \mathrm{kJ}\\&= 3.99\end{aligned}$$

So the change in internal energy comes out to approximately 3.99 kJ (about 4 kJ).

Option (D)

For an ideal gas, enthalpy also depends only on temperature. At constant pressure, we use:

$$\begin{aligned} \Delta \mathrm{H} &= \mathrm{nC_p} \Delta\ \mathrm{T} \\ &= 1 \times \frac{5}{2} \times \frac{8.314 \times 337}{1000}\ \mathrm{kJ} \\ &= 7\ \mathrm{kJ} \end{aligned}$$

So, $\Delta H = 7$ kJ.

Question5

A cup of water at $5^{\circ}\mathrm{C}$ (system) is placed in a microwave oven and the oven is turned on for one minute during which the water begins to boil. Which of the following option is true?

JEE Main 2026 (Online) 23rd January Morning Shift

Options:

A.

$q = +ve, w = -ve, \Delta U = +ve$

B.

$q = +ve, w = 0, \Delta U = -ve$

C.

$q = +ve, w = -ve, \Delta U = -ve$

D.

$q = -ve, w = -ve, \Delta U = -ve$

Answer: A

Solution:

$$\underset{5^{\circ}\mathrm{C}}{\mathrm{H_2O}(\ell)} \longrightarrow \underset{100^{\circ}\mathrm{C}}{\mathrm{H_2O}(\ell)} \rightleftarrows \underset{100^{\circ}\mathrm{C}}{\mathrm{H_2O}(\mathrm{g})}$$

When water is heated in the microwave, its temperature increases from $5^{\circ}\mathrm{C}$ to $100^{\circ}\mathrm{C}$. After reaching $100^{\circ}\mathrm{C}$, it starts changing from liquid to vapour (boiling), so liquid water and water vapour can exist together.

During boiling, the water changes into steam and the volume increases (expansion).

Because of expansion, the system does work on the surroundings, so work done by the system is negative.

$\mathrm{w} = -\mathrm{ve}$

Since the microwave supplies energy to the water, heat is absorbed by the system, so q is positive.

Also, water vapour has more internal energy than liquid water at the same temperature, so the internal energy of the system increases. Therefore, $\Delta \mathrm{U}$ is positive.

So, $q = +\mathrm{ve}$ and $\Delta \mathrm{U} = +\mathrm{ve}$.

Question6

Match the LIST-I with LIST-II

List-I Isothermal process for ideal gas system		List-II Work done ($V_f > V_i$)	
A.	Reversible expansion	I.	$\mathrm{w} = 0$
B.	Free expansion	II.	$\mathrm{w} = -\mathrm{nR}T \ln \frac{\mathrm{V_f}}{\mathrm{V_i}}$
C.	Irreversible expansion	III.	$\mathrm{w} = -\mathrm{p_{ex}}(\mathrm{V_f} - \mathrm{V_i})$
D.	Irreversible compression	IV.	$\mathrm{w} = -\mathrm{p_{ex}}(\mathrm{V_i} - \mathrm{V_f})$

Choose the correct answer from the options given below:

JEE Main 2026 (Online) 24th January Morning Shift

Options:

A.

A-IV, B-I, C-III, D-II

B.

A-I, B-III, C-II, D-IV

C.

A-II, B-I, C-III, D-IV

D.

A-IV, B-II, C-III, D-I

Answer: C

Solution:

For any process, NCERT uses the expression for work (done on the system):

$w = -\int_{V_i}^{V_f} P_{\text{ext}} \; dV$

A. Reversible isothermal expansion (ideal gas)

In reversible isothermal change, $P_{\text{ext}} \; = P = \dfrac{nRT}{V}$.

$w = -\int_{V_i}^{V_f} \frac{nRT}{V} \, dV = -nRT \ln\left(\frac{V_f}{V_i}\right)$

So, **A** $\rightarrow$ **II**.

B. Free expansion

In free expansion, $P_{\text{ext}} \; = 0$.

$w = -\int_{V_i}^{V_f} 0 \, dV = 0$

So, **B** $\rightarrow$ **I**.

C. Irreversible expansion (against constant external pressure)

Here P_{ext} is constant.

$w = -P_{\text{ext}} \; (V_f - V_i)$

So, **C** $\rightarrow$ **III**.

D. Irreversible compression (against constant external pressure)

Compression means work done on the system is positive. Using the same NCERT form:

$w = -P_{\text{ext}} \; (V_f - V_i) = P_{\text{ext}} \; (V_i - V_f)$

This corresponds to the given form **IV** (written as $-P_{\text{ext}}\ (V_i - V_f)$ in the list, which becomes positive when $V_i > V_f$ for compression).

So, **D** $\rightarrow$ **IV**.

Final matching

$A\text{-}II,\ B\text{-}I,\ C\text{-}III,\ D\text{-}IV$

Correct option: C

Question7

The heat of atomisation of methane and ethane are ' x ' kJmol^{-1} and ' y ' kJmol^{-1} respectively. The longest wavelength (λ) of light capable of breaking the $\mathrm{C}-\mathrm{C}$ bond can be expressed in SI unit as :

JEE Main 2026 (Online) 24th January Evening Shift

Options:

A.

$\frac{\mathrm{N_A hc}}{250(y-6x)}$

B.

$\mathrm{N_A hc}\left(y - \frac{6x}{4}\right)^{-1}$

C.

$\frac{\mathrm{hc}}{1000}\left(\frac{y-6x}{4}\right)^{-1}$

D.

$\frac{\mathrm{N_A hc}}{250(4y-6x)}$

Answer: D

Solution:

For methane:

$\Delta H_{\text{atom}}\ (CH_4) = x =\ 4D(\text{C–H})\ \Rightarrow\ D(\text{C–H}) = \frac{x}{4}$

For ethane:

$\Delta H_{\text{atom}}\ (C_2H_6) = y = 6D(\text{C–H}) + D(\text{C–C})$

$\Rightarrow\ D(\text{C–C}) = y - 6\left(\frac{x}{4}\right) = y - \frac{6x}{4} = \frac{4y-6x}{4}\quad (\text{kJ mol}^{-1})$

Energy per molecule needed to break the C–C bond:

$E = \frac{D(\text{C–C}\ )\times 1000}{N_A}\quad (\text{J})$

Longest wavelength corresponds to minimum photon energy:

$\frac{hc}{\lambda} = E \Rightarrow \lambda = \frac{N_A hc}{1000\ D(\text{C–C}\ )}$

$\lambda = \frac{N_A hc}{1000\left(\frac{4y-6x}{4}\right)} = \frac{N_A hc}{250(4y-6x)}$

Correct option: D.

Question8

$20.0 \mathrm{dm}^3$ of an ideal gas ' X ' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until pressure of the gas is 0.2 MPa . Which of the following option is correct?

(Given: $\log 2=0.3010$ and $\log 5=0.6989$)

JEE Main 2026 (Online) 28th January Morning Shift

Options:

A.

$\mathrm{w}=-9.1 \mathrm{~kJ}, \Delta \mathrm{U}=0, \Delta \mathrm{H}=0, \mathrm{q}=9.1 \mathrm{~kJ}$

B.

$\mathrm{w}=9.1 \mathrm{~J}, \Delta \mathrm{U}=9.1 \mathrm{~J}, \Delta \mathrm{H}=0 ; \mathrm{q}=0$

C.

$\mathrm{w}=-3.9 \mathrm{~kJ}, \Delta \mathrm{U}=0, \Delta \mathrm{H}=0 ; \mathrm{q}=3.9 \mathrm{~kJ}$

D.

$\mathrm{w}=+4.1 \mathrm{~kJ}, \Delta \mathrm{U}=0, \Delta \mathrm{H}=0 ; \mathrm{q}=-4.1 \mathrm{~kJ}$

Answer: A

Solution:

For an **ideal gas** undergoing **isothermal reversible expansion**:

- Temperature is constant $\Rightarrow \Delta U=0$ (for ideal gas, U depends only on T)
- Hence $\Delta H=0$ (since H also depends only on T for ideal gas)

Step 1: Given data (convert units)

- $V_1=20.0 \, \mathrm{dm}^3=20.0 \times 10^{-3} \, \mathrm{m}^3=0.020 \, \mathrm{m}^3$
- $P_1=0.5 \, \mathrm{MPa}=0.5 \times 10^6 \, \mathrm{Pa}=5.0 \times 10^5 \, \mathrm{Pa}$
- $P_2=0.2 \, \mathrm{MPa}$
- $T=600 \, \mathrm{K}$

Step 2: Find number of moles using $PV=nRT$

$n=\frac{P_1 V_1}{R T}=\frac{\left(5.0 \times 10^5\right)(0.020)}{(8.314)(600)}=\frac{10000}{4988.4} \approx 2.0 \mathrm{~mol}$

Step 3: Work done in reversible isothermal expansion

For reversible isothermal process:

$w=-n R T \ln \left(\frac{V_2}{V_1}\right)=-n R T \ln \left(\frac{P_1}{P_2}\right)$

Convert to base-10 log form (NCERT style):

$w=-2.303 \, n R T \, \log \left(\frac{P_1}{P_2}\right)$

Now,

$\frac{P_1}{P_2}=\frac{0.5}{0.2}=2.5$

Given:

$\log(2.5) = \log\left(\frac{5}{2}\right) = \log 5 - \log 2 = 0.6989 - 0.3010 = 0.3979$

So,

$w = -2.303(2.0)(8.314)(600)(0.3979)$

First calculate nRT:

$nRT = 2.0 \times 8.314 \times 600 \approx 9976.8 \text{ J}$

Then,

$w \approx -2.303 \times 9976.8 \times 0.3979 \approx -9130 \text{ J} \approx -9.1 \text{ kJ}$

Step 4: Heat and energy changes

For isothermal ideal gas:

$\Delta U = 0$

First law (NCERT sign convention):

$\Delta U = q + w$

So,

$0 = q + (-9.1) \Rightarrow q = +9.1 \text{ kJ}$

Also,

$\Delta H = 0$

Correct option

Option A:

$w = -9.1\,\text{kJ}, \quad \Delta U = 0, \quad \Delta H = 0, \quad q = +9.1\,\text{kJ}$

Question9

The plot of $\log_{10} K$ vs $\frac{1}{T}$ gives a straight line. The intercept and slope respectively are (where K is equilibrium constant).

JEE Main 2026 (Online) 28th January Evening Shift

Options:

A.

$-\frac{\Delta S^{\circ} R}{2.303}, \frac{\Delta H^{\circ} R}{2.303}$

B.

$-\frac{\Delta H^{\circ}}{2.303R}, \frac{\Delta S^{\circ}}{2.303R}$

C.

$\frac{\Delta S^{\circ}}{2.303R}, -\frac{\Delta H^{\circ}}{2.303R}$

D.

$\frac{2.303R}{\Delta H^{\circ}}, \frac{2.303R}{\Delta S^{\circ}}$

Answer: C

Solution:

$\log_{10} K = -\frac{\Delta H^o}{2.303RT} + \frac{\Delta S^o}{2.303R}$

Compare this with the straight line form $y = mx + c$ where $y = \log_{10} K$ and $x = \frac{1}{T}$.

Then the constant term is the y-intercept: $\frac{\Delta S^o}{2.303R}$

And the coefficient of $\frac{1}{T}$ is the slope: $-\frac{\Delta H^o}{2.303R}$

Question10

Consider the following data.

(i) $2Al(s) + 6HCl(aq) \rightarrow Al_2Cl_6(aq) + 3H_2(g) + 1200\ kJ/mol$

(ii) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g) + 164\ kJ/mol$

(iii) $HCl(g) + aq \rightarrow HCl(aq) + 83\ kJ/mol$

(iv) $Al_2Cl_6(s) + aq \rightarrow Al_2Cl_6(aq) + 663\ kJ/mol$

The enthalpy of formation of anhydrous solid Al_2Cl_6 is :

JEE Main 2026 (Online) 2nd April Morning Shift

Options:

A.

$-648\ kJ\ mol^{-1}$

B.

$-1350\ kJ\ mol^{-1}$

C.

$-2002\ kJ\ mol^{-1}$

D.

$-1527\ kJ\ mol^{-1}$

Answer: D

Solution:

Write the given reaction data.

(i) $2Al(s) + 6HCl(aq) \rightarrow Al_2Cl_6(aq) + 3H_2(g)$; $\Delta_r H_1 = -1200\ kJ/mol$

(ii) $H_2(g) + Cl_2(g) \rightarrow 2HCl(g)$; $\Delta_r H_2 = -164\ kJ/mol$

(iii) $HCl(g) + aq \rightarrow HCl(aq)$; $\Delta_r H_3 = -83\ kJ/mol$

(iv) $Al_2Cl_6(s) + aq \rightarrow Al_2Cl_6(aq)$; $\Delta_r H_4 = +663\ kJ/mol$

Adjust the equations so that we can connect them to the target reaction.

Target reaction: $2Al(s) + 3Cl_2(g) \rightarrow Al_2Cl_6(s)$

The given first reaction produces $Al_2Cl_6(aq)$ from 2Al and 6HCl(aq) . We need to convert all species to match the target reaction, step by step.

Use Hess's Law

According to Hess's Law, the net enthalpy change for the target reaction is the sum of the enthalpy changes for all the necessary intermediate reactions.

First, multiply equation (ii) by 3 to get 6 moles of HCl(g):

$3[\mathrm{H}_2 + \mathrm{Cl}_2 \rightarrow 2\mathrm{HCl(g)}]; \Delta_\mathrm{r} H_2 = -164 \times 3$ kJ/mol

Then, multiply equation (iii) by 6 to convert 6 moles of HCl(g) to HCl(aq):

$6[\mathrm{HCl(g)} + \mathrm{aq} \rightarrow \mathrm{HCl(aq)}]; \Delta_\mathrm{r} H_3 = -83 \times 6$ kJ/mol

We also reverse equation (iv) to obtain $\mathrm{Al_2Cl_6(aq)} \rightarrow \mathrm{Al_2Cl_6(s)}$, giving $\Delta_\mathrm{r} H_4 = +663$ kJ/mol .

Add up all enthalpy changes.

Using Hess's Law:

$\Delta_\mathrm{r} H = \Delta_\mathrm{r} H_1 + \Delta_\mathrm{r} H_2 + \Delta_\mathrm{r} H_3 + \Delta_\mathrm{r} H_4$

Substitute the values:

$\Delta_\mathrm{r} H = (-1200) - (164 \times 3) - (83 \times 6) + 663$

Simplify the calculation.

$\Delta_\mathrm{r} H = -1200 - 492 - 498 + 663 = -1527$ kJ/mol

Final Answer: The enthalpy of formation of solid anhydrous $\mathrm{Al_2Cl_6}$ is $\boxed{-1527 \text{ kJ/mol}}$.

Question11

Gas 'A' undergoes change from state 'X' to state 'Y'. In this process, the heat absorbed and work done by the gas is 10 J and 18 J respectively. Now gas is brought back to state 'X' by another process during which 6 J of heat is evolved. In the reverse process of 'Y' to 'X',

JEE Main 2026 (Online) 2nd April Evening Shift

Options:

A.

18 J of the work is done by the gas 'A'.

B.

2 J of the work is done by the gas 'A'.

C.

12 J of the work is done on the gas 'A' by the surrounding.

D.

14 J of the work is done on the gas 'A' by the surrounding.

Answer: D

Solution:

For a thermodynamic process, using the NCERT sign convention:

- Heat absorbed by the gas: $Q > 0$
- Heat evolved by the gas: $Q < 0$

- Work done by the gas: $W > 0$

Also, from the first law of thermodynamics:

$\Delta U = Q - W$

Step 1: Process from X to Y

Given:

- Heat absorbed, $Q_{XY} = +10$ J
- Work done by gas, $W_{XY} = +18$ J

So,

$\Delta U_{XY} = Q_{XY} - W_{XY}$

$\Delta U_{XY} = 10 - 18 = -8$ J

Thus,

$U_Y - U_X = -8$ J

Step 2: Reverse process from Y to X

For the reverse process,

$\Delta U_{YX} = -\Delta U_{XY} = +8$ J

Also, during this process, 6 J of heat is evolved, so

$Q_{YX} = -6$ J

Using first law again:

$\Delta U_{YX} = Q_{YX} - W_{YX}$

Substitute the values:

$8 = -6 - W_{YX}$

$-W_{YX} = 14$

$W_{YX} = -14$ J

Negative sign means work is done on the gas by the surroundings.

So, in the process from Y to X,

14 J

of work is done on the gas by the surroundings.

Correct option: D

$\boxed{\text{14 J of work is done on the gas by the surrounding}}$

Question12

Given below are two statements :

Statement I : For an ideal gas, heat capacity at constant volume is always greater than the heat capacity at constant pressure.

Statement II : In a constant volume process, no work is produced and all the heat withdrawn goes into the chaotic motion and is reflected by a temperature increase of the ideal gas.

In the light of the above statements, choose the correct answer from the options given below

Options:

A.

Both Statement I and Statement II are true

B.

Both Statement I and Statement II are false

C.

Statement I is true but Statement II is false

D.

Statement I is false but Statement II is true

Answer: D

Solution:

For an ideal gas, we know that

$C_p - C_v = R$

where $R > 0$.

So,

$C_p > C_v$

This means **Statement I is false**, because it says that heat capacity at constant volume is greater than heat capacity at constant pressure.

Now look at **Statement II**.

In a constant volume process,

$\Delta V = 0$

So the work done is

$W = P\Delta V = 0$

From the first law of thermodynamics,

$Q = \Delta U + W$

Since $W = 0$,

$Q = \Delta U$

For an ideal gas, internal energy depends only on temperature. Therefore, the heat supplied or withdrawn at constant volume changes the internal energy, which is associated with the random or chaotic motion of molecules, and this is reflected in a change in temperature.

So **Statement II is true** in the NCERT sense.

Hence, the correct answer is:

$\boxed{\text{Option D}}$

That is, **Statement I is false but Statement II is true**.

Question13

The correct order of molar heat capacities measured at 298 K and 1 bar is :

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

Copper(s) $>$ Bromine(l) $>$ Helium(g)

B.

Bromine(l) > Copper(s) > Helium(g)

C.

Helium(g) > Bromine(l) > Copper(s)

D.

Helium(g) > Bromine(l) = Copper(s)

Answer: B

Solution:

1. Helium (g):

Helium is a monoatomic gas. In the gaseous state, monoatomic molecules only possess 3 translational degrees of freedom.

According to the principle of equipartition of energy, the molar heat capacity at constant volume (C_V) is given by:

$C_V = \frac{3}{2}R$

For a gas, the molar heat capacity at constant pressure (C_P) is higher than C_V by the gas constant R:

$C_P = C_V + R = \frac{3}{2}R + R = \frac{5}{2}R = 2.5R$

2. Copper (s):

Copper is a solid metal composed of single atoms in a crystal lattice. In solids, atoms do not translate or rotate, but they vibrate about their mean positions. They have 3 vibrational degrees of freedom in three dimensions.

According to the Dulong-Petit law for solid elements, each vibrational degree of freedom contributes R to the molar heat capacity (since vibration involves both kinetic and potential energy).

Therefore, the molar heat capacity for monoatomic solids is approximately:

$C_m \approx 3R$

3. Bromine (l):

Bromine exists as a diatomic molecule (Br_2) in the liquid state. The molar heat capacity here will be significantly higher for a few fundamental reasons:

Firstly, because it is diatomic, a single mole of Br_2 contains 2 moles of atoms. By simple extension of solid-state vibration, this alone would contribute roughly $2 \times 3R = 6R$.

Secondly, in the liquid state, molecules have translational, rotational, and vibrational degrees of freedom, and the intermolecular forces are continuously broken and formed, which allows the liquid to absorb a large amount of thermal energy. Hence, the molar heat capacity of $Br_2(l)$ is considerably higher than $3R$.

Comparing the Values:

- Helium (g): $2.5R$
- Copper (s): $\approx 3R$
- Bromine (l): $> 6R$

When we arrange them in decreasing order of their molar heat capacities, we get:

$Bromine(l) > Copper(s) > Helium(g)$

Therefore, the correct option is **Option B**.

Question14

Consider the following data for the reaction

$X_2(g) + Y_2(g) \rightleftharpoons 2XY(g)$

at $600\ \mathrm{K}^{\circ}$. The $\Delta_{\mathrm{r}} \mathrm{G}^{\ominus}$ (in kJmol^{-1}) for the reaction is :

Compound	$\Delta_f \mathrm{H}_{600\ \mathrm{K}}^{\ominus}\left(\mathrm{kJ\ mol}^{-1}\right)$	$\mathrm{S}^{\ominus}{}_{600\ \mathrm{K}}\left(\mathrm{J\ mol}^{-1}\ \mathrm{K}^{-1}\right)$
$\mathrm{XY}(\mathrm{g})$	42	200
$\mathrm{X}_2(\mathrm{g})$	8	140
$\mathrm{Y}_2(\mathrm{g})$	80	250

JEE Main 2026 (Online) 5th April Evening Shift

Options:

A.

-21000

B.

-10

C.

-1000

D.

-9.012

Answer: B

Solution:

For the reaction

$X_2(g)+Y_2(g) \rightleftharpoons 2XY(g)$

we use

$\Delta_r G^{\circ}=\Delta_r H^{\circ}-T \Delta_r S^{\circ}$

Using standard enthalpies of formation,

$\Delta_r H^{\circ}=\sum \nu \Delta_f H^{\circ}(\text{products})-\sum \nu \Delta_f H^{\circ}(\text{reactants})$

So,

$\Delta_r H^{\circ}=2(42)-[8+80]$

$\Delta_r H^{\circ}=84-88=-4\ \mathrm{kJ\ mol}^{-1}$

Using standard molar entropies,

$\Delta_r S^{\circ}=\sum \nu S^{\circ}(\text{products})-\sum \nu S^{\circ}(\text{reactants})$

$\Delta_r S^{\circ}=2(200)-[140+250]$

$\Delta_r S^\circ = 400 - 390 = 10 \text{ J mol}^{-1}\text{K}^{-1}$

Convert into kJ mol $^{-1}$K^{-1}:

$10 \text{ J mol}^{-1}\text{K}^{-1} = 0.010 \text{ kJ mol}^{-1}\text{K}^{-1}$

At $T = 600$ K,

$\Delta_r G^\circ = -4 - 600(0.010)$

$\Delta_r G^\circ = -4 - 6 = -10 \text{ kJ mol}^{-1}$

Hence, the correct answer is

$\boxed{-10 \text{ kJ mol}^{-1}}$

So, **Option B** is correct.

Question15

Arrange the following isothermal processes in order of the magnitude of the work $(\mathrm{p}-\mathrm{V})$ involved between states 1 and 2.

A. Expansion in single stage $\mathrm{w_A}$

B. Expansion in multi stages w_B

C. Compression in single stage $\mathrm{w_C}$

D. Compression in multi stages w_D

Choose the correct option.

JEE Main 2026 (Online) 6th April Morning Shift

Options:

A.

$|w_B| > |w_A| > |w_C| > |w_D|$

B.

$|w_C| > |w_D| > |w_A| > |w_B|$

C.

$|w_C| > |w_D| > |w_B| > |w_A|$

D.

$|w_B| > |w_A| > |w_D| > |w_C|$

Answer: C

Solution:

Work done in a $p-V$ process is given by

$\omega = -P\Delta V$

So, between states 1 and 2, if the external pressure is constant,

$\omega = -p(v_2 - v_1)$

For **compression**, volume decreases: $V_1 \longrightarrow V_2$ and $V_2

So, $(v_2 - v_1)$ is negative, and hence

$\omega = +$ (Positive)

For **expansion**, volume increases: $v_1 \longrightarrow v_2$ and $v_2 > v_1$.

So, $(v_2 - v_1)$ is positive, and hence

$\omega = -$ (Negative)

Therefore, in magnitude and sign, the work done in compression is greater than the work done in expansion (compression work is positive, expansion work is negative in this convention).

Now compare **single-step** and **multi-step** processes using the area under the $(P - V)$ graph.

In **multi-step compression**, at each small step, P_{ext} is kept very close to P_{sys}, so the process is closer to reversible and the area under the curve (work done) is smaller.

But in **single-step compression**, the external pressure is applied in one go (effectively larger pressure difference), so the area under the $(P - V)$ graph is more. Hence, work done is more in single-step compression than in multi-step compression.

Similarly, in **multi-step expansion**, P_{ext} stays close to P_{gas}, so the area under the curve is larger (more negative work by the system).

But in **single-step expansion**, the external pressure is suddenly lowered, so the area under the $(p - v)$ graph is smaller. Hence, the magnitude of work is less in single-step expansion than in multi-step expansion.

So the overall order is: compression $>$ expansion, and within each, single-step has larger magnitude than multi-step.

$\omega_C > \omega_D > \omega_B > \omega_A$

Question16

Match List - I with List - II.

Given V_1 and V_2 are initial and final volumes respectively.

List - I (Isothermal process)		List - II (Expression)	
A.	Reversibleexpansion	I.	$\mathrm{q} = 0$
B.	Free expansion	II.	$\mathrm{q} = \mathrm{nRT} \ln \frac{\mathrm{V}_2}{\mathrm{V}_1}$
C.	IrreversibleCompression	III.	$\mathrm{w} = -\mathrm{p}_{\text{ext}}(\mathrm{V}_1 - \mathrm{V}_2)$
D.	Cyclic reversible	IV.	$\frac{\mathrm{q}_{\text{rev}}}{\mathrm{T}} = 0$

Choose the correct answer from the options given below :

JEE Main 2026 (Online) 6th April Evening Shift

Options:

A.

A-II, B-III, C-I, D-IV

B.

A-II, B-I, C-IV, D-III

C.

A-II, B-I, C-III, D-IV

D.

A-I, B-II, C-III, D-IV

Answer: C

Solution:

(A) Reversible isothermal expansion

For an ideal gas in a *reversible* and *isothermal* expansion:

$w=-nRT\ln\frac{V_2}{V_1}$

Since the process is isothermal,

$\Delta U=0 \implies q=-w$

Hence,

$q=nRT\ln\frac{V_2}{V_1}$

Therefore, A corresponds to **II.**

(B) Free expansion

In *free expansion*, gas expands against zero external pressure:

$p_{\text{ext}}=0$

Thus,

$w=0$ and $q=0$

(Also, for an ideal gas, } ΔU = 0.)

Therefore, B corresponds to **I.**

(C) Irreversible isothermal compression

For *irreversible compression* (against constant external pressure):

$w=-p_{\text{ext}}\left(V_1-V_2\right)$

Therefore, C corresponds to **III.**

(D) Cyclic reversible process

For a *cyclic reversible* process:

$\oint\frac{q_{\text{rev}}}{T}=0$

Therefore, D corresponds to **IV.**

Correct Option: (C)

$\boxed{A-II,\ B-I,\ C-III,\ D-IV}$

Question17

Use the following data :

Substance	$\frac{\Delta_f H^{\ominus}(500\ \mathrm{K})}{\mathrm{kJ\ mol^{-1}}}$	$\frac{S^{\ominus}(500\ \mathrm{K})}{\mathrm{JK^{-1}\ mol^{-1}}}$
AB(g)	32	222
A_2(g)	6	146
B_2(g)	x	280

One mole each of $A_2(g)$ and $B_2(g)$ are taken in a 1 L closed flask and allowed to establish the equilibrium at 500 K .

$$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$$

The value of x $\left(\text{inkJmol}^{-1}\right)$ is _____ . (Nearest integer)

(Given : $\log K = 2.2 \quad R = 8.3\ J\ K^{-1}\ mol^{-1}$ **)**

JEE Main 2026 (Online) 21st January Morning Shift

Answer: 70

Solution:

For the reaction at 500 K:

$A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$

1) Calculate ΔH° of reaction at $500\ K$

Using standard enthalpies of formation:

$\Delta H^\circ = \sum \nu \Delta_f H^\circ(\text{products}) - \sum \nu \Delta_f H^\circ(\text{reactants})$

$\Delta H^\circ = 2(32) - (6 + x) = 64 - 6 - x = (58 - x)$ kJ mol $^{-1}$

2) Calculate ΔS° of reaction at $500\ K$

$\Delta S^\circ = 2(222) - (146 + 280) = 444 - 426 = 18$ J K^{-1}mol^{-1}

3) Calculate ΔG° using $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

Convert $T\Delta S^\circ$ into kJ:

$T\Delta S^\circ = 500 \times 18 = 9000$ J mol^{-1} $= 9$ kJ mol $^{-1}$

So,

$\Delta G^\circ = (58 - x) - 9 = (49 - x)$ kJ mol $^{-1}$

4) Use $\Delta G^\circ = -RT \ln K$

Given $\log K = 2.2$ (base 10),

$\ln K = 2.2 \ln 10 = 2.2 \times 2.303 = 5.0666$

Now,

$\Delta G^\circ = -RT \ln K = -(8.3)(500)(5.0666)$ J mol^{-1}

$\Delta G^\circ = -21026$ J mol^{-1} ≈ -21.0 kJ mol $^{-1}$

5) Equate and solve for x

$49 - x = -21$

$x = 70$ kJ mol $^{-1}$

Nearest integer: $\boxed{70}$

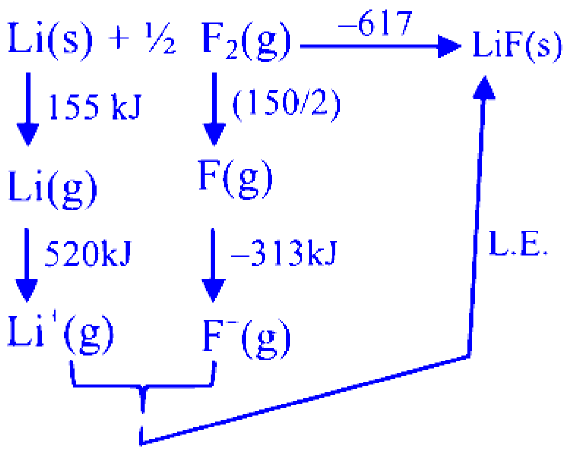
Question18

If the enthalpy of sublimation of Li is 155 kJ mol^{-1}, enthalpy of dissociation of F_2 is 150 kJ mol^{-1}, ionization enthalpy of Li is 520 kJ mol^{-1}, electron gain enthalpy of F is -313 kJ mol^{-1}, standard enthalpy of formation of LiF is -594 kJ mol^{-1}. The magnitude of lattice enthalpy of LiF is ____ kJmol^{-1}. (Nearest Integer)

JEE Main 2026 (Online) 22nd January Evening Shift

Answer: 1031

Solution:



Question19

At the transition temperature T, $A \rightleftharpoons B$ and $\Delta G^0 = 105 - 35 \log T$ where A and B are two states of substance X. The transition temperature in $^\circ\text{C}$ when pressure is 1 atm is ________. (Nearest integer)

JEE Main 2026 (Online) 2nd April Morning Shift

Answer: 727

Solution:

At the transition temperature, the two states A and B are in equilibrium.

So, for the equilibrium condition,

$\Delta G^0 = 0$

Given:

$\Delta G^0 = 105 - 35 \log T$

At transition temperature,

$105 - 35 \log T = 0$

Now solve:

$35 \log T = 105$

$\log T = \frac{105}{35} = 3$

So,

$T = 10^3 = 1000 \mathrm{~K}$

Now convert into degree Celsius:

$T(^{\circ}\mathrm{C}) = 1000 - 273 = 727^{\circ}\mathrm{C}$

Nearest integer:

$\boxed{727}$

Question20

If 3.365 g of ethanol (l) is burnt completely in a bomb calorimeter at 298.15 K , the heat produced is 99.472 kJ . The $|\Delta \mathrm{H_f}^{\circ}|$ of ethanol at 298.15 K is

____ $\times 10^2 \mathrm{~kJ~mol}^{-1}$. (Nearest integer)

Given: Standard enthalpy for combustion of graphite $= -393.5 \mathrm{~kJ~mol}^{-1}$

Standard enthalpy of formation of water $(\mathrm{l}) = -285.8 \mathrm{~kJ~mol}^{-1}$

Molar mass in gmol^{-1} of $\mathrm{C}, \mathrm{H}, \mathrm{O}$ are 12,1 and 16 respectively

JEE Main 2026 (Online) 4th April Evening Shift

Answer: 3

Solution:

For complete combustion of ethanol,

$\mathrm{C_2H_5OH(l)} + 3\mathrm{O_2(g)} \rightarrow 2\mathrm{CO_2(g)} + 3\mathrm{H_2O(l)}$

First, calculate moles of ethanol burnt:

Molar mass of ethanol, $\mathrm{C_2H_5OH}$:

$2(12) + 6(1) + 16 = 46 \mathrm{~g~mol}^{-1}$

So, moles of ethanol:

$n = \frac{3.365}{46} = 0.07315 \mathrm{~mol}$

Given heat produced for 0.07315 mol ethanol is 99.472 kJ.

Hence, molar heat of combustion at constant volume:

$\Delta U_{\text{comb}} = \frac{-99.472}{0.07315}$

$\Delta U_{\text{comb}} \approx -1359.8 \mathrm{~kJ~mol}^{-1}$

Now use the relation:

$\Delta H = \Delta U + \Delta n_g RT$

Here, in the combustion reaction:

- Gaseous reactants = 3 mol O_2
- Gaseous products = 2 mol CO_2

So,

$\Delta n_g = 2 - 3 = -1$

Thus,

$\Delta H_{\text{comb}} = \Delta U_{\text{comb}} + (-1)RT$

At 298.15 K,

$RT = \frac{8.314 \times 298.15}{1000} \approx 2.478 \text{ kJ mol}^{-1}$

Therefore,

$\Delta H_{\text{comb}} = -1359.8 - 2.478 = -1362.3 \text{ kJ mol}^{-1}$

Now apply Hess's law:

$\Delta H^\circ_{\text{comb}} = \left[2\Delta H_f^\circ(CO_2) + 3\Delta H_f^\circ(H_2O)\right] - \Delta H_f^\circ(C_2H_5OH)$

Given:

$\Delta H_f^\circ(CO_2) = -393.5 \text{ kJ mol}^{-1}$

$\Delta H_f^\circ(H_2O(l)) = -285.8 \text{ kJ mol}^{-1}$

So,

$-1362.3 = [2(-393.5) + 3(-285.8)] - \Delta H_f^\circ(\text{ethanol})$

$-1362.3 = [-787.0 - 857.4] - \Delta H_f^\circ(\text{ethanol})$

$-1362.3 = -1644.4 - \Delta H_f^\circ(\text{ethanol})$

$\Delta H_f^\circ(\text{ethanol}) = -1644.4 + 1362.3 = -282.1 \text{ kJ mol}^{-1}$

Therefore,

$\Delta H_f^\circ = 282.1 \text{ kJ mol}^{-1} = 2.821 \times 10^2 \text{ kJ mol}^{-1}$

Nearest integer:

$\boxed{3}$

Question21

Consider the reaction $X \rightleftharpoons Y$ at 300 K . If ΔH^θ and K are $28.40 \text{ kJ mol}^{-1}$ and 1.8×10^{-7} at the same temperature, then the magnitude of ΔS^θ for the reaction in $\text{JK}^{-1} \text{ mol}^{-1}$ is _____ . (Nearest integer)

(Given : $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}, \ln 10 = 2.3, \log 3 = 0.48, \log 2 = 0.30$)

JEE Main 2026 (Online) 6th April Morning Shift

Answer: 34

Solution:

For the reaction,

$X \rightleftharpoons Y$

we use the relation

$\Delta G^{\theta} = \Delta H^{\theta} - T\Delta S^{\theta}$

and also,

$\Delta G^{\theta} = -RT \ln K$

So,

$\Delta H^{\theta} - T\Delta S^{\theta} = -RT \ln K$

Hence,

$\Delta S^{\theta} = \frac{\Delta H^{\theta} + RT \ln K}{T}$

Now substitute the given values:

$\Delta H^{\theta} = 28.40 \text{ kJ mol}^{-1} = 28400 \text{ J mol}^{-1}$

$T = 300 \text{ K}, \quad R = 8.3 \text{ J K}^{-1}\text{mol}^{-1}, \quad K = 1.8 \times 10^{-7}$

First find $\ln K$:

$\ln(1.8 \times 10^{-7}) = \ln 1.8 + \ln 10^{-7}$

Using

$\ln 1.8 = \ln\left(\frac{18}{10}\right) = \ln 18 - \ln 10$

Now,

$\log 18 = \log 2 + 2\log 3 = 0.30 + 2(0.48) = 1.26$

So,

$\log 1.8 = 1.26 - 1 = 0.26$

Therefore,

$\ln 1.8 = 2.3 \times 0.26 = 0.598$

Also,

$\ln 10^{-7} = -7 \ln 10 = -7(2.3) = -16.1$

Thus,

$\ln K = 0.598 - 16.1 = -15.502$

Now calculate $RT \ln K$:

$RT \ln K = 8.3 \times 300 \times (-15.502)$

$= 2490 \times (-15.502) \approx -38600$

Now,

$\Delta S^{\theta} = \frac{28400 + (-38600)}{300}$

$= \frac{-10200}{300} = -34 \text{ J K}^{-1}\text{mol}^{-1}$

So the magnitude of ΔS^{θ} is

$\boxed{34}$

Question22

Consider the reaction

$2H_2S(g) + 3O_2(g) \rightarrow 2H_2O(l) + 2SO_2(g)$

The magnitude of enthalpy change for the reaction in kJmol^{-1} is ____ . (Nearest integer)

$$Given : \Delta_f H^\ominus (H_2 S) = -20.1 \text{ kJ mol}^{-1}$$
$$\Delta_f H^\ominus (H_2O) = -286.0 \text{ kJ mol}^{-1}$$
$$\Delta_f H^\ominus (SO_2) = -297.0 \text{ kJ mol}^{-1}$$

JEE Main 2026 (Online) 8th April Evening Shift

Answer: 1126

Solution:

We are given the reaction:

$2H_2S(g) + 3O_2(g) \rightarrow 2H_2O(l) + 2SO_2(g)$

We are also given the standard enthalpies of formation:

$\Delta_f H^\circ(H_2S) = -20.1 \text{ kJ mol}^{-1}$
$\Delta_f H^\circ(H_2O) = -286.0 \text{ kJ mol}^{-1}$
$\Delta_f H^\circ(SO_2) = -297.0 \text{ kJ mol}^{-1}$

According to NCERT convention,

$\Delta_r H^\circ = \sum \Delta_f H^\circ(\text{products}) - \sum \Delta_f H^\circ(\text{reactants})$

Products:

$2H_2O(l) + 2SO_2(g)$

$\sum \Delta_f H^\circ(\text{products}) = [2(-286.0) + 2(-297.0)] \text{ kJ}$

$= (-572.0 - 594.0) = -1166.0 \text{ kJ}$

Reactants:

$2H_2S(g) + 3O_2(g)$

$\sum \Delta_f H^\circ(\text{reactants}) = [2(-20.1) + 3(0)] \text{ kJ} = -40.2 \text{ kJ}$

$\Delta_r H^\circ = (-1166.0) - (-40.2) = -1125.8 \text{ kJ}$

The **magnitude of enthalpy change** is

$|\Delta_r H^\circ| = 1126 \text{ kJ mol}^{-1} \quad \text{(nearest integer)}$

Final Answer

$\boxed{1126 \text{ kJ mol}^{-1}}$

Question23

A liquid when kept inside a thermally insulated closed vessel at 25°C was mechanically stirred from outside. What will be the correct option for the following thermodynamic parameters ?

JEE Main 2025 (Online) 22nd January Morning Shift

Options:

A. $\Delta U = 0, q < 0, w > 0$

B. $\Delta \mathrm{U}>0, \mathrm{q}=0, \mathrm{w}>0$

C. $\Delta \mathrm{U}=0, \mathrm{q}=0, \mathrm{w}=0$

D. $\Delta \mathrm{U}<0, \mathrm{q}=0, \mathrm{w}>0$

Answer: B

Solution:

Let’s analyze the situation step by step:

System: The liquid inside a **thermally insulated** (adiabatic) and **closed** vessel.

Process: The liquid is **mechanically stirred** from outside.

Heat Exchange (q)

Because the vessel is **thermally insulated**, there is **no heat exchange** between the system and surroundings. Hence,

$q=0$.

Work (w)

The external mechanical stirring does **work on the system** by agitating the liquid. Work done **on** the system is conventionally taken as **positive**:

$w>0$.

Change in Internal Energy (ΔU)

From the First Law of Thermodynamics,

$\Delta U \ = \ q \ + \ w.$

Since $q=0$ and $w>0$, we get

$\Delta U=w>0$.

Hence:

$\boxed{\Delta U>0, \quad q=0, \quad w>0.}$

Answer: Option B is correct.

Question24

Match List - I with List - II.

	List - I (Partial Derivatives)		List - II (Thermodynamic Quantity)
(A)	$\left(\frac{\partial \mathrm{G}}{\partial \mathrm{T}}\right)_{\mathrm{P}}$	(I)	Cp
(B)	$\left(\frac{\partial \mathrm{H}}{\partial \mathrm{T}}\right)_{\mathrm{P}}$	(II)	$-$S
(C)	$\left(\frac{\partial \mathrm{G}}{\partial \mathrm{P}}\right)_{\mathrm{T}}$	(III)	Cv
(D)	$\left(\frac{\partial \mathrm{U}}{\partial \mathrm{T}}\right)_{\mathrm{V}}$	(IV)	V

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 22nd January Evening Shift

Options:

A. (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

B. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

C. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

D. (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

Answer: C

Solution:

We want to match each partial derivative in List–I to the appropriate thermodynamic quantity in List–II:

---

1. $\left(\dfrac{\partial G}{\partial T}\right)_P$

A well-known thermodynamic identity is:

$\left(\frac{\partial G}{\partial T}\right)_P = -S.$

Hence,

$(A) \quad \longrightarrow \quad -S \quad (\text{II}).$

---

2. $\left(\dfrac{\partial H}{\partial T}\right)_P$

By definition of the heat capacity at constant pressure,

$\left(\frac{\partial H}{\partial T}\right)_P = C_P.$

Hence,

$(B) \quad \longrightarrow \quad C_P \quad (\text{I}).$

---

3. $\left(\dfrac{\partial G}{\partial P}\right)_T$

A standard thermodynamic relation (for a single-component system) is:

$\left(\frac{\partial G}{\partial P}\right)_T = V.$

Hence,

$(C) \quad \longrightarrow \quad V \quad (\text{IV}).$

---

4. $\left(\dfrac{\partial U}{\partial T}\right)_V$

By definition of the heat capacity at constant volume,

$\left(\frac{\partial U}{\partial T}\right)_V = C_V.$

Hence,

$(D) \quad \longrightarrow \quad C_V \quad (\text{III}).$

Final Matching

$(A) \to (\text{II}), \quad (B) \to (\text{I}), \quad (C) \to (\text{IV}), \quad (D) \to (\text{III}).$

Which corresponds to:

Option C: (A)-(II), (B)-(I), (C)-(IV), (D)-(III).

Question25

Ice at $-5^{\circ}C$ is heated to become vapor with temperature of $110^{\circ}C$ at atmospheric pressure. The entropy change associated with this process can be obtained from

JEE Main 2025 (Online) 23rd January Morning Shift

Options:

A. $\int_{268\ K}^{273\ K} C_{p,m} dT + \frac{\Delta H_{m,\ fusion}}{T_f} + \frac{\Delta H_{m,\ vaporisation}}{T_b} + \int_{273\ K}^{373\ K} C_{p,m} dT + \int_{373\ K}^{383\ K} C_{p,m} dT$

B. $\int_{268\ K}^{383\ K} C_p dT + \frac{\Delta H_{melting}}{273} + \frac{\Delta H_{boiling}}{373}$

C. $\int_{268\ K}^{383\ K} C_p dT + \frac{q_{rev}}{T}$

D. $\int_{268\ K}^{273\ K} \frac{C_{p,m}}{T} dT + \frac{\Delta H_{m,\ fusion}}{T_f} + \frac{\Delta H^{373\ K}_{m,\ vaporisation}}{T_b} + \int_{273\ K} \frac{C_{p,m} dT}{T} + \int_{373\ K}^{383\ K} \frac{C_{p,m} dT}{T}$

Answer: D

Solution:

$$\underbrace{\underset{268\ K}{\text{Ice}} \rightarrow \underset{273\ K}{\text{Ice}}}_{(1)} \underbrace{\rightleftharpoons \underset{273\ K}{\text{Water}}}_{(2)} \underbrace{\rightarrow \underset{273\ K}{\text{Water}}}_{(3)} \underbrace{\rightleftharpoons \underset{373\ K}{\text{Water vapour}}}_{(4)} \underbrace{\rightarrow \underset{383\ K}{\text{Water vapour}}}_{(5)}$$

$\Delta S_{overall} \quad = \Delta S_1 + \Delta S_2 + \Delta S_3 + \Delta S_4 + \Delta S_5$

$\Delta\ S_2 = \frac{\Delta H_{m\ fusion}}{273} \quad T_f = 273'K'$

$\Delta S_3 = \int_{273}^{373} \frac{C_{p,\ m} dT}{T}$

$\Delta\ S_4 = \frac{\Delta H_{m\ vaporisation}}{373} \quad T_b = 373'K'$

$\Delta S_5 = \int_{373}^{383} \frac{C_{p,\ m} dT}{T}$

Answer $= (2)$

Question26

The effect of temperature on spontaneity of reactions are represented as :

	ΔH	ΔS	Temperature	Spontaneity
(A)	$+$	$-$	any T	Non spontaneous
(B)	$+$	$+$	low T	spontaneous
(C)	$-$	$-$	low T	Non spontaneous
(D)	$-$	$+$	any T	spontaneous

The incorrect combinations are :

Options:

A. (A) and (C) only

B. (B) and (D) only

C. (A) and (D) only

D. (B) and (C) only

Answer: D

Solution:

Let's analyze each combination using the Gibbs free energy equation:

$\Delta G = \Delta H - T\Delta S$

A reaction is spontaneous when $\Delta G < 0$.

Now, consider each case:

(A) $\Delta H > 0$ and $\Delta S < 0$

Here, $\Delta G = +(\text{positive}) - T(-(\text{positive})) = \text{positive} + T(\text{positive})$

This is positive for any temperature.

Thus, the reaction is non-spontaneous for any T, which is correct.

(B) $\Delta H > 0$ and $\Delta S > 0$

Now, $\Delta G = +(\text{positive}) - T(\text{positive})$.

For ΔG to be negative, the temperature has to be **high** enough such that $T\Delta S > \Delta H$.

The table incorrectly states that the reaction is spontaneous at low T.

(C) $\Delta H < 0$ and $\Delta S < 0$

In this case, $\Delta G = -(\text{positive}) - T(-(\text{positive})) = -(\text{positive}) + T(\text{positive})$.

At very low temperatures, the term $T\Delta S$ is small, making ΔG negative (spontaneous).

The table mistakenly indicates that the reaction is non-spontaneous at low T.

(D) $\Delta H < 0$ and $\Delta S > 0$

Then, $\Delta G = -(\text{positive}) - T(\text{positive})$.

This expression is negative at all temperatures, so the reaction is correctly marked as spontaneous.

Thus, the incorrect combinations are:

(B): It should be spontaneous at high T, not low T.

(C): It should be spontaneous at low T, not non-spontaneous.

Therefore, the incorrect combinations are (B) and (C) only.

The correct answer is Option D.

Question27

Let us consider an endothermic reaction which is non-spontaneous at the freezing point of water. However, the reaction is spontaneous at boiling point of water. Choose the correct option.

JEE Main 2025 (Online) 24th January Morning Shift

Options:

A. Both ΔH and ΔS are (-ve)

B. ΔH is $(+\text{ve})$ but ΔS is (-ve)

C. ΔH is $(-\text{ve})$ but ΔS is (+ve)

D. Both ΔH and ΔS are (+ve)

Answer: D

Solution:

Reaction is spontaneous at relatively high temperature and non-spontaneous at low temperature $\Delta G = \Delta H - T\Delta S$It is only possible when ΔH and ΔS both are positive.

Option (1)

Question28

Which of the following mixing of 1 M base and 1 M acid leads to the largest increase in temperature?

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. 50 mL HCl and 20 mL NaOH

B. 30 mL HCl and 30 mL NaOH

C. 45 mL CH_3COOH and 25 mL NaOH

D. 30 mL CH_3COOH and 30 mL NaOH

Answer: B

Solution:

The rise in temperature of neutralization reaction will be maximum for maximum number of moles of strong acid and strong base neutralized and lower volume of final solution.

$$\begin{array}{lllll} (B) & HCl & + NaOH & \rightarrow & NaCl + H_2O \\ \text{mmol} & 30 & 30 & & \\ & - & - & & \end{array}$$

Option (D) and (C) have weak acids and in option (A) only 20 mmol of HCl will be neutralized with 70 mL final volume.

Question29

$$S(g)+\frac{3}{2}O_2(g) \rightarrow SO_3(g)+2x\text{kcal}$$

$$SO_2(g)+\frac{1}{2}O_2(g) \rightarrow SO_3(g)+y\text{kcal}$$

The heat of formation of $SO_2(g)$ is given by :

JEE Main 2025 (Online) 24th January Evening Shift

Options:

A. $2x+y$ kcal

B. $\frac{2x}{y}$ kcal

C. $y-2x$ kcal

D. $x+y$ kcal

Answer: C

Solution:

To determine the heat of formation for $SO_2(g)$, we need to consider the given reactions and their enthalpy changes.

The reactions are:

$S(g)+\frac{3}{2}O_2(g) \rightarrow SO_3(g)+2x\text{kcal}$

$SO_2(g)+\frac{1}{2}O_2(g) \rightarrow SO_3(g)+y\text{kcal}$

For reaction 2, we have:

$\Delta H_r=(\Delta H_f)_{SO_3}-(\Delta H_f)_{SO_2}=-y$

Given that:

$-y=-2x-(\Delta H_f)_{SO_2}$

Rearranging this equation gives:

$(\Delta H_f)_{SO_2}=y-2x$

Thus, the heat of formation for $SO_2(g)$ is $y-2x$ kcal.

Question30

Ice and water are placed in a closed container at a pressure of 1 atm and temperature 273.15 K . If pressure of the system is increased 2 times, keeping temperature constant, then identify correct observation from following

JEE Main 2025 (Online) 28th January Morning Shift

Options:

A. Liquid phase disappears completely.

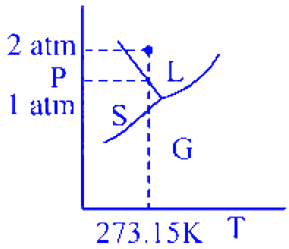
B. The amount of ice decreases.

C. The solid phase (ice) disappears completely.

D. Volume of system increases .

Answer: C

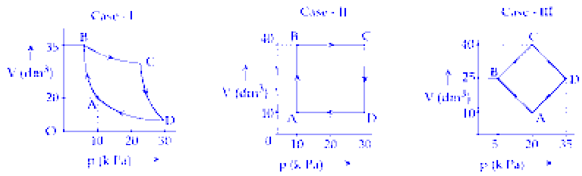
Solution:



Phase diagram of H_2O

If pressure is made two time then mixture of ice and water will completely convert into water (liquid) form.

Question31



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A→B→C→D→A as shown in the three cases above.

Choose the correct option regarding ΔU :

JEE Main 2025 (Online) 28th January Evening Shift

Options:

A.

$\Delta \mathrm{U}(\text{ Case-III }) > \Delta \mathrm{U}(\text{ Case-II }) > \Delta \mathrm{U}(\text{ Case-I })$

B. $\Delta \mathrm{U}(\text{ Case-I }) > \Delta \mathrm{U}(\text{ Case-II }) > \Delta \mathrm{U}(\text{ Case-III })$

C. $\Delta \mathrm{U}(\text{ Case-I }) = \Delta \mathrm{U}(\text{ Case-II }) = \Delta \mathrm{U}(\text{ Case-III })$

D. $\Delta \mathrm{U}(\text{ Case-I }) > \Delta \mathrm{U}(\text{ Case-III }) > \Delta \mathrm{U}(\text{ Case-II })$

Answer: C

Solution:

As internal energy ' U ' is a state function, its cyclic integral must be zero in a cyclic process

$\therefore \Delta U \text{ case } (I) = \Delta U \text{ case } (II) = \Delta U \text{ case (III)}$

Question32

500 J of energy is transferred as heat to 0.5 mol of Argon gas at 298 K and 1.00 atm. The final temperature and the change in internal energy respectively are: Given: R = 8.3 J K^{-1} mol^{-1}

JEE Main 2025 (Online) 29th January Morning Shift

Options:

A.

368 K and 500 J

B.

348 K and 300 J

C.

378 K and 300 J

D.

378 K and 500 J

Answer: B

Solution:

Amount of energy transpired as hat, $Q = 500\text{ J}$

Number of moles of Argon, $n = 0.5\text{ mol}$

Temperature (Initial), $T_i = 298\text{ K}$

Pressure, $P = 1.00\text{ atm}$

Final temperature, $T_F = ?$

Change in internal energy, $\Delta V = ?$

$R = 8.3\text{ J K}^{-1}\text{ mol}^{-1}$

For a constant pressure process, the heat added is related to the temperature changes as

$Q = nC_p\Delta T$ .... (1)

Also, the change in internal energy formula is
$\Delta U = nC_V\Delta T \quad -(2)$ .... (2)

For monoatomic gas (Argon),

$$C_v = \frac{3}{2}R$$

So, $C_p = C_v + R \quad (C_p - C_v = R)$

$$= \frac{3}{2}R + R = \frac{5}{2}R$$

In equation (1) $\Rightarrow$

$Q = nC_p\Delta T$

$Q = nC_p(T_F - T_i)$

$$500\text{ J} = 0.5\text{ mol} \times \frac{5}{2}R(T_F - T_i) \quad R = 8.3\text{JK}^{-1}\text{ mol}^{-1}$$

$$500\text{ J} = 0.5\text{ mol} \times \frac{5}{2} \times 8.3\text{ J}K^{-1}\text{ mol}^{-1}(T_F - T_i)$$

$$T_F - T_i = \frac{500\text{ J}}{0.5\text{mol} \times \frac{5}{2} \times 8.3\text{ J K}^{\ 1}\text{ mol}^{-1}}$$

$$
\begin{aligned}
&= \frac{500}{0.5 \times \frac{5}{2} \times 8.3}\mathrm{k} \\
&= 48.2\mathrm{k} \\
T_F &= 48.2\mathrm{k} + 298\mathrm{k} \\
&= 346.2\mathrm{k} \\
&\approx 348\mathrm{k}
\end{aligned}
$$

In equation (2) $\Rightarrow$

$$
\begin{aligned}
\Delta U &= nC_V\Delta T \\
&= 0.5\ \mathrm{mol} \times \frac{3}{2}R \times 48.2\ \mathrm{K} \\
&= 0.5\ \mathrm{mol} \times \frac{3}{2} \times 8.3\ \mathrm{J\ K^{-1} mol^{-1}} \times 48.2\ \mathrm{K} \\
&= 300.045\ \mathrm{J} \approx 300\ \mathrm{J}
\end{aligned}
$$

The final temperature is, 348 k

change in internal energy is 300 J

So, correct answer is Option 2) 348 K and 300 J

Question33

If C (diamond) $\rightarrow C$ (graphite) $+X$ kJ mol^{-1}

C (diamond) $+\mathrm{O}_2(\mathrm{~g}) \rightarrow \mathrm{CO}_2(\mathrm{~g})+\mathrm{YkJmol}^{-1}$

C (graphite) $+\mathrm{O}_2(\mathrm{~g}) \rightarrow \mathrm{CO}_2(\mathrm{~g})+\mathrm{ZkJmol}^{-1}$

at constant temperature. Then

JEE Main 2025 (Online) 29th January Evening Shift

Options:

A.

$-\mathrm{X}=\mathrm{Y}+\mathrm{Z}$

B.

$\mathrm{X}=\mathrm{Y}-\mathrm{Z}$

C.

$\mathrm{X}=-\mathrm{Y}+\mathrm{Z}$

D.

$\mathrm{X}=\mathrm{Y}+\mathrm{Z}$

Answer: B

Solution:

Given,

$C_{(diamond)} \longrightarrow C_{(graphite)} + X\,kJ\,mol^{-1}$ .... (1)

$C_{(diamond)} + O_2(g) \longrightarrow CO_2(g) + Y\,kJ\,mol^{-1}$ .... (2)

$C_{(graphite)} + O_2(g) \longrightarrow CO_2(g) + Z\,kJ\,mol^{-1}$ .... (3)

Condition for temperature : constant

Hess's law is applied here. (Hess's law of constant heat summation)

The given reactions (2) and (3),

(2) - (3) $\Rightarrow$

$C_{(diamond)} + O_2(g) - \left(C_{(graphite)} + O_2(g)\right) \longrightarrow CO_2(g) + Y - (CO_2(g) + Z)$

$C_{(diamond)} + O_2(g) - C_{(grpahite)} - O_2(g) \longrightarrow CO_2(g) + Y - CO_2(g) - Z$

$C_{(diamond)} - C_{(graphite)} \rightarrow Y - Z$

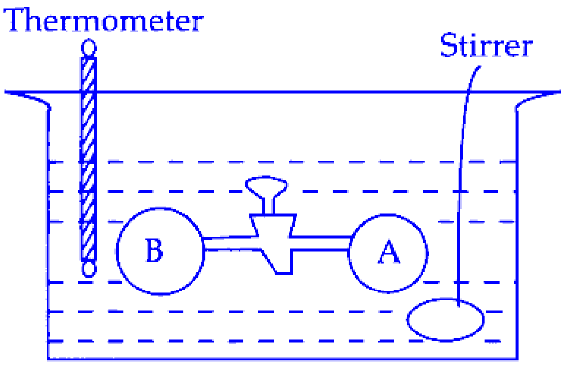
$C_{(diamond)} \rightarrow C_{(graphite)} + (Y - Z)$ .... (4)

Comparing (1) and (4),

$X = Y - Z$

So, the correct answer is option (2) $X = Y - Z$

Question34



Two vessels A and B are connected via stopcock. The vessel A is filled with a gas at a certain pressure. The entire assembly is immersed in water and is allowed to come to thermal equilibrium with water. After opening the stopcock the gas from vessel A expands into vessel B and no change in temperature is observed in the thermometer. Which of the following statement is true ?

JEE Main 2025 (Online) 2nd April Morning Shift

Options:

A. dq $\neq$ 0

B. The pressure in the vessel B before opening the stopcock is zero

C. dw $\neq$ 0

D. dU $\neq$ 0

Answer: B

Solution:

It is free expansion of gas $\Rightarrow P_{\text{ext}} = 0$

Where $\mathrm{w} = 0, \mathrm{q} = 0$ and $\Delta \mathrm{U} = 0$

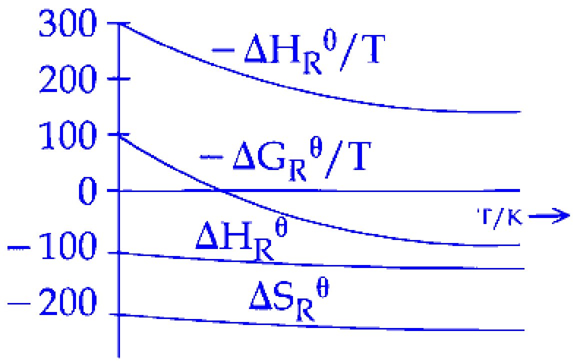
Question35

Which of the following graphs correctly represents the variation of thermodynamic properties of Haber's process?

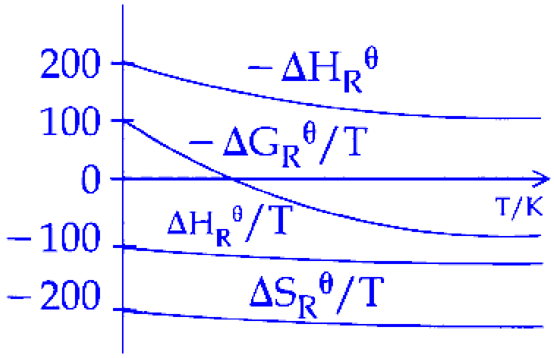
JEE Main 2025 (Online) 2nd April Evening Shift

Options:

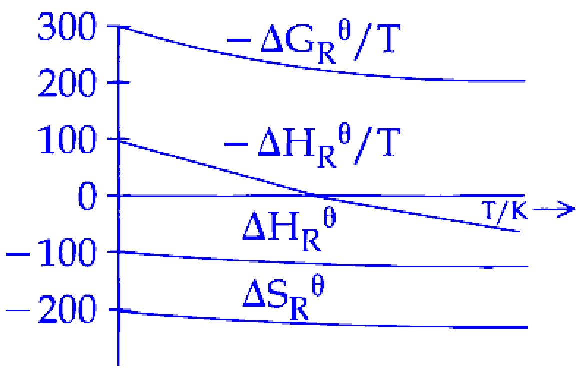
A.



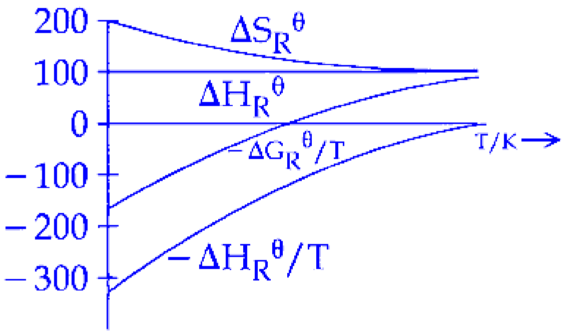
B.



C.



D.



Answer: A

Solution:

In the Haber process, the chemical reaction is as follows:

$N_2(g) + 3H_2(g) \rightarrow 2NH_3$

The change in enthalpy (ΔH°) is negative, indicating that the reaction is exothermic.

The change in entropy (ΔS°) is also negative because the number of moles of gas decreases during the reaction.

Key points to consider:

As temperature increases, the term $-\frac{\Delta H_R^0}{T}$ decreases.

The free energy change is given by:

$\Delta G^\circ = -RT \ln K_{eq}$

Rearranging gives:

$R \ln K_{eq} = -\frac{\Delta G^0}{T}$

For an exothermic reaction, as temperature increases, the equilibrium constant (K_{eq}) decreases.

Both ΔH° and ΔS° are almost constant with respect to temperature.

Question36

Arrange the following in order of magnitude of work done by the system/on the system at constant temperature.

(a) $|w_{\text{reversible}}|$ for expansion in infinite stages.

(b) $|w_{\text{irreversible}}|$ for expansion in single stage.

(c) $|w_{\text{reversible}}|$ for compression in infinite stages.

(d) $|w_{\text{irreversible}}|$ for compression in single stage.

Choose the correct answer from the options given below :

JEE Main 2025 (Online) 2nd April Evening Shift

Options:

A.

$d > c = a > b$

B.

$c = a > d > b$

C.

$a > c > b > d$

D.

$a > b > c > d$

Answer: A

Solution:

For isothermal process

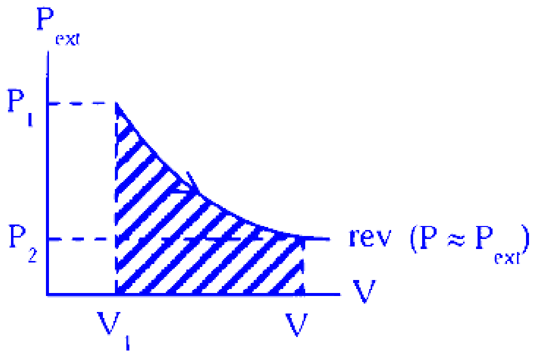
$$|W_{\text{reversible}}|_{\text{expansion}} = |W_{\text{reversible}}|_{\text{compression}}$$

$$= -nRT \ln \frac{V_f}{V_i}$$

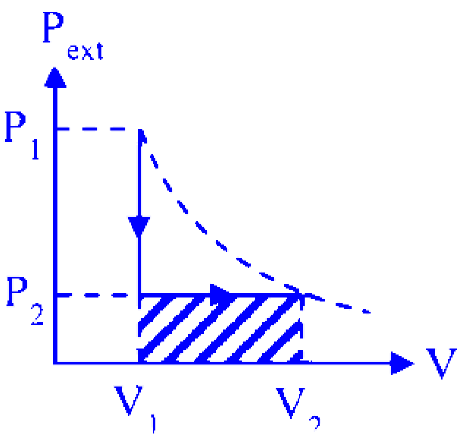
$$|W_{\text{irreversible}}|_{\text{expansion}} < |W_{\text{irreversible}}|_{\text{compression}}$$

$$d > c = a > b$$

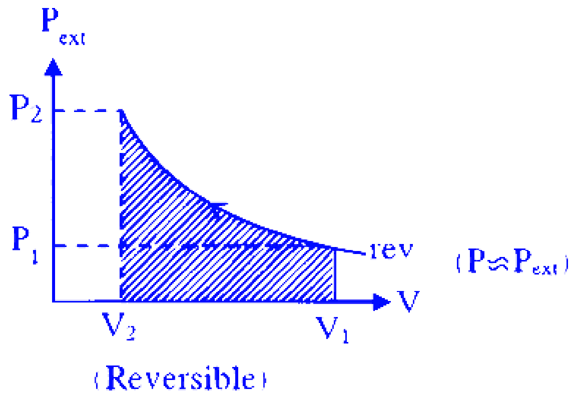
$$|W_{\text{irreversible}}|_{\text{expansion}} = -P_{ext}\ (V_f - V_i)$$

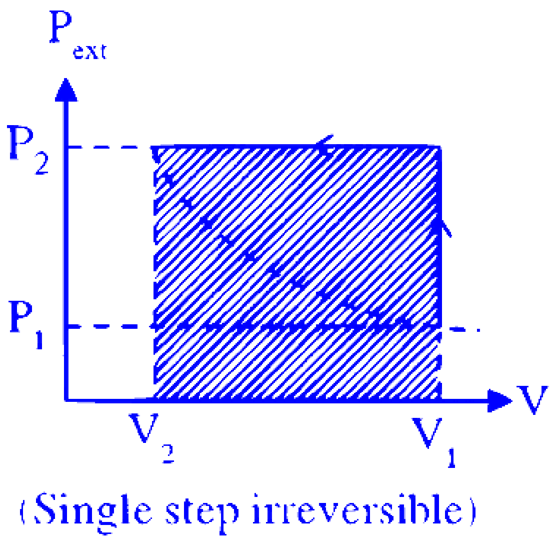


(Reversible expansion)



(Single step irreversible)





Graphical representation

We can compare work by area of PV graph.

Question37

Given below are two statements:

Statement I : When a system containing ice in equilibrium with water (liquid) is heated, heat is absorbed by the system and there is no change in the temperature of the system until whole ice gets melted.

Statement II : At melting point of ice, there is absorption of heat in order to overcome intermolecular forces of attraction within the molecules of water in ice and kinetic energy of molecules is not increased at melting point.

In the light of the above statements, choose the correct answer from the options given below

JEE Main 2025 (Online) 3rd April Evening Shift

Options:

A. Both Statement I and Statement II are false

B. Statement I is true but Statement II is false

C. Both Statement I and Statement II are true

D. Statement I is false but Statement II is true

Answer: C

Solution:

The correct choice is Option C: Both Statement I and Statement II are true.

Explanation:

Statement I is true because

During the ice–water phase change at 0 °C, any heat added goes into the latent heat of fusion.

The temperature remains fixed at 0 °C until all the ice has melted.

Statement II is also true because

The heat absorbed at the melting point is used to overcome (break) the hydrogen-bonding forces between water molecules in ice—in other words, it increases the system's potential energy.

Since no heat goes into raising the kinetic energy of the molecules, the temperature (which measures average kinetic energy) does not increase during melting.

Mathematically, for a mass m of ice, the heat needed is

$Q = m\,L_f$

where L_f is the latent heat of fusion of ice. This heat does not change the temperature but is entirely consumed in changing the phase (i.e., increasing potential energy).

Question38

Let us consider a reversible reaction at temperature, T.In this reaction, both $\Delta \mathrm{H}$ and $\Delta \mathrm{S}$ were observed to have positive values. If the equilibrium temperature is Te , then the reaction becomes spontaneous at:

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. $\mathrm{Te} > \mathrm{T}$

B. $\mathrm{T} > \mathrm{Te}$

C. $\mathrm{T} = \mathrm{Te}$

D. $\mathrm{Te} = 5\ \mathrm{T}$

Answer: B

Solution:

For reaction to be spontaneous according to 2^{nd} law:

$$
\begin{aligned}
& \Delta \mathrm{G} < 0 \\
& \Rightarrow \Delta \mathrm{H} - \mathrm{T}\Delta\ \mathrm{S} < 0 \\
& \Rightarrow\ \mathrm{T} > \left(\frac{\Delta \mathrm{H}}{\Delta\ \mathrm{S}}\right) = \mathrm{T_e} \\
& \Rightarrow\ \mathrm{T} > \mathrm{T_e}
\end{aligned}
$$

Question39

One mole of an ideal gas expands isothermally and reversibly from $10\mathrm{dm}^3$ to $20\mathrm{dm}^3$ at 300 K . $\Delta \mathrm{U}, \mathrm{q}$ and work done in the process respectively are

Given: $\mathrm{R}=8.3 \mathrm{~J} \mathrm{~K}^{-1} \mathrm{~mol}^{-1}$

$\ln 10=2.3$

$\log 2=0.30$

$\log 3=0.48$

JEE Main 2025 (Online) 4th April Morning Shift

Options:

A. $0, 21.84 \mathrm{~kJ},-1.726 \mathrm{~J}$

B. $0, 21.84 \mathrm{~kJ}, 21.84 \mathrm{~kJ}$

C. $0, 1.718 \mathrm{~kJ},-1.718 \mathrm{~kJ}$

D. $0,-17.18 \mathrm{~kJ}, 1.718 \mathrm{~J}$

Answer: C

Solution:

$$
\begin{aligned}
& (10 \mathrm{~L}, 300 \mathrm{~K}) \xrightarrow{\mathrm{n}=1}(20 \mathrm{~L}, 300 \mathrm{~K}) \\
& -\mathrm{q}=\mathrm{w}=-\mathrm{nRT} \ln \frac{\mathrm{V}_2}{\mathrm{~V}_1} \\
& =-8.3 \times 300 \times \ln \left(\frac{20}{10}\right) \\
& =-1.718 \mathrm{~kJ} \\
& \Rightarrow \mathrm{q}=1.718 \mathrm{~kJ} \\
& \mathrm{w}=-1.718 \mathrm{~kJ} \\
& \Delta \mathrm{U}=0(\because \Delta \mathrm{~T}=0)
\end{aligned}
$$

Question40

Consider the given data :

(a) $\mathrm{HCl}(\mathrm{g})+10 \mathrm{H}_2 \mathrm{O}(\mathrm{l}) \rightarrow \mathrm{HCl} .10 \mathrm{H}_2 \mathrm{O} \Delta \mathrm{H}=-69.01 \mathrm{~kJ} \mathrm{~mol}^{-1}$

(b) $\mathrm{HCl}(\mathrm{g})+40 \mathrm{H}_2 \mathrm{O}(\mathrm{l}) \rightarrow \mathrm{HCl} .40 \mathrm{H}_2 \mathrm{O} \Delta \mathrm{H}=-72.79 \mathrm{~kJ} \mathrm{~mol}^{-1}$

Choose the correct statement :

JEE Main 2025 (Online) 4th April Evening Shift

Options:

A. The heat of dilution for the HCl $\left(\mathrm{HCl} .10 \mathrm{H}_2 \mathrm{O}\right.$ to $\left.\mathrm{HCl} .40 \mathrm{H}_2 \mathrm{O}\right)$ is $3.78 \mathrm{~kJ} \mathrm{~mol}^{-1}$.

B. Dissolution of gas in water is an endothermic process.

C. The heat of solution depends on the amount of solvent.

D. The heat of formation of HCl solution is represented by both (a) and (b).

Answer: C

Solution:

From the given information

ΔH is negative so it means dissolution of gas HCl(g) is exothermic.

$\mathrm{HCl(\,g)+10H_2O(l) \rightarrow HCl.10H_2O}$ ..... (1)

$\Delta H_1 = -69.01 \frac{\mathrm{kJ}}{\mathrm{mol}}$

$\mathrm{HCl(\,g)+40H_2O(l) \rightarrow HCl.40H_2O}$ ..... (2)

$\Delta H_2 = -72.79 \frac{\mathrm{kJ}}{\mathrm{mol}}$

Hence heat of solution depends upon amount of solvent

By equation .... (2) $-$ equation .... (1)

$\mathrm{HCl.10H_2O + 30H_2O(\ell) \rightarrow HCl.40H_2O}$

So Heat of dilution $= -72.79 - (-69.01)$

$= -3.78 \frac{\mathrm{kJ}}{\mathrm{mol}}$

Hence option (3) is incorrect.

For heat of formation reactant should be in elemental form hence option (4) is incorrect

----

Question41

Total enthalpy change for freezing of 1 mol of water at $10^{\circ}\mathrm{C}$ to ice at $-10^{\circ}\mathrm{C}$ is _______

(Given: $\Delta_{\text{fus}}\ \mathrm{H} = x\,\mathrm{kJ/mol}$

$$\mathrm{C_p\left[H_2O(l)\right]} = y\,\mathrm{J\,mol^{-1}\,K^{-1}}$$
$$\mathrm{C_p\left[H_2O(\,s)\right]} = z\,\mathrm{J\,mol^{-1}\,K^{-1}}$$

JEE Main 2025 (Online) 7th April Morning Shift

Options:

A. $-x-10y-10z$

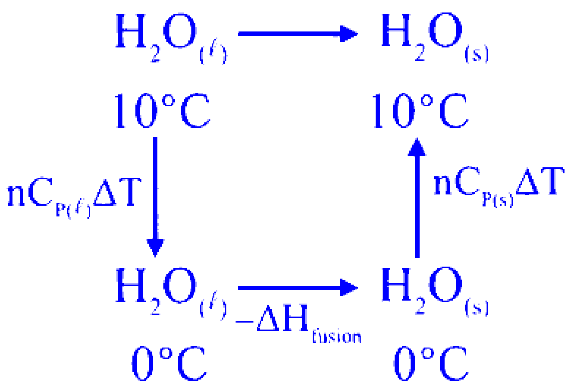
B. $x-10y-10z$

C. $-10(100x+y+z)$

D. $10(100\mathrm{x}+y+z)$

Answer: C

Solution:



$\Delta H = 1 \times y(0 - 10) - x \times 1000 + 1 \times z\,(-10^\circ - 0^\circ)$
$\Delta H = -10(100x + y + z)$ Joule.

Question42

The correct statement amongst the following is :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

The standard state of a pure gas is the pure gas at a pressure of 1 bar and temperature 273 K.

B.

$\Delta_f H^\circ_{500}$ is zero for $O_2(g)$.

C.

$\Delta_f H^\circ_{298}$ is zero for $O(g)$.

D.

The term 'standard state' implies that the temperature is 0°C.

Answer: B

Solution:

Option A:

It claims that the standard state for a pure gas is defined at 1 bar and 273 K.

In modern thermochemistry, the standard state generally means the pure substance at 1 bar, but the standard temperature most commonly used is 298 K (25°C), not 273 K (0°C).

Thus, Option A is incorrect.

Option B:

It states that the standard enthalpy of formation $\Delta_f H^\circ_{500}$ is zero for $O_2(g)$.

By convention, the standard enthalpy of formation for an element in its most stable form is defined as zero. Even though typical tables list $\Delta_f H^\circ_{298}$ values, if one were to evaluate the formation enthalpy at any temperature where the element remains in its standard state, the reaction

$O_2(g) \rightarrow O_2(g)$

has no change in enthalpy.

Therefore, by definition, $\Delta_f H^\circ_{500}(O_2(g)) = 0$.

This makes Option B correct.

Option C:

It claims that $\Delta_f H^\circ_{298}$ is zero for $O(g)$ (atomic oxygen).

However, the standard state of oxygen is diatomic $O_2(g)$, not atomic oxygen. Since $O(g)$ is not the most stable form of oxygen under standard conditions, its formation enthalpy is not zero.

Hence, Option C is incorrect.

Option D:

It asserts that the term "standard state" implies that the temperature is 0°C (273 K).

In fact, "standard state" typically specifies a standard pressure (1 bar) and a chosen temperature—commonly 298 K for tabulated thermodynamic data—not necessarily 0°C.

Thus, Option D is also incorrect.

To summarize:

The only correct statement is Option B.

Therefore, the correct answer is: Option B.

Question43

The hydration energies of K^+ and Cl^- are $-x$ and $-y$ kJ/mol respectively. If the lattice energy of KCl is $-z$ kJ/mol, then the heat of solution of KCl is :

JEE Main 2025 (Online) 7th April Evening Shift

Options:

A.

$x+y+z$

B.

$z-(x+y)$

C.

$-z-(x+y)$

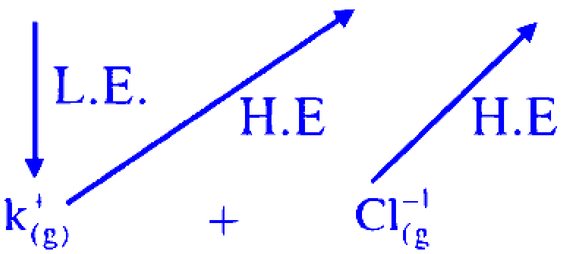
D.

$x-y-z$

Answer: B

Solution:

$\mathrm{KCl_{(s)}} + \mathrm{H_2O} \xrightarrow{\Delta\mathrm{H\ sol.}} \mathrm{K^+_{(aq)}} + \mathrm{Cl^{-1}_{(aq)}}$



$$\begin{aligned}\Delta H_{Sol^n.} &= L\cdot E\cdot + (H\cdot E)_{K^+_{(g)}} + (HE)_{Cl^{-1}_{(g)}}\\ &= Z - x - y\\ &= z - (x+y)\end{aligned}$$

Question44

Consider the following cases of standard enthalpy of reaction (ΔH_r° in $kJmol^{-1}$)

$$\begin{aligned}&C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)\Delta H_1^\circ = -1550\\ &C(\text{graphite}) + O_2(g) \rightarrow CO_2(g)\Delta H_2^\circ = -393.5\\ &H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)\Delta H_3^\circ = -286\end{aligned}$$

The magnitude of $\Delta H^\circ_{fC_2H_6(g)}$ is _________ $kJmol^{-1}$ (Nearest integer).

JEE Main 2025 (Online) 22nd January Evening Shift

Answer: 95

Solution:

$$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(I)\Delta H_1^\circ = -1550\ldots \text{(i)}$$

$$C \text{ (graphite) } + O_2(g) \rightarrow CO_2(g)\Delta H_2^\circ = -393.5\ldots \text{(ii)}$$

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(I)\Delta H_3^\circ = -286\ldots\ldots(iii)$$

From $2 \times eq^n(ii) + 3 \times eq^n(iii) - eq^n(i)$

$$2C_{\text{(graphite)}} + 3H_2(g) \rightarrow C_2H_6(g) : \Delta H^\circ_{fC_2H_6}$$

$$(\Delta H_f^\circ)_{C_2H_6} = 2 \times (-393.5) + 3 \times (-286) - (-1550)$$

$$= -95\ kJ/mol$$

Question45

The standard enthalpy and standard entropy of decomposition of N_2O_4 to NO_2 are $55.0\ kJ\ mol^{-1}$ and $175.0\ J/K/mol$ respectively. The standard free energy change for this reaction at $25^\circ C$ in $J\ mol^{-1}$ is ________ (Nearest integer)

JEE Main 2025 (Online) 23rd January Morning Shift

Answer: 2850

Solution:

$\Delta H^{\circ}_{rxn} = 55\ kJ/mol, \quad T = 298\ K$

$\Delta S^{o}_{rxn} = 175\ J/mol$

$\Delta G^{o}_{rxn} = \Delta H^{o}_{rxn} - T\Delta S^{o}_{rxn}$

$\Rightarrow \Delta G^{o}_{rxn} = 55000\ J/mol - 298 \times 175\ J/mol$

$\Rightarrow \Delta G^{o}_{rxn} = 55000 - 52150$

$\Rightarrow \Delta G^{o}_{rxn} = 2850\ J/mol$

Question46

The bond dissociation enthalpy of $X_2 \Delta H^{\circ}_{bond}$ calculated from the given data is ____________ $kJmol^{-1}$. (Nearest integer)

$M^+X^-(s) \rightarrow M^+(g) + X^-(g) \Delta H^{\circ}_{lattice} = 800\ kJ\ mol^{-1}$

$M(s) \rightarrow M(g) \Delta H^{\circ}_{sub} = 100\ kJ\ mol^{-1}$

$M(g) \rightarrow M^+(g) + e^-(g) \Delta H^{\circ}_{i} = 500\ kJ\ mol^{-1}$

$X(g) + e^-(g) \rightarrow X^-(g) \Delta H^{\circ}_{eg} = -300\ kJ\ mol^{-1}$

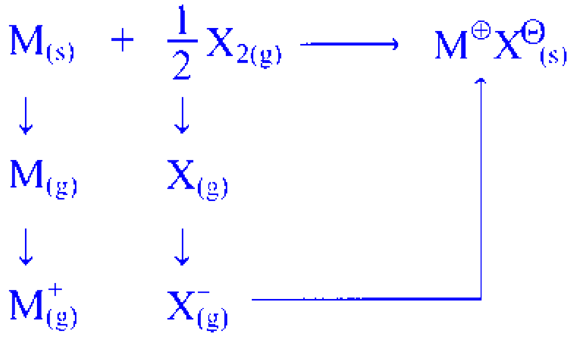
$M(s) + \frac{1}{2} X_2(g) \rightarrow M^+X^-(s) \Delta H^{\circ}_{f} = -400\ kJ\ mol^{-1}$

[Given : M^+X^- is a pure ionic compound and X forms a diatomic molecule X_2 in gaseous state]

JEE Main 2025 (Online) 23rd January Evening Shift

Answer: 200

Solution:



$\therefore \Delta H_f(MX) = \Delta H_{sub}\ (M) + \text{ I.E. } (M) + \frac{1}{2}[\text{ B.E. } (X - X)]$
$\quad + EG(X) + \text{ L.E. } (MX)$

$-400 = (100) + (500) + \frac{1}{2}(\text{ B.E. }) + (-300) + (-800)$

$\therefore \text{ B.E. } = 200\ kJ\ mole^{-1}$

Question47

Standard entropies of X_2, Y_2 and XY_5 are 70, 50 and $110 \ J \ K^{-1} \ mol^{-1}$ respectively. The temperature in Kelvin at which the reaction

$$\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightleftharpoons XY_5 \Delta H^{\ominus} = -35 \ kJ \ mol^{-1}$$

will be at equilibrium is __________ (Nearest integer)

JEE Main 2025 (Online) 24th January Morning Shift

Answer: 700

Solution:

To determine the temperature at which the given reaction is at equilibrium, we need to apply the concept of Gibbs free energy change (ΔG^0) at equilibrium, where ΔG^0 is equal to zero.

The given reaction is:

$\frac{1}{2}X_2 + \frac{5}{2}Y_2 \rightleftharpoons XY_5$

We are given the standard entropies and enthalpy change ($\Delta H^{\ominus}$):

Standard entropies:

$X_2 = 70 \, J \, K^{-1} \, mol^{-1}$

$Y_2 = 50 \, J \, K^{-1} \, mol^{-1}$

$XY_5 = 110 \, J \, K^{-1} \, mol^{-1}$

Standard enthalpy change:

$\Delta H^{\ominus} = -35 \, kJ/mol$

First, calculate the standard entropy change (ΔS^0_{Rxn}) of the reaction:

$\Delta S^0_{Rxn} = 110 - \left(\frac{1}{2} \times 70 + \frac{5}{2} \times 50\right)$

$= 110 - (35 + 125)$

$= 110 - 160$

$= -50 \, J \, K^{-1} \, mol^{-1}$

At equilibrium, $\Delta G^0 = 0$, and the relation $\Delta G^0 = \Delta H^0 - T\Delta S^0$ applies. Thus:

$0 = -35000 \, J/mol - T(-50 \, J \, K^{-1} \, mol^{-1})$

Solving for T:

$T \cdot 50 = 35000$

$T = \frac{35000}{50}$

$T = 700 \, K$

Therefore, the temperature at which the reaction is at equilibrium is 700 Kelvin.

Question48

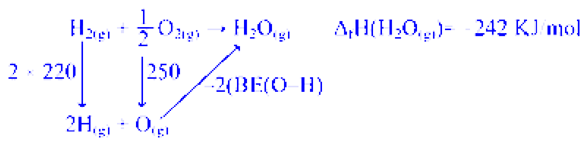
The formation enthalpies, $\Delta H_f^{\ominus}$ for $H_{(g)}$ and $O_{(g)}$ are 220.0 and $250.0 \text{ kJ mol}^{-1}$, respectively, at 298.15 K , and $\Delta H_f^{\ominus}$ for $H_2O_{(g)}$ is $-242.0 \text{ kJ mol}^{-1}$ at the same temperature. The average bond enthalpy of the $O - H$ bond in water at 298.15 K is _______ kJ mol^{-1} (nearest integer).

JEE Main 2025 (Online) 28th January Morning Shift

Answer: 466

Solution:

$\frac{1}{2} H_{2(g)} \rightarrow H(g) \quad ; \Delta_f H \left(H_{(g)}\right) = 220 \text{KJ/mol}$
$\frac{1}{2} O_{2(g)} \rightarrow O(g) \quad ; \Delta_f H \left(O_{(g)}\right) = 250 \text{KJ/mol}$



$\Delta H_f \left(H_2O_{(l)}\right) = -242 = 440 + 250 - 2(\text{ B.E.} (O - H))$
$\text{BE}(O - H) = 466 \text{KJ/mol}$

Question49

Consider the following data :

Heat of formation of $CO_2(g) = -393.5 \text{ kJ mol}^{-1}$

Heat of formation of $H_2O(l) = -286.0 \text{ kJ mol}^{-1}$

Heat of combustion of benzene $= -3267.0 \text{ kJ mol}^{-1}$

The heat of formation of benzene is __________ kJmol^{-1}. (Nearest integer)

JEE Main 2025 (Online) 28th January Evening Shift

Answer: 48

Solution:

To determine the heat of formation of benzene ($\Delta H_f[C_6H_6]$), we use the given data:

Heat of formation of $CO_2(g) = -393.5$ kJ/mol

Heat of formation of $H_2O(l) = -286.0$ kJ/mol

Heat of combustion of benzene = -3267.0 kJ/mol

The reaction for the combustion of benzene is:

$$C_6H_6 + \frac{15}{2}O_2(g) \rightarrow 6CO_2(g) + 3H_2O(l)$$

Using the formula for the enthalpy change of the reaction (ΔH_R):

$$\Delta H_R = \Delta H_C = \Sigma\Delta H_f(\text{Products}) - \Sigma\Delta H_f(\text{Reactants})$$

Substitute values into the equation:

$$-3267 = 6 \times (-393.5) + 3 \times (-286) - \Delta H_f[C_6H_6]$$

Solving this equation, we find:

$$\Delta H_f[C_6H_6] = 48 \text{ kJ/mol}$$

Thus, the heat of formation of benzene is 48 kJ/mol.

Question50

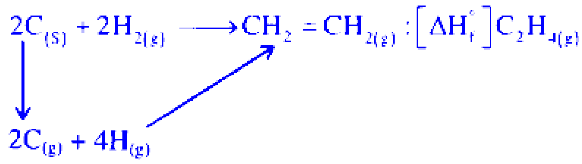
$\Delta H^\ominus{}_{\text{sub}} \ [\text{C (graphite)}] = 710 \text{ kJ mol}^{-1}$
$\Delta_{C-H}H^\ominus = 414 \text{ kJ mol}^{-1}$
$\Delta_{H-H}H^\ominus = 436 \text{ kJ mol}^{-1}$
$\Delta_C = CH^\ominus = 611 \text{ kJ mol}^{-1}$

The $\Delta H_f\ominus$ for $CH_2 = CH_2$ is_______ kJmol^{-1} (nearest integer value)

JEE Main 2025 (Online) 3rd April Morning Shift

Answer: 25

Solution:



$$[\Delta H_f^o]_{C_2H_{4(g)}} = 2 \times [\Delta H_{sub}^o]_{C_{(s)}} + 2 \times \Delta H_{H-H}^o - 1 \times \Delta H_{C=C}^o - 4 \times \Delta H_{C-H}^o$$
$$\Rightarrow \quad [\Delta H_f^o]_{C_2H_{4(g)}} = (2 \times 710) + (2 \times 436) - 611 - 4 \times 414$$
$$\Rightarrow \quad [\Delta H_f^o]_{C_2H_{4(g)}} = 25 \text{ kJ/mol}$$

Question51

A sample of n -octane (1.14 g) was completely burnt in excess of oxygen in a bomb calorimeter, whose heat capacity is 5 kJ K^{-1}. As a result of combustion reaction, the temperature of the calorimeter is increased by 5 K . The magnitude of the heat of combustion of octane at constant volume is__________ kJmol^{-1} (nearest integer).

JEE Main 2025 (Online) 3rd April Evening Shift

Answer: 2500

Solution:

Calculate the moles of octane:

Moles of octane $= \frac{1.14 \text{ g}}{114 \text{ g/mol}} = 0.01$ moles

Determine the heat evolved during combustion:

The heat capacity of the calorimeter is 5 kJ/K and the temperature increase is 5 K.

Heat evolved $= C \times \Delta T = 5 \text{ kJ/K} \times 5 \text{ K} = 25 \text{ kJ}$

Calculate the magnitude of the heat of combustion:

The heat of combustion per mole of octane is found by dividing the total heat evolved by the moles of octane combusted:

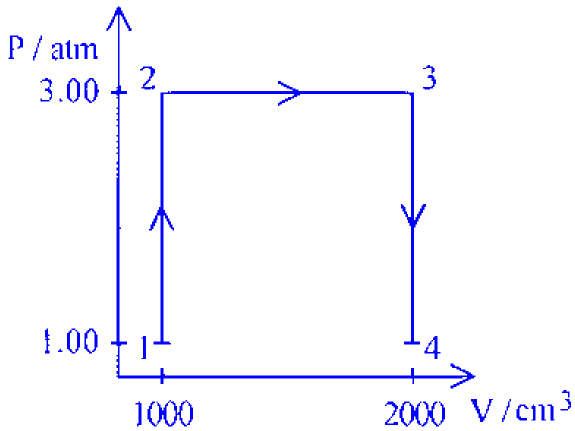
Magnitude of Heat of combustion $= \frac{25 \text{ kJ}}{0.01 \text{ moles}} = 2500 \text{ kJ/mol}$

Therefore, the magnitude of the heat of combustion of octane at constant volume is 2500 kJ/mol.

Question52

A perfect gas (0.1 mol) having $\overline{\mathrm{C}}_v = 1.50\mathrm{R}$ (independent of temperature) undergoes the above transformation from point 1 to point 4. If each step is reversible, the total work done (w) while going from point 1 to point 4 is $(-)$___________J (nearest integer)

[Given: $\mathrm{R} = 0.082 \text{ L atm K}^{-1} \text{ mol}^{-1}$]



JEE Main 2025 (Online) 3rd April Evening Shift

Answer: 304

Solution:

To determine the total work done when a perfect gas undergoes a transformation from point 1 to point 4, we can analyze each step of the process:

Given Data

Moles of gas, $n = 0.1$ mol

Specific heat at constant volume, $\overline{\mathrm{C}}_v = 1.50\mathrm{R}$

Gas constant, $\mathrm{R} = 0.082\,\mathrm{L\,atm\,K^{-1}\,mol^{-1}}$

Work Done Calculation

Step 1: From Point 1 to Point 2 ($W_{1\to2}$)

Since this is an isochoric process (constant volume), the work done, $W_{1\to2}$, is zero.

$\mathrm{W}_{1\to2} = 0$

Step 2: From Point 2 to Point 3 ($W_{2\to3}$)

This step is isobaric (constant pressure), and the work done can be calculated as:

$\mathrm{W}_{2\to3} = -P\Delta V = -P(V_3 - V_2)$

Given:

$P = 3$ atm

Change in volume, $\Delta V = V_3 - V_2 = 2 - 1 = 1$ L

$\mathrm{W}_{2\to3} = -3$ atm $\times\, 1$ L $= -3$ L atm

Converting to Joules (using 1 L atm $= 101.3$ J):

$\mathrm{W}_{2\to3} = -3 \times 101.3$ J $= -304$ J

Step 3: From Point 3 to Point 4 ($W_{3\to4}$)

This is another isochoric process, so the work done, $W_{3\to4}$, is zero.

$\mathrm{W}_{3\to4} = 0$

Total Work Done

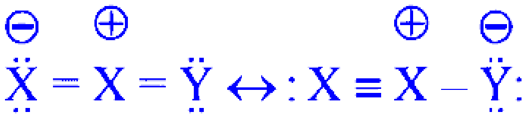
The total work done from point 1 to point 4 is the sum of work done in each step:

Total work done $= W_{1\to2} + W_{2\to3} + W_{3\to4} = 0 + (-304) + 0 = -304$ J

Thus, the total work done during this transformation is -304 J.

Question53

Resonance in $\mathrm{X_2Y}$ can be represented as



The enthalpy of formation of X_2Y $\left(X=X(g)+\frac{1}{2}Y=Y(g)\rightarrow X_2Y(g)\right)$ is 80 kJ mol^{-1}. The magnitude of resonance energy of X_2Y is __ kJ mol^{-1} (nearest integer value).

Given: Bond energies of $X\equiv X$, $X=X$, $Y=Y$ and $X=Y$ are 940, 410, 500, and 602 kJ mol $^{-1}$ respectively.
valence X: 3, Y: 2

JEE Main 2025 (Online) 8th April Evening Shift

Answer: 98

Solution:

$\Delta\mathrm{H_{R.E}}=\Delta\mathrm{H_{f(exp)}}-\Delta\mathrm{H_{f(\,Theo\,)}}$

$\Delta\mathrm{H_{f(exp)}}$ for $\mathrm{X_2Y_{(g)}}=80\ \mathrm{kJ/mole}$

for $\Delta\mathrm{H_{f\,(Theo)}}$

$\mathrm{X_{2(\,g)}}+\frac{1}{2}\mathrm{Y_{2(\,g)}}\rightarrow\mathrm{X_2Y_{(g)}}\Delta\mathrm{H_f}=?$

$\Delta\mathrm{H_{f(\,Theo\,)}}=\left(\mathrm{BE_{X=X}}+\frac{1}{2}\mathrm{BE_{Y=Y}}\right)-\left(\mathrm{BE_{X=X}}+\mathrm{BE_{X=Y}}\right)$

$=\left(940+\frac{1}{2}\times 500\right)-(410+602)$

$=178\ \mathrm{kJ/mole}$

$\Delta\mathrm{H_{R.E}}=80-178$

$=-98\ \mathrm{kJ/mol}$

$|\Delta\mathrm{H_{R.E}}|=98$

Question54

If three moles of an ideal gas at 300K expand isotherrnally from 30dm^3 to 45dm^3 against a constant opposing pressure of 80kPa, then the amount of heat transferred is ______J.

[27-Jan-2024 Shift 1]

Answer: 1200

Solution:

Using, first law of thermodynamics,

$\Delta U = Q + W,$

$\Delta U = 0$: Process is isothermal

$Q = -W$

$W = -P_{ext} \Delta V$: Irreversible

$= -80 \times 10^3 (45 - 30) \times 10^{-3}$

$= -1200J$

Question55

For a certain thermochemical reaction $M \rightarrow N$ at $T = 400K$, $\Delta H^{\ominus} = 77.2kJmol^{-1}$, $\Delta S = 122JK^{-1}$, log equilibrium constant (logK) is _____ $\times 10^{-1}$.

[27-Jan-2024 Shift 2]

Answer: 37

Solution:

$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$

$= 77.2 \times 10^3 - 400 \times 122 = 28400J$

$\Delta G^\circ = -2.303\, RT \log K$

$\Rightarrow 28400 = -2.303 \times 8.314 \times 400 \log K$

$\Rightarrow \log K = -3.708 = -37.08 \times 10^{-1}$

Question56

Which of the following is not correct?

[29-Jan-2024 Shift 1]

Options:

A.

ΔG is negative for a spontaneous reaction

B.

ΔG is positive for a spontaneous reaction

C.

ΔG is zero for a reversible reaction

D.

ΔG is positive for a non-spontaneous reaction

Answer: B

Solution:

$(\Delta G)_{P,\,T}$ = (+)ve for non-spontaneous process

Question57

Standard enthalpy of vapourisation for CCl_4 is 30.5kJmol^{-1}. Heat required for vapourisation of 284g of CCl_4 at constant temperature is____ kJ. (Given molar mass in gmol^{-1} ; C = 12, Cl = 35.5)

[29-Jan-2024 Shift 2]

Answer: 56

Solution:

$\Delta H^{\circ}_{vap}\ CCl_4 = 30.5\ kJ/mol$

Mass of $CCl_4 = 284$ gm

Molar mass of $CCl_4 = 154 g/mol$

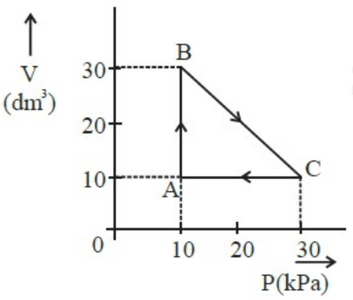
Moles of $CCl_4 = \frac{284}{154} = 1.844$ mol

ΔH°_{vap} for 1 mole = 30.5 kJ/mol

ΔH°_{vap} for 1.844 mol = 30.5 × 1.844

= 56.242 kJ

Question58



An ideal gas undergoes a cyclic transformation starting from the point A and coming back to the same point by tracing the path A → B → C → A as shown in the diagram. The total work done in the process is _____J.

[30-Jan-2024 Shift 1]

Answer: 200

Solution:

Work done is given by area enclosed in the P vs V cyclic graph or V vs P cyclic graph.

Sign of work is positive for clockwise cyclic process for V vs P graph.

$$W = \frac{1}{2} \times (30 - 10) \times (30 - 10) = 200\,\text{kPa} - \text{dm}^3$$

$$= 200 \times 1000\,\text{Pa} - \text{L} = 2\text{L} - \text{bar} = 200\text{J}$$

Question59

Two reactions are given below:

$$2Fe_{(s)} + \frac{3}{2}O_{2(g)} \rightarrow Fe_2O_{3(s)},\ \Delta H^o = -822\,\text{kJ/mol}$$

$$C_{(s)} + \frac{1}{2}O_{2(g)} \rightarrow CO_{(g)},\ \Delta H^o = -110\,\text{kJ/mol}$$

Then enthalpy change for following reaction

$$3C_{(s)} + Fe_2O_{3(s)} \rightarrow 2Fe_{(s)} + 3CO_{(g)}$$

[30-Jan-2024 Shift 2]

Answer: 492

Solution:

$$2Fe_{(s)} + \frac{3}{2}O_{2(g)} \longrightarrow Fe_2O_{3(s)},\ \Delta H^o = -822\,\text{kJ/mol} \ ....(1)$$

$$C_{(s)} + \frac{1}{2}O_{2(g)} \longrightarrow CO_{(g)},\ \Delta H^\circ = -110\,\text{kJ/mol} \ ....(2)$$

$$3C_{(s)} + Fe_2O_{3(s)} \longrightarrow 2Fe_{(s)} + 3CO_{(g)},\ \Delta H_3 = ?$$

$$(3) = 3 \times (2) - (1)$$

$$\Delta H_3 = 3 \times \Delta H_2 - \Delta H_1$$

$$= 3(-110) + 822$$

$$= 492\,\text{kJ/mole}$$

Question60

Consider the following reaction at 298K.

$$\frac{3}{2}O_{2(g)} \rightleftharpoons O_{3(g)} \cdot K_P = 2.47 \times 10^{-29} .$$

$\Delta_r G^\ominus$ for the reaction is ______kJ. (Given R $= 8.314\text{JK}^{-1}\text{mol}^{-1}$)

[31-Jan-2024 Shift 1]

Answer: 163

Solution:

$\frac{3}{2}O_{2(g)} \rightleftharpoons O_{3(g)} \cdot K_P = 2.47 \times 10^{-29}$.

$\Delta_r G^{\ominus} = -RT \ln K_P$

$= -8.314 \times 10^{-3} \times 298 \times \ln(2.47 \times 10^{-29})$

$= -8.314 \times 10^{-3} \times 298 \times (-65.87)$

$= 163.19 \text{ kJ}$

Question61

If 5 moles of an ideal gas expands from 10L to a volume of 100L at 300K under isothermal and reversible condition then work, w, is −xJ. The value of x is_____

(Given R = 8.314$JK^{-1}mol^{-1}$)

[31-Jan-2024 Shift 2]

Answer: None

Solution:

It is isothermal reversible expansion, so work done negative

$W = -2.303nRT \log\left(\frac{V_2}{V_1}\right)$

$= -2.303 \times 5 \times 8.314 \times 300 \log\left(\frac{100}{10}\right)$

$= -28720.713 \text{J}$

$\equiv -28721 \text{J}$

Question62

Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following :

[1-Feb-2024 Shift 1]

Options:

A.

$q = 0, \Delta T \neq 0, w = 0$

B.

$q = 0, \Delta T < 0, w \neq 0$

C.

$q \neq 0, \Delta T = 0, w = 0$

D.

$q = 0, \Delta T = 0, w = 0$

Answer: D

Solution:

During free expansion of an ideal gas under adiabatic condition $q = 0, \Delta T = 0, w = 0$.

Question63

For a certain reaction at 300K, K = 10, then ΔG° for the same reaction is______ $\times 10^{-1}$ $kJmol^{-1}$. (Given $R = 8.314 JK^{-1} mol^{-1}$)

[1-Feb-2024 Shift 2]

Answer: 57

Solution:

$\Delta G^\circ = -RT \ell n K$

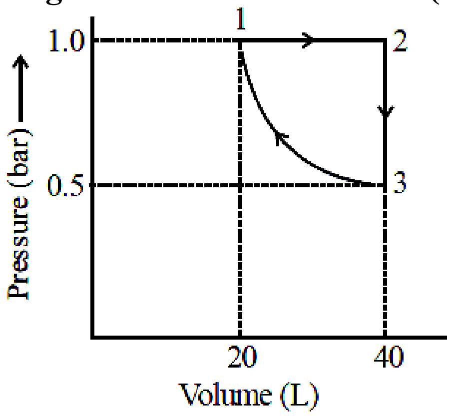
$= -8.314 \times 300 \ln(10)$

$= 5744.14 J/mole$

$= 57.44 \times 10^{-1} kJ/mole$

Question64

One mole of an ideal monoatomic gas is subjected to changes as shown in the graph. The magnitude of the work done (by the system or on the system) is J (nearest integer).



Given : log 2 = 0.3, ln 10 = 2.3
[24-Jan-2023 Shift 2]

Answer: 620

Solution:

$$\begin{aligned}
&1 \rightarrow 2 \Rightarrow \text{ Isobaric process} \\
&2 \rightarrow 3 \Rightarrow \text{ Isochoric process} \\
&3 \rightarrow 1 \Rightarrow \text{ Isothermal process} \\
&W = W_{1\rightarrow 2} + W_{2\rightarrow 3} + W_{3\rightarrow 1} \\
&= \left(-P(V_2 - V_1) + 0\left[-P_1V_1 \ln\left(\frac{V_2}{V_1}\right)\right]\right) \\
&= \left[-1 \times (40-20) + 0 + \left[-1 \times 20 \ln\left(\frac{20}{40}\right)\right]\right] \\
&= -20 + 20 \ln 2 \\
&= -20 + 20 \times 2.3 \times 0.3 \\
&= -6.2 \text{ bar L} \\
&|W| = 6.2 \text{ bar } 1 = 620\text{J}
\end{aligned}$$

Question65

An athlete is given 100g of glucose ($C_6H_{12}O_6$) for energy. This is equivalent to 1800 kJ of energy. The 50% of this energy gained is utilized by the athlete for sports activities at the event. In order to avoid storage of energy, the weight of extra water he would need to perspire is _______ g (Nearest integer)
Assume that there is no other way of consuming stored energy.
Given : The enthalpy of evaporation of water is 45 kJ mol^{-1}
Molar mass of C, H&O are 12.1 and 16gmol^{-1}.
[25-Jan-2023 Shift 1]

Answer: 360

Solution:

$C_6H_{12}O_6(s) + 6O_2 \rightarrow 6CO_2(g) + 6H_2O(l)$
Extra energy used to convert $H_2O(l)$ into $H_2O(l)$ into $H_2O(g)$

$$\begin{aligned}
&= \frac{1800}{2} = 900 \text{ kJ} \\
&\Rightarrow 900 = n_{H_2O} \times 45 \\
&n_{H_2O} = \frac{900}{45} = 20 \text{ mole} \\
&W_{H_2O} = 20 \times 18 = 360\text{g}
\end{aligned}$$

Question66

28.0L of CO_2 is produced on complete combustion of 16.8L gaseous mixture of ethene and methane at 25°C and 1 atm. Heat evolved during the combustion process is _________ kJ.
Given : $\Delta H_C(CH_4) = -900$ kJ mol^{-1}

$\Delta H_C(C_2H_4) = -1400$ kJ mol^{-1}
[25-Jan-2023 Shift 2]

Answer: 847

Solution:

Let, Volume of C_2H_4 is x litre
Total volume of $CO_2 = 2x + 16.8 - x$
$\Rightarrow \ 28 = 16.8 + x$
$x = 11.2L$
$n_{CH_4} = \frac{PV}{RT} = \frac{1 \times 5.6}{0.082 \times 298} = 0.229$ mole
$n_{C_2H_2} = \frac{11.2}{0.082 \times 298} = 0.458$ mole
$\therefore$ Heat evolved $= 0.229 \times 900 + 0.458 \times 1400$
$= 206.1 + 641.2$
$= 847.3$ kJ

Question67

Which of the following relations are correct?
(A) $\Delta U = q + p\,\Delta V$
(B) $\Delta G = \Delta H - T\,\Delta S$
(C) $\Delta S = \frac{q_{rev}}{T}$
(D) $\Delta H = \Delta U - \Delta nRT$
Choose the most appropriate answer from the options given below :
[29-Jan-2023 Shift 2]

Options:

A. C and D only

B. B and C only

C. A and B only

D. B and D only

Answer: B

Solution:

Only (B) and (C) are correct.
(B) $G = H - TS$
At constant T
$\Delta G = \Delta H - T\,\Delta S$
(A) First law is given by
$\Delta U = Q + W$
If we apply constant P and reversible work.
$\Delta U = Q - P\,\Delta V$
(C)By definition of entropy change
$dS = \frac{dq\,q_{rev}}{T}$
At constant T

$\Delta S = \frac{q_{rev}}{T}$
(D) $H = U + PV$
For ideal gas
$H = U + nRT$
At constant T
$\Delta H = \Delta U + \Delta nRT$

Question68

When 2 litre of ideal gas expands isothermally into vacuum to a total volume of 6 litre, the change in internal energy is ______ J. (Nearest integer)
[30-Jan-2023 Shift 1]

Answer: 0

Solution:

For ideal gas $U = f(T)$
and for isothermal process, $\Delta U = 0$

Question69

1 mole of ideal gas is allowed to expand reversibly and adiabatically from a temperature of $27^{\circ}C$. The work done is $3\ kJ\ mol^{-1}$. The final temperature of the gas is _______ K (Nearest integer). Given $C_v = 20 J mol^{-1} K^{-1}$.
[30-Jan-2023 Shift 2]

Answer: 150

Solution:

$q = 0$
$\Delta U = W$
$1 \times 20 \times [T_2 - 300] = -3000$
$T_2 - 300 = -150$
$T_2 = 150K$

Question70

The enthalpy change for the conversion of
$\frac{1}{2}Cl_2(g)$ to $Cl^-(aq)$ is $(-)$ ________
$kJ\ mol^{-1}$ (Nearest$^-$. integer)
Given : $\Delta_{dis} H^{\circ}_{Cl_{2(g)}} = 240 kJ mol^{-1}$

$\Delta_{eg} H_{Cl_{(g)}}^{\circ} = -350 kJmol^{-1}$

$\Delta_{hyd} H_{Cl_{(g)}^{\circ}}^{\circ} = -380 kJmol^{-1}$

[31-Jan-2023 Shift 1]

Answer: 610

Solution:

$$\frac{1}{2}Cl_{2(g)} \rightarrow Cl_{(g)} \rightarrow Cl_{(g)}^{-} \rightarrow Cl_{(aq.)}^{-}$$
$$\Delta H^{\circ} = \frac{1}{2} \times 240 + (-350) + (-380)$$
$$= -610 \text{ ans.}$$

Question71

Enthalpies of formation of $CCl_4(g)$, $H_2O(g)$, $CO_2(g)$ and $HCl(g)$ are -105, -242, -394 and -92 kJ mol^{-1} respectively. The magnitude of enthalpy of the reaction given below is ________ kJ mol^{-1} (nearest integer)

$CCl_4(g) + 2H_2O(g) \rightarrow CO_2(g) + 4HCl(g)$

[31-Jan-2023 Shift 2]

Answer: 173

Solution:

$$\Delta_r H = \sum H_p - \sum H_R$$
$$= (-394 + 4 \times -92) - (-105 + (2 \times -242))$$
$$= -173 \text{ kJ/mol}$$

Question72

0.3g of ethane undergoes combustion at $27^{\circ}C$ in a bomb calorimeter. The temperature of calorimeter system (including the water) is found to rise by $0.5^{\circ}C$. The heat evolved during combustion of ethane at constant pressure is _______ kJ mol^{-1}.
(Nearest integer)
[Given: The heat capacity of the calorimeter system is 20 kJ K^{-1}, $R = 8.3 JK^{-1}mol^{-1}$.
Assume ideal gas behaviour.
Atomic mass of C and H are 12 and 1 $gmol^{-1}$ respectively]
[1-Feb-2023 Shift 2]

Answer: 1006

Solution:

(Bomb calorimeter → const volume
Heat released
By combustion of 1 mole

$C_2H_6(\Delta U) = -\dfrac{20 \times 0.5}{0.3} \times 30 = -1000\,kJ$

$C_2H_6(g) + 7/2O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$

$\Delta ng = 2 - (2 + 7/2) = -(7/2)$

$\Delta H = \Delta U + \Delta nRT$

$= -1000 - 7/2 \times 8.3 \times 300\,kJ$

$= -1000 - 6.225$

$= -1006\,kJ$

So heat released $= 1006\,kJ\,mol^{-1}$

Question73

The value of log K for the reaction A ⇋ B at 298K is___
(Nearest integer)
Given: $\Delta H^0 = -54.07\,kJ\,mol^{-1}$
$\Delta S^\circ = 10 JK^{-1} mol^{-1}$
(Take $2.303 \times 8.314 \times 298 = 5705$)
[6-Apr-2023 shift 1]

Answer: 10

Solution:

$\Delta G^0 = \Delta H^0 - T\Delta S$

$\Rightarrow \Delta G^0 = (-54070 - 10 \times 298)$

Also, $\Delta G^0 = (-2.303\,RT \log K)$

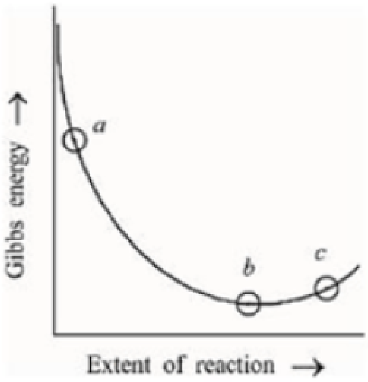
$\Rightarrow (-54070 - 10 \times 298)$

$= (-2.303 \times 8.134 \times 298 \log K)$

$\Rightarrow \log K = 10$ Ans: 10

Question74

Consider the graph of Gibbs free energy G vs Extent of reaction. The number of statement/s from the following which are true with respect to points (a), (b) and (c) is____



A. Reaction is spontaneous at (a) and (b)
B. Reaction is at equilibrium at point (b) and nonspontaneous at point (c)
C. Reaction is spontaneous at (a) and nonspontaneous at (c)
D. Reaction is non-spontaneous at (a) and (b)
[6-Apr-2023 shift 1]

Answer: 2

Solution:

For, Spontaneous process $dG < 0$
For, Equilibrium $dG = 0$
For, Nonspontaneous process $dG > 0$
$\therefore$ A Wrong
B Correct
C Correct
D Wrong

Question75

Consider the following date
Heat of combustion of $H_2(g) = -241.8\ kJ\ mol^{-1}$
Heat of combustion of $C(s) = -393.5\ kJ\ mol^{-1}$
Heat of combustion of $C_2H_5OH(1) = -1234.7\ kJ\ mol^{-1}$.
The heat of formation of $C_2H_5OH(1)$ is (-)_____ kJ mol^{-1} (Nearest integer)
[6-Apr-2023 shift 2]

Answer: 278

Solution:

$$2C_{(s)} + 3H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow C_2H_5OH_{(l)}$$
$$(\Delta H_f)_{C_2H_5OH_{(l)}} = \Sigma(\Delta H_{comb})_{reactant} - \sum(\Delta H_{comb})_{product}$$
$$= 2 \times (-393.5) + 3(-241.8) - (-1234.7)$$
$$= -277.7\ kJ/mol$$

Question76

When a 60W electric heater is immersed in a gas for 100 s in a constant volume container with adiabatic walls, the temperature of the gas rises by $5^{\circ}C$. The heat capacity of the given gas is _______ JK^{-1} (Nearest integer)
[8-Apr-2023 shift 1]

Answer: 1200

Solution:

Adiabatic wall \{no heat exchange between system and surrounding\}
$C_v \times \Delta T = P \times t/ \sec$
$C_v \times 5 = 60 \times 100$
$C_v = 1200$

Question77

For complete combustion of ethane,
$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(l)$
The amount of heat produced as measured in bomb calorimeter is 1406 $KJ\ mol^{-1}$ at 300K. The minimum value of $T \Delta S$ needed to reach equilibrium is _______ (−) KJ (Nearest integer)
Given : $R = 8.3JK^{-1}mol^{-1}$
[8-Apr-2023 shift 2]

Answer: 1411

Solution:

$\Delta G = \Delta H - T \Delta S$ at equilibrium:-
$\Delta G = 0$
$T \Delta S = \Delta H = \Delta U + \Delta ngRT = -1406 + (-2) \times 8.3 \times 300 \times 10^{-3} = -1410.98 \approx 1411$

Question78

The number of incorrect statement from the following is _______
[8-Apr-2023 shift 2]

Options:

A. The electrical work that a reaction can perform at constant pressure and temperature is equal to the reaction Gibbs energy

B. E_{cell}^{0} is dependent on the pressure

C. $\frac{dE_{cell}^{o}}{dT} = \frac{\Delta_r S^{\circ}}{nF}$

D. A cell is operating reversibly if the cell potential is exactly balanced by an opposing source of potential difference

Answer: A

Solution:

$$dG = vdp - sdT$$
$$dG = -sdT$$
$$\frac{dG}{dT} = -S \Rightarrow \frac{d\Delta G}{dT} = -\Delta S$$
$$\frac{dE^{\circ}}{dT} = \frac{-\Delta S}{-nF}$$

Question79

The enthalpy change for the adsorption process and micelle formation respectively are [10-Apr-2023 shift 1]

Options:

A. $\Delta H_{ads} < 0$ and $\Delta H_{mic} < 0$

B. $\Delta H_{ads} > 0$ and $\Delta H_{mic} < 0$

C. $\Delta H_{ads} < 0$ and $\Delta H_{mic} > 0$

D. $\Delta H_{ads} > 0$ and $\Delta H_{mic} > 0$

Answer: C

Solution:

Adsorption → Exothermic ($\Delta H_{ads} = -ve$)
Micelle formation → Endothermic ($\Delta H_{mic} = +ve$)
$\Delta H_{ads} < O$ and $\Delta H_{mic} > O$

Question80

Given
(A) $2CO(g) + O_2(g) \rightarrow 2CO_2(g)\ \Delta H_1^{o} = -x\ kJ\ mol^{-1}$

(B) C (graphite) $+ O_2(g) \rightarrow CO_2(g)\quad \Delta H_2^{\circ} = -y\ KJ\ mol^{-1}$
The ΔH° for the reaction C(graphite) $+ \frac{1}{2}O_2(g) \rightarrow CO(g)$ is
[10-Apr-2023 shift 1]

Options:

A. $\frac{x-2y}{2}$

B. $\frac{x+2y}{2}$

C. $\frac{2x-y}{2}$

D. $2y-x$

Answer: A

Solution:

$$2\,CO(g)+O_2(g) \rightarrow 2CO_2(g) \quad \Delta H_1^\circ = y\,kJ/mol\,..........(2)$$
$$C(\text{ graphite })+O_2(g) \rightarrow CO_2(g) \quad \Delta H_2^\circ = -y\ \ kJ/mol.........\ (2)$$

$$C(\text{ graphite })+{}^{1/2}O_2(g) \rightarrow CO(g) \quad \Delta H_3^\circ = ?$$
$$\Delta H_3^\circ = H_2^\circ - \frac{H_1^\circ}{2} = -y - \frac{-x}{2}$$
$$\Delta H_3^\circ = \frac{x}{2} - y = \frac{x-2y}{2}$$

Question81

$FeO_4^{2-} \xrightarrow{+2.2V} Fe^{3+} \xrightarrow{+0.70V} Fe^{2+} \xrightarrow{-0.45V} Fe^0$

$E^\circ_{FeO_4^{2-}/Fe^{2+}}$ is $x \times 10^{-3}$V.

The value of x is _______

[10-Apr-2023 shift 1]

Answer: 1825

Solution:

$$FeO_4^{2-} + 3e^{\ominus} \rightarrow Fe^{+3}\ \Delta G_1$$
$$Fe^{+3} + e^{\ominus} \rightarrow Fe^{+2}\ \Delta G_2$$
$$FeO_4^{-2} + 4e^{\ominus} \rightarrow Fo^{+2}\ \Delta G_3$$
$$\Delta G_3 = \Delta G_1 + \Delta G_2$$
$$(-)4E_3^\circ F = (-)3 \times 2.2 \times F + (-)1 \times 0.7 \times f$$
$$4E_3^\circ = 6.6 + 0.7 = 7.3$$
$$E_3^\circ = \frac{7.3}{4} = 1.825 = 1825 \times 10^{-3}$$

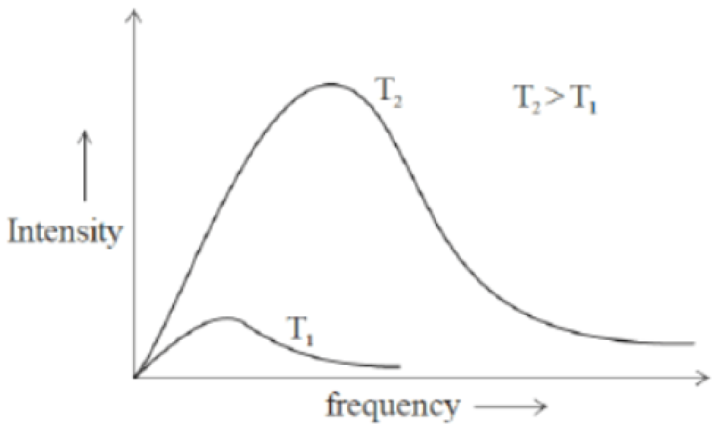
Question82

The number of incorrect statement/s about the black body from the following is ________
(A) Emit or absorb energy in the form of electromagnetic radiation
(B) Frequency distribution of the emitted radiation depends on temperature
(C) At a given temperature, intensity vs frequency curve passes through a maximum value
(D) The maximum of the intensity vs frequency curve is at a higher frequency at higher

temperature compared to that at lower temperature
[10-Apr-2023 shift 1]

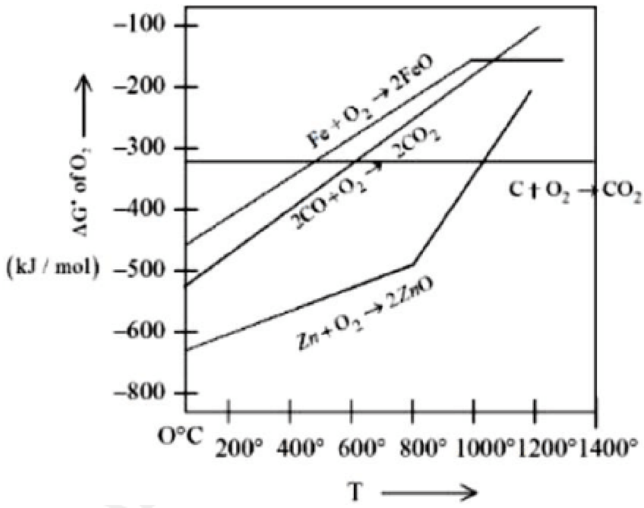
Answer: 0

Solution:



Question83

Gibbs energy vs T plot for the formation of oxides is given below.



For the given diagram, the correct statement is -
[10-Apr-2023 shift 2]

Options:

A. At 600°C, C can reduce ZnO

B. At 600°C, C can reduce FeO

C. At 600°C, CO cannot reduce FeO

D. At 600°C, CO can reduce ZnO

Answer: B

Solution:

$FeO + C \rightarrow Fe + CO_2$

At $600^{\circ}C$ ΔG of Reaction is $-Ve$

Question84

$A(g) \rightleftharpoons 2B(g) + C(g)$
For the given reaction, if the initial pressure is 450 mmHg and the pressure at time t is 720 mmHg at a constant temperature T and constant volume V. The fraction of A(g) decomposed under these conditions is $x \times 10^{-1}$. The value of x is _________ (nearest integer)
[10-Apr-2023 shift 2]

Answer: 3

Solution:

$A(g) \rightleftharpoons 2B(g) + C(g)$
$t = 0 \quad 450$
$\text{time} \quad 450 - x \quad 2x$
$P_T = P_A + P_B + P_C$
$720 = 450 - x + 2x + x$
$2x = 270$
$x = 135$

Fraction of A decomposed $= \dfrac{135}{450} = 0.3 = 3 \times 10^{-1}$

So, $x = 3$

Question85

The number of endothermic process/es from the following is
A. $I_2(g) \rightarrow 2I(g)$
B. $HCl(g) \rightarrow H(g) + Cl(g)$
C. $H_2O(l) \rightarrow H_2O(g)$
D. $C(s) + O_2(g) \rightarrow CO_2(g)$
E. Dissolution of ammonium chloride in water
[10-Apr-2023 shift 2]

Answer: 4

Solution:

$A \rightarrow$ Endothermic (Atomisation) $B \rightarrow$ Endothermic (Atomisation)
$C \rightarrow$ Endothermic (Vapourisation) $D \rightarrow$ Exothermic (Combustion)
$E \rightarrow$ Endothermic (Dissolution)

Question86

Solid fuel used in rocket is a mixture of Fe_2O_3 and Al (in ratio 1 : 2) the heat evolved (KJ) per gram of the mixture is ________ (Nearest integer)
Givne $\Delta H_f^{\ominus}(Al_2O_3) = -1700$ KJ mol^{-1}
$\Delta H_f^{\ominus}(Fe_2O_3) = -840$ KJ mol^{-1}
[11-Apr-2023 shift 1]

Answer: 4

Solution:

$Fe_2O_3 + 2\,Al \rightarrow Al_2O_3 + 2\,Fe$
$\Delta H_r = (\Delta H_f)Al_2O_3 - \Delta\, H_f^{\circ}(Fe_2O_3)$
$= -1700 - (-840)$
$= -860$ kJ
Fe_2O_3 & $Al \rightarrow 1 : 2$
$Fe_2O_3 = 1$ mole $= (2 \times 25 + 48)$
$= 112 + 48 = 160$ gm
$Al = 2$ mole $= 2 \times 27 = 54$ gm
Total mass $= 160 + 54 = 214$ gm
Heat evolved per gm $= \frac{-860}{214}$ kJ $= -4.01 \approx 4$ kJ

Question87

The total number of intensive properties from the following is ______ new line volume, Molar heat capacity, Molarity, E^{θ} cell, Gibbs free energy change, Molar mass, Mole
[11-Apr-2023 shift 2]

Answer: 4

Solution:

Extensive ⇒ Mole, Volume, Gibbs free energy.
Intensive ⇒ Molar mass, Molar heat capacity, Molarity, E^{θ} cell.

Question88

One mole of an ideal gas at 350K is in a 2.0L vessel of thermally conducting walls, which are in contact with the surroundings. It undergoes isothermal reversible expansion from 2.0L to 3.0L against a constant pressure of 4 atm. The change in entropy of the surroundings (ΔS) is _______ JK^{-1} (Nearest integer)

Given: $R = 8.314 JK^{-1} mol^{-1}$
[12-Apr-2023 shift 1]

Answer: 3

Solution:

$$\Delta S_{System} = nR\ell\left(\frac{V_2}{V_1}\right) = 1 \times 8.314 \ln\left(\frac{3}{2}\right)$$
$$\Delta S_{System} = 3.37$$
$$\Delta S_{Sur.} = 3.37$$

Question89

$A_2 + B_2 \rightarrow 2\,AB \cdot \Delta H_f^0 = -200\,kJ\,mol^{-1}$ new line AB, A_2 and B_2 are diatomic molecules. If the bond enthalpies of A_2, B_2 and AB are in the ratio 1 : 0.5 : 1, then the bond enthalpy of A_2 is ________ kJ mol^{-1}. (Nearest integer)
[13-Apr-2023 shift 1]

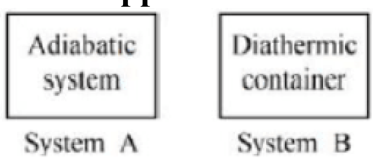
Answer: 800

Solution:

$A_2 + B_2 \rightarrow 2\,AB \quad \Delta H_f^\circ = -200\,kJ/mol$
Bond enthalpy of $A_2 = x$
Bond enthalpy of $B_2 = 0.5x$
Bond enthalpy of $AB = x$
$\Delta H_f^\circ = x + 0.5x - 2x = -2(200)$
$-0.5x = -400$
$x = \frac{400}{0.5} = 800\,kJ/mol$
Bond enthalpy of $A_2 = x = 800\,kJ/mol$

Question90

What happens when methane undergoes combustion in systems A and B respectively?



[13-Apr-2023 shift 2]

System A	System B
Temperature remains same	Temperature rises

B.

System A	System B
Temperature falls	Temperature rises

C.

System A	System B
Temperature falls	Temperature remains same

D.

System A	System B
Temperature rises	Temperature remains same

Answer: D

Solution:

Adiabatic boundary does not allow heat exchange thus heat generated in container can't escape out thereby increasing the temperature. In case of Diathermic container, heat flow can occur to maintain the constant temperature.

Question91

30.4 kJ of heat is required to melt one mole of sodium chloride and the entropy change at the melting point is _______ $28.4 JK^{-1} mol^{-1}$ at 1 atm. The melting point of sodium chloride is K. (Nearest Integer)
[15-Apr-2023 shift 1]

Answer: 1070

Solution:

$\Delta S = \frac{\Delta H}{T_{mp}}$

$28.4 = \frac{30.4 \times 1000}{T_{mp}}$

$T_{mp} = 1070.422K.$

Question92

$2O_3(g) \rightleftharpoons 3O_2(g)$
At 300K, ozone is fifty percent dissociated. The standard free energy change at this temperature and 1 atm pressure is (−)______$Jmol^{-1}$. (Nearest integer)
[Given : In 1.35 = 0.3 and .R = $8.3JK^{-1}mol^{-1}$]
[24-Jun-2022-Shift-1]

Answer: 747

Solution:

$$\underset{1-x}{2O_3(g)} \rightleftharpoons \underset{\frac{3x}{2}}{3O_2(g)}$$
Given, $x = 0.5$
$$\therefore k_p = \frac{[3(0.5)]^3 \times 1}{[2]^3 \times (0.5)^2 \times 1.25}$$
$$\therefore k_p = \frac{27}{8} \times \frac{0.5}{1.25} = 1.35$$
$$\Delta G^\circ = -2.303\,RT \log k_p$$
$$= -2.303 \times 8.3 \times 300 \log 1.35$$
$$= -8.3 \times 300 \ln(1.35)$$
$$= -747 Jmol^{-1}$$

Question93

The standard entropy change for the reaction
$4\,Fe(s) + 3O_2(g) \rightarrow 2Fe_2O_3(s)$ is $-550 JK^{-1}$ at 298K.
[Given : The standard enthalpy change for the reaction is $-165\,kJ\,mol^{-1}$].
The temperature in K at which the reaction attains equilibrium is____(Nearest Integer)
[25-Jun-2022-Shift-1]

Answer: 300

Solution:

$\Delta G = \Delta H - T\,\Delta\,S = 0$ at equilibrium
$\Rightarrow -165 \times 10^3 - T \times (-505) = 0$
$\Rightarrow T = 300K$

Question94

At 25°C and 1 atm pressure, the enthalpy of combustion of benzene (I) and acetylene (.g) are $-3268\,kJ\,mol^{-1}$ and $-1300\,kJ\,mol^{-1}$, respectively. The change in enthalpy for the reaction $3C_2H_2(g) \rightarrow C_6H_6(I)$, is
[25-Jun-2022-Shift-2]

Options:

A. $+324\ \mathrm{kJ\ mol^{-1}}$

B. $+632\ \mathrm{kJ\ mol^{-1}}$

C. $-632\ \mathrm{kJ\ mol^{-1}}$

D. $-732\ \mathrm{kJ\ mol^{-1}}$

Answer: C

Solution:

$\Delta H = \Sigma\,\Delta H_{\text{Combustion}}$ (Reactant) $-\Sigma\,\Delta H_{\text{Combustion}}$ (Product)

$= 3 \times (-1300) - [-3268]$

$= -632\ \mathrm{kJ\ mol^{-1}}$

Question95

For complete combustion of methanol

$$CH_3OH(l) + \tfrac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

the amount of heat produced as measured by bomb calorimeter is 726 $\mathrm{kJ\ mol^{-1}}$ at 27°C. The enthalpy of combustion for the reaction is $-x\ \mathrm{kJ\ mol^{-1}}$, where x is____ (Nearest integer)
(Given : $R = 8.3\mathrm{JK^{-1}mol^{-1}}$)
[26-Jun-2022-Shift-1]

Answer: 727

Solution:

$CH_3OH(l) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

$\Delta H = \Delta U + \Delta n_g RT$

$= -726\ \mathrm{kJ} + \left(\frac{-1}{2}\right) \times 8.3 \times 300$

$\simeq -727\ \mathrm{kJ\ mol^{-1}}$

Question96

CNG is an important transportation fuel. When 100g CNG is mixed with 208g oxygen in vehicles, it leads to the formation of CO 2 and H_2O and produces large quantity of heat during this combustion, then the amount of carbon dioxide, produced in grams is ____ [nearest integer]
[Assume CNG to be methane]
[26-Jun-2022-Shift-2]

Answer: 143

Solution:

$$\begin{array}{lcccc} & CH_4 & +2O_2 \longrightarrow & CO_2 & +2H_2O \\ \text{Initially:} & 6.25 & 6.5 & 0 & 0 \\ \text{finally:} & 6.25-\frac{6.5}{2} & 0 & \frac{6.5}{2} & 6.5 \end{array}$$

Initially,
Mass of $CNG(CH_4) = 100\text{ gm}$
∴ Moles of $CH_4 = \frac{100}{16} = 6.25$ moles
Mass of $O_2 = 208\text{ gm}$
∴ Moles of $O_2 = \frac{208}{32} = 6.5$ moles
From reaction you can see,
1 mole of CH_4 react with 2 mole of O_2
∴6.25 mole of CH_4 react with $6.25 \times 2 = 12.5$ moles of O_2
But here only 6.5 moles of O_2 present
So, O_2 will act as limiting reagent.
∴ Produced CO_2 will depends on moles of O_2.
From 2 moles of $O_2$1 mole of CO_2
∴ From 6.5 moles of O_2 $\frac{6.5}{2}$ moles of CO_2 produced.
∴ Weight of $CO_2 = \frac{6.5}{2} \times 44 = 143\text{ gm}$

Question97

A fish swimming in water body when taken out from the water body is covered with a film of water of weight 36g. When it is subjected to cooking at $100^{\circ}C$, then the internal energy for vaporization in $kJ\,mol^{-1}$ is _____[nearest integer]
[Assume steam to be an ideal gas. Given $\Delta_{vap}H^{\Theta}$ for water at 373K and 1 bar is $41.1\ kJ\,mol^{-1}$; $R = 8.31 JK^{-1}mol^{-1}$]
[26-Jun-2022-Shift-2]

Answer: 38

Solution:

$H_2O(l) \rightarrow H_2O(g)$
$n = \frac{36}{18} = 2\text{ mol}$
$\Delta U = \Delta H - \Delta n_g RT$
$= 41.1 - \frac{1 \times 8.31 \times 373}{1000}\ kJ/mol$
$= 38\ kJ/mol$

Question98

40% of HI undergoes decomposition to H_2 and I_2 at 300K. $\Delta G^{\ominus}$ for this decomposition reaction at one atmosphere pressure is ____ $Jmol^{-1}$. [nearest integer]
(Use .$R = 8.31 JK^{-1} mol^{-1}$; $\log 2 = 0.3010$, $\ln 10 = 2.3$, $\log 3 = 0.477$)
[26-Jun-2022-Shift-2]

Answer: 2735

Solution:

$$\begin{array}{llll} & HI & \rightleftharpoons \frac{1}{2}H_2 + & \frac{1}{2}I_2 \\ \text{At } t=0: & 1 & 0 & 0 \\ \text{At } t=t_{eq}: & 1-\alpha & \frac{\alpha}{2} & \frac{\alpha}{2} \end{array}$$

$$\therefore K = \frac{\left(\frac{\alpha}{2}\right)^{1/2} \times \left(\frac{\alpha}{2}\right)^{1/2}}{(1-\alpha)}$$

Given $\alpha = \frac{40}{100} = 0.4$

$$\therefore K = \frac{\left(\frac{0.4}{2}\right)^{1/2} \times \left(\frac{0.4}{2}\right)^{1/2}}{(1-0.4)}$$

$= \frac{1}{3}$

We know,

$\Delta G^{\circ} = -RT \ln K$

$= -RT \ln\left(\frac{1}{3}\right)$

$= +RT \ln 3$

$= +8.314 \times 300 \times \ln 3$

$= 2735 J/mol$

Question99

Choose the correct answer from the options given below:

	List - I		List - II
(A)	Spontaneous process	(I)	$\Delta H < 0$
(B)	Process with $\Delta P = 0$, $\Delta T = 0$	(II)	$\Delta G_{T,P} < 0$
(C)	$\Delta H_{\text{reaction}}$	(III)	Isothermal and isobaric process
(D)	Exothermic Process	(IV)	[Bond energies of molecules in reactants]-[Bond energies of product molecules

[27-Jun-2022-Shift-1]

Options:

A. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)

B. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

C. (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

D. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

Answer: B

Solution:

(A) For a spontaneous process $\Delta G_{T, P} < 0$
(B) $\Delta P = 0 \rightarrow$ Isobaric process, $\Delta T = 0 \rightarrow$ Isothermal process
(C) $\Delta H_{\text{reaction}} = (\Sigma$ Bond energies of reactants $) - (\Sigma$ bond energies of products $)$
(D) $\Delta H < 0$ is for exothermic reaction

Question100

When 5 moles of He gas expand isothermally and reversibly at 300K from 10 litre to 20 litre, the magnitude of the maximum work obtained is____ J. [nearest integer] (Given : $R = 8.3 JK^{-1} mol^{-1}$ and $\log 2 = 0.3010$)
[27-Jun-2022-Shift-2]

Answer: 8630

Solution:

$$W_{\text{rev}} = -2.303\ nRT \log_{10}\left(\frac{V_2}{V_1}\right)$$
$$= -2.303 \times 5 \times 8.3 \times 300 \times \log_{10}\left(\frac{20}{10}\right)$$
$$\simeq -8630 J$$

Question101

4.0L of an ideal gas is allowed to expand isothermally into vacuum until the total volume is 2.0L. The amount of heat absorbed in this expansion is _____L atm.
[28-Jun-2022-Shift-1]

Answer: 0

Solution:

Work done $= -P_{\text{ext}}\ \Delta v$
$\because P_{\text{ext}} = 0$ (vacuum)
$\therefore w = 0, \quad \Delta U = 0$ (as the process is isothermal)
So, $q = 0$

Question102

For combustion of one mole of magnesium in an open container at 300K and 1 bar pressure, $\Delta_C H^{\ominus} = -601.70$ kJ mol $^{-1}$, the magnitude of change in internal energy for the reaction is _____ kJ. (Nearest integer)
(Given : $R = 8.3 JK^{-1} mol^{-1}$)
[28-Jun-2022-Shift-2]

Answer: 600

Solution:

$Mg(s) + \frac{1}{2} O_2(g) \longrightarrow MgO(s)$
$\Delta H = \Delta U + \Delta ngRT$
$\Delta ng = -\frac{1}{2}$
$-601.70 = \Delta U - \frac{1}{2}(8.3)(300) \times 10^{-3}$
$\Delta U = -601.70 + 1.245$
$\Delta U \simeq -600$ kJ
The magnitude of change in internal energy is 600 kJ.

Question103

17.0g of NH_3 completely vapourises at $-33.42^{\circ}C$ and 1 bar pressure and the enthalpy change in the process is 23.4 kJ mol $^{-1}$. The enthalpy change for the vapourisation of 85g of NH_3 under the same conditions is ____kJ.
[29-Jun-2022-Shift-1]

Answer: 117

Solution:

Given data is for 1 moles and asked for 5 moles so
value is 23.4 × 5 = 117 kJ

Question104

2.2g of nitrous oxide (N_2O) gas is cooled at a constant pressure of 1 atm from 310K to 270K causing the compression of the gas from 217.1 mL to 167.75 mL. The change in internal energy of the process, ΔU is ' $-x$' J . The value of ' x ' is ______[nearest integer]
(Given : atomic mass of N = $14 g mol^{-1}$ and of O = $16 g mol^{-1}$. Molar heat capacity of N_2O is

$100 JK^{-1} mol^{-1}$)
[29-Jun-2022-Shift-2]

Answer: 195

Solution:

$$\begin{aligned}
\Delta T &= -40K \\
\Delta U &= q + w \\
&= \frac{100 \times 2.2}{44}(-40) - (-49.39) \times 10^{-3} \times 101.325 \\
&= -200 + 5 \\
&= -195J
\end{aligned}$$

$x = 195$

Question105

The enthalpy of combustion of propane, graphite and dihydrogen at 298K are $-2220.0 kJ\ mol^{-1}$, $-393.5 kJ\ mol^{-1}$ and $-285.8 kJ\ mol^{-1}$ respectively. The magnitude of enthalpy of formation of propane (C_3H_8) is kJ mol^{-1}. (Nearest integer)
[25-Jul-2022-Shift-1]

Answer: 104

Solution:

Enthalpy of combustion of propane, graphite and H_2 at 298K are
$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(I), \Delta H_1 = -2220\ kJ\ mol^{-1}$
$C(\text{ graphite }) + O_2(g) \rightarrow CO_2(g), \quad \Delta H_2 = -393.5\ kJ\ mol^{-1}$
$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(I), \quad \Delta H_3 = -285.8\ kJ\ mol^{-1}$
The desired reaction is
$3C\ (\text{graphite}) + 4H_2(g) \rightarrow C_3H_8(g)$
$\Delta H_f = 3\Delta H_2 + 4\Delta H_3 - \Delta H_1$
$= 3(-393.5) + 4(-285.8) - (-2220)$
$= -103.7\ kJ\ mol^{-1}$
$|\Delta H_f| \simeq 104\ kJ\ mol^{-1}$

Question106

While performing a thermodynamics experiment, a student made the following observations.
$HCl + NaOH \rightarrow NaCl + H_2O\ \Delta H = -57.3 kJ\ mol^{-1}$
$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O\ \Delta H = -55.3 kJ\ mol^{-1}$
The enthalpy of ionization of CH_3COOH as calculated by the student is _____kJ mol^{-1}.

(nearest integer)
[25-Jul-2022-Shift-2]

Answer: 2

Solution:

(I) $HCl + NaOH \rightarrow NaCl + H_2O$
$\Delta H_1 = -57.3 KJmol^{-1}$
(II) $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$
$\Delta H_2 = -55.3 KJmol^{-1}$
Reaction (I) can be written as
(III) $NaCl + H_2O \rightarrow HCl + NaOH$
$\Delta H_3 = 57.3 KJmol^{-1}$
By adding (II) and (III)
$CH_3COOH + NaCl \rightarrow CH_3COONa + HCl \quad \Delta H_r$
$\Delta H_r = \Delta H_3 + \Delta H_2 \quad = 57.3 - 55.3$
$= 2 \text{ kJ mol}^{-1}$

Question107

2.4g coal is burnt in a bomb calorimeter in excess of oxygen at 298K and 1 atm pressure. The temperature of the calorimeter rises from 298K to 300K. The enthalpy change during the combustion of coal is $-x \text{ kJ mol}^{-1}$. The value of x is ________. (Nearest Integer)
(Given : Heat capacity of bomb calorimeter 20.0 kJ K^{-1}. Assume coal to be pure carbon)
[26-Jul-2022-Shift-1]

Answer: 200

Solution:

Q(Heat evolved $) = -\dfrac{C_{system} \Delta T}{n}$
$n_{coal} = \dfrac{2.4}{12}$
$Q = \dfrac{-20(300-298)}{0.2}$
$Q = -200 \text{ kJ/mol}$
$x = 200$

Question108

For the reaction
$H_2F_2(g) \rightarrow H_2(g) + F_2(g)$
$\Delta U = -59.6 \text{ kJ mol}^{-1}$ at $27^{\circ}C$
The enthalpy change for the above reaction is (−) _______ kJ mol^{-1} [nearest integer]

Given: $R = 8.314 JK^{-1} mol^{-1}$
[26-Jul-2022-Shift-2]

Answer: 57

Solution:

$H_2F_2(g) \longrightarrow H_2(g) + F_2(g)$
$\Delta U = -59.6\,kJ\,mol^{-1}$ at $27^{\circ}C$
$\Delta H = \Delta U + \Delta n_g RT$
$= -59.6 + \dfrac{1 \times 8.314 \times 300}{1000}$
$= -57.10\,kJ\,mol^{-1}$

Question109

The molar heat capacity for an ideal gas at constant pressure is $20.785 JK^{-1} mol^{-1}$. The change in internal energy is 5000J upon heating it from 300K to 500K. The number of moles of the gas at constant volume is _______. [[Nearest integer] (Given: $R = 8.314 JK^{-1} mol^{-1}$)
[27-Jul-2022-Shift-1]

Answer: 2

Solution:

$C_p = 20.785 JK^{-1} mol^{-1}$
and $\Delta U = nCv\,\Delta\,T$
$\therefore\ nC_v = \dfrac{5000}{200} = 25$
and we know that
$C_p - C_v = R$
$20.785 - \dfrac{25}{n} = 8.314$
$n = \dfrac{25}{(20.785 - 8.314)} = 2$

Question110

A gas (Molar mass $= 280 g mol^{-1}$) was burnt in excess O_2 in a constant volume calorimeter and during combustion the temperature of calorimeter increased from 298.0K to 298.45K. If the heat capacity of calorimeter is $2.5\,kJ\,K^{-1}$ and enthalpy of combustion of gas is $9\,kJ\,mol^{-1}$ then amount of gas burnt is ________ g. (Nearest Integer)
[27-Jul-2022-Shift-2]

Answer: 35

Solution:

Let xg is burnt

$\text{moles} = \frac{x}{280}$

heat released by $\frac{x}{280}$ mole $= 2.5 \times 0.45\,kJ$

heat released by 1 mole $= \frac{2.5 \times 0.45 \times 280}{x} kJ$

$\Delta H = \Delta U + \Delta\, ngRT$

$\Delta H \simeq \Delta U$

$9 = \frac{2.5 \times 280 \times 0.45}{x}$

$x = 35g$

Question111

Which of the following relation is not correct?
[28-Jul-2022-Shift-1]

Options:

A. $\Delta H = \Delta U - P \Delta V$

B. $\Delta U - q + W$

C. $\Delta S_{sys} + \Delta S_{surr} \geq sl\,ant0$

D. $\Delta G = \Delta H - T \Delta S$

Answer: A

Solution:

If $U + Pv$ (By definition)

$\Delta 14 = \Delta U + \Delta\,(Pr)$ at constant pressure

$\Delta H = \Delta U + P \Delta V$

Question112

Given below are two statements: One is labelled as Assertion AA and the other is labelled as Reason RR
Assertion A: The reduction of a metal oxide is easier if the metal formed is in liquid state than solid state.
Reason RR : The value of $\Delta G^{\ominus}$ Δ G becomes more on negative side as entropy is higher in liquid state than solid state.
In the light of the above statements, choose the most appropriate answer from the options given below
[28-Jul-2022-Shift-2]

Options:

A. Both A and R are correct and R is the correct explanation of A

B. Both A and R are correct but R is NOT the correct explanation of A

C. A is correct but R is not correct

D. A is not correct but R is correct

Answer: A

Solution:

Reduction of a metal oxide is easier if the metal is formed in a liquid state at the temperature of reduction because the entropy is higher if the metal is in a liquid state.

Question113

Among the following the number of state variables is _______.
Internal energy (U)
Volume (V)
Heat (q)
Enthalpy (H)
[28-Jul-2022-Shift-2]

Answer: 3

Solution:

State variables are internal energy (U), Volume (V) and Enthalpy (H).

Question114

When 600 mL of $0.2\mathrm{M}\,HNO_3$ is mixed with 400 mL of 0.1M NaOH solution in a flask, the rise in temperature of the flask is _______ $\times 10^{-2}{}^{\circ}\mathrm{C}$
(Enthalpy of neutralisation $= 57\ \mathrm{kJ\ mol^{-1}}$ and Specific heat of water $= 4.2\ \mathrm{JK^{-1}g^{-1}}$) (Neglect heat capacity of flask)
[29-Jul-2022-Shift-1]

Answer: 54

Solution:

HNO_3
$600\ \mathrm{mL} \times 0.2\mathrm{M} = 120\ \mathrm{m\,mol}$
NaOH
$400\ \mathrm{mL} \times 0.1\mathrm{M} = 40\ \mathrm{m\,mol}$

	HNO_3	+ NaOH	$\rightarrow NaNO_3 + H_2O$
Bef.	120	40	
Aft.	80	0	40m mol

Heat liberated from reaction
$= 40 \times 10^{-3} \times 57 \times 10^{3}$ J Heat gained by solution $= mC\,\Delta T$
$m =$ mass of solution $= V \times d = 1000 \times 1$
$= 1000 g$
Heat gained by solution $= 1000 \times 4.2 \times \Delta T \ldots$ (2)
From (1) and (2)
Heat liberated = Heat gained
$40 \times 10^{-3} \times 57 \times 10^{3} = 1000 \times 4.2 \times \Delta T$
$\Delta T = 54 \times 10^{-2}\,°C$
(Rounded off to the nearest integer)

Question115

Five moles of an ideal gas at 293K is expanded isothermally from an initial pressure of 2.1M Pa to 1.3M Pa against at constant external pressure 4.3M Pa. The heat transferred in this process is .... kJ mol $^{-1}$. (Rounded off to the nearest integer) [R = 8.314J mol $^{-1}$K $^{-1}$]
[25 Feb 2021 Shift 2]

Answer: 15

Solution:

The gas performs isothermal irreversible work (W).
where, $\Delta U = 0$ (change in internal energy)
From, 1st law of thermodynamics,
$\Rightarrow \Delta U = \Delta Q + W$
$\Rightarrow 0 = \Delta Q + W$
$\Rightarrow \Delta Q = -W$
Now, $W = -p_{ext}(V_2 - V_1)$
$= -p_{ext}\left(\frac{nRT}{p_2} - \frac{nRT}{p_1}\right) = -p_{ext} \times nRT\left(\frac{1}{p_2} - \frac{1}{p_1}\right)$
Given, $p_{ext} = 4.3 M\,Pa$, $p_1 = 2.1 M\,Pa$, $p_2 = 1.3 M\,Pa$,
$n = 5 mol$, $T = 293K$ and $R = 8.314 J\,mol^{-1}K^{-1}$
$= -4.3 \times 5 \times 8.314 \times 293\left(\frac{1}{1.3} - \frac{1}{2.1}\right)$
$= -15347.70 J\,mol^{-1}$
$= -15.347 kJ\,mol^{-1} \sim eq - 15 kJ\,mol^{-1}$
$\Rightarrow \Delta Q = 15 kJ\,mol^{-1}$

Question116

The ionisation enthalpy of Na^+ formation from Na(g) is 495.8kJ mol $^{-1}$, while the electron gain enthalpy of Br is −325.0kJ mol $^{-1}$. Given, the lattice enthalpy of NaBr is −728.4kJ mol $^{-1}$. The energy for the formation of NaBr ionic solid is (−)............ × 10^{-1}kJ mol $^{-1}$.
[25 Feb 2021 Shift 1]

Answer: 5576

Solution:

$Na(g) \xrightarrow{IE_1} Na^+(g)[IE_1 = 495.8 kJ\ mol^{-1}]$

$Br(g) \xrightarrow[\text{gain enthalpy}]{\text{Electron}} (EG) Br^-(g)$[Gain enthalpy $= -325 kJ\ mol^{-1}$]

$Na^+ + Br^- \xrightarrow[\text{energy}]{\text{Lattice}} (LE) NaBr(s)$[Lattice energy $= -728.4 kJ\ mol^{-1}$]

$\Delta H_{Formation} = IE_1 +$ Gain enthalpy + Lattice energy $\Delta H = 495.8 + (-325) + (-728.4)$

$= -557.6 kJ / mol$

$= -5576 \times 10^{-1} kJ$

Question117

The average S − F bond energy in kJ mol^{-1} of SF$_6$ is .... (Rounded off to the nearest integer) [Given, the values of standard enthalpy of formation of SF$_6$(g), S(g) and F (g) are −1100, 275 and 80kJ mol^{-1} respectively.]
[26 Feb 2021 Shift 2]

Answer: 309

Solution:

So, $\Delta_f H^\circ[S, g] + 6 \times \Delta_f H^\circ[F, g] = \Delta_f H^\circ[SF_6, g] + 6 \times E_{S-F}$

[$\therefore E_{S-F}$ = Average S − F bond energy in SF_6]

$275 + 6 \times 80 = -1100 + 6 \times E_{S-F}$

$\Rightarrow E_{S-F} = \dfrac{275 + 6 \times 80 + 1100}{6}$

$= 309.16 kJ\ mol^{-1} = 309 kJ\ mol^{-1}$

Question118

At 25°C, 50g of iron reacts with HCl to form FeCl$_2$. The evolved hydrogen gas expands against a constant pressure of 1 bar. The work done by the gas during this expansion is ......J. (Round off to the nearest integer) [Given, R = 8.314J mol^{-1}K^{-1}. Assume, hydrogen is an ideal gas]
[Atomic mass off Fe is 55.85u]
[16 Mar 2021 Shift 2]

Answer: 2218

Solution:

$\underset{50g}{Fe(s)} + HCl(aq) \longrightarrow FeCl_2(aq) + H_2(g)$

(Moles of Fe = Moles of H_2)

moles $= \dfrac{50}{55.85}$ 0.8952mol 0.8952mol

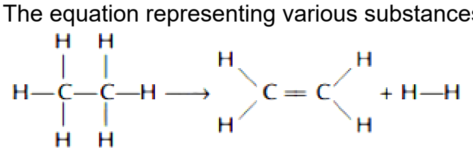
$= 0.8952 \text{mol}$
Work done can be given by
$W = -p_{ext} \Delta V$
$W = -\Delta n_g RT$
where, Δn_g = { gaseous moles on product side } - {gaseous mole on reactant side}
$= 0.8952 - 0$
$W = -0.8952 \times 8.314 \times 298$
$= -2217.92 \text{J}$
Closest integer $= 2218$

Question119

For the reaction,
$C_2H_6 \longrightarrow C_2H_4 + H_2$ (Round off to the nearest integer). [Given : Bond enthalpies in kJ mol m^{-1} : $C-C = 347$, $C=C = 611$; $C-H = 414$; $H-H = 436$
[18 Mar 2021 Shift 1]

Answer: 128

Solution:

The equation representing various substances involved in chemical reaction is


Given bond enthalpies of various bonds in kJ mol^{-1} i.e.
$C-C = 347$, $C=C = 611$, $C-H = 414$, $H-H = 436$
This reaction involves the breaking of $2C-H$ bonds and formation, of $1C=C$ bond and $1H-H$ bond.
$\Delta_r H =$ [Sum of bond enthalpies of reactants] - [Sum of bond enthalpies of products]
$= [1\,\Delta_{C-C} H + 6\,\Delta_{C-H} H] - [1\Delta_C = CH + 4\,\Delta_{C-H} H + 1\,\Delta_{H-H} H]$
$= (357 + 6 \times 414) - [611 + 4 \times 414 + 436]$
$= 2831 - 2703$
$= 128 \text{kJ/mol}$

Question120

The standard enthalpies of formation of Al_2O_3 and CaO are -1675kJ mol^{-1} and -625kJ mol^{-1} respectively. For the reaction, $3CaO + 2Al \longrightarrow 3Ca + Al_2O_3$ the standard reaction enthalpy $\Delta_t H^\circ$........ kJ . (Round off to the nearest integer).
[17 Mar 2021 Shift 1]

Answer: 230

Solution:

Given, $\Delta_f H_{Al_2O_3}^{o} = -1675 kJ/mol$

$\Delta_f H_{CaO}^{\circ} = -625 kJ/mol$

To find $\Delta_r H^{\circ}$ for the reaction

$3CaO + 2Al \longrightarrow 3Ca + Al_2O_3; \Delta_r H^{\circ} = ?$

$\Delta_r H^{\circ} = \sum_{\Delta_f} H_{\text{product}}^{\circ} - \sum_{\Delta_f} H_{\text{Reactant}}^{\circ}$

$= \left\{\Delta_f H_{Al_2O_3}^{\circ} + 3\,\Delta_f H_{Ca}^{\circ}\right\} - \{3\,\Delta_f H_{CaO}^{\circ} + 2\,\Delta_f H_{Al}^{\circ}\}$

$= \Delta H_f^{\circ}(Al_2O_3) - 3 \times \Delta H_f^{a}(CaO)$

We know, $\Delta_f H^{\circ}$ for elemental state $= 0$

$\Rightarrow \Delta_f H_{Ca}^{o} = 0, \Delta_f H_{Al}^{o} = 0$

Putting the value,

$\Delta_t H^{\circ} = \{-1675 + 0\} - \{3(-625) + 0\}$

$= -1675 + 1905$

$= 230 kJ$

Question121

During which of the following processes, does entropy decrease?

A. Freezing of water to ice at $0^{\circ}C$.

B. Freezing of water to ice at $-10^{\circ}C$.

C. $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g)$

D. Adsorption of CO(g) and lead surface.

E. Dissolution of NaCl in water.

[17 Mar 2021 Shift 2]

Options:

A. A, B, C and D

B. B and C

C. A and E

D. A, C and E

Answer: A

Solution:

Entropy will decrease in A, B, C and D processes.

A, B$\rightarrow$ Freezing of water will decrease entropy as particles will move closer and forces of attraction will increase. This leads to a decrease in randomness. So, entropy decreases.

(A) Water $\xrightarrow{0^{\circ}C}$ ice; $\Delta S = -$ ve

(B) Water $\xrightarrow{-10^{\circ}C}$ ice; $\Delta S = -$ ve

(C) $N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g); \Delta S = -$ ve

Number of moles are decreasing

$n = 2 - (3 + 1)$

$n = -2$

So, entropy decreases.

(D) Adsorption; $\Delta S = -$ ve

Adsorption will lead to a decrease in the randomness of gaseous particles.

So, entropy decreases.

(E) $NaCl(s) \longrightarrow Na^{+}(aq) + Cl^{-}(aq); \Delta S > 0$

The number of species on product side is more than the number of species on reactant side. So, entropy increases on dissolution of NaCl in water.

Question122

When 400mL of 0.2M H_2SO_4 solution is mixed with 600mL of 0.1M N aOH solution, the increase in temperature of the final solution is $-10^{-2}K$. (Round off to the nearest integer).
[Use : $H^+(aq) + OH^-(aq) \rightarrow H_2O$:
$\Delta_\gamma H = -57.1 kJ\ mol^{-1}$]
Specific heat of $H_2O = 4.18 J\ K^{-1} g^{-1}$
density of $H_2O = 1.0 gcm^{-3}$
Assume no change in volume of solution on mixing.
[27 Jul 2021 Shift 2]

Answer: 82

Solution:

$n_{H^+} = \frac{400 \times 0.2}{1000} \times 2 = 0.16$
$n_{OH^-} = \frac{600 \times 0.1}{1000} = 0.06 (L.R)$
Now, heat liberated from reaction = heat gained by solutions
or, $0.06 \times 57.1 \times 10^3$
$= (1000 \times 1.0) \times 4.18 \times \Delta T$
$\therefore \Delta T = 0.8196 K$
$= 81.96 \times 10^{-2} K \approx 82 \times 10^{-2} K$

Question123

For water at 100°C and 1 bar,
$\Delta_{vap} H - \Delta_{vap} U = \rule{2em}{0.4pt} \times 10^2 J\ mol^{-1}$.
(Round off to the Nearest Integer)
[Use : $R = 8.31 J\ mol^{-1} K^{-1}$]
[Assume volume of $H_2O(l)$ is much smaller than volume of $H_2O(g)$. Assume $H_2O(g)$ treated as an ideal gas]
[27 Jul 2021 Shift 1]

Answer: 31

Solution:

$H_2O_{(l)} \rightleftharpoons H_2O_{(v)}$
$\Delta H = \Delta U + \Delta n_g RT$
for 1 mole waters ;
$\Delta n_g = 1$
$\therefore \Delta n_g RT = 1 mol \times 8.31 J/mol - k \times 373 K$
$= 3099.63 J \cong 31 \times 10^2 J$

Question124

A system does 200 J of work and at the same time absorbs 150 J of heat. The magnitude of the change in internal energy is __________ J. (Nearest integer)
[25 Jul 2021 Shift 2]

Answer: 50

Solution:

$w = -200J, q = +150 : \Delta U = q + w$
$\Delta U = 150 - 200 = -50J$: magnitude $= 50J = |\Delta U|$

Question125

At 298K, the enthalpy of fusion of a solid (X) is 2.8~kJ mol $^{-1}$ and the enthalpy of vaporisation of the liquid (X) is 98.2kJ mol $^{-1}$. The enthalpy of sublimation of the substance (X) in kJ mol $^{-1}$ is ________ . (in nearest integer)
[25 Jul 2021 Shift 1]

Answer: 101

Solution:

$\Delta H_{sub} = \Delta H_{fivs.} + \Delta H_{vap.}$
$= 2.8 + 98.2$
$= 101kJ/mol$

Question126

At 298.2K the relationship between enthalpy of bond dissociation (in kJ mol $^{-1}$) for hydrogen (E_H) and its isotope, deuterium (E_D), is best described by:
[25 Jul 2021 Shift 1]

Options:

A. $E_H = \frac{1}{2}E_D$

B. $E_H = E_D$

C. $E_H \simeq E_D - 7.5$

D. $E_H = 2E_D$

Answer: C

Solution:

Enthalpy of bond dissociation (kJ/mole) at 298.2K
For , hydrogen = 435.88
For , Deuterium = 443.35
$\therefore E_H \simeq E_D - 7.5$

Question127

If the standard molar enthalpy change for combustion of graphite powder is -2.48×10^2kJ mol $^{-1}$,the amount of heat generated on combustion of 1g. of graphite powder is _______ kJ . (Nearest integer)
[22 Jul 2021 Shift 2]

Answer: 21

Solution:

1mol graphite $= 12\text{gmC}$
Ans. $= \frac{248}{12} = 20.67\text{kJ} / \text{gm}$ heat evolved

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Question128

For a given chemical reaction A → B at 300K the free energy change is −49.4kJ mol $^{-1}$ and the enthalpy of reaction is 51.4kJ mol $^{-1}$. The entropy change of the reaction is J K $^{-1}$mol $^{-1}$.
[20 Jul 2021 Shift 2]

Answer: 336

Solution:

Given chemical reaction:
$A \xrightarrow[T\ 300K]{} B [\Delta G]_{P,T} = -49.4$ kJ/mol
$\Delta H_{rxn} = 51.4$ kJ/mol
$\Delta S_{rxn} = ?$
⇒ From the relation $[\Delta G]_{P,T} = \Delta H - T\Delta S$
$\Rightarrow \Delta S_{rxn} = \frac{\Delta H_{rn} - [\Delta G]_{P,T}}{T}$
$= \frac{[51.4 - (-49.4)] \times 1000}{300} \frac{J}{mol\ K}$
$\Rightarrow \Delta S_{rxn} = 336 \frac{J}{mol\ K}$

Question129

For water $\Delta_{vap}H = 41 \text{ kJ mol}^{-1}$ at 373K and 1 bar pressure. Assuming that water vapour is an ideal gas that occupies a much larger volume than liquid water, the internal energy change during evaporation of water is .... kJ mol^{-1}.
[Use $R = 8.3 \text{Jmol}^{-1}\text{K}^{-1}$]
[26 Aug 2021 Shift 2]

Answer: 38

Solution:

$H_2O(l) \rightarrow H_2O(g)$
$\because \Delta H = \Delta U + \Delta n_g RT$
ΔH = enthalpy of vaporisation
ΔU = change in internal energy
Δn_g = number of moles of water vapour
$R = 8.3 \text{Jmol}^{-1}\text{K}^{-1}$
T = 373 K
$\therefore 41 \text{ kJ} = \Delta U + RT$
$\Delta U = 41 - 8.3 \times 373 \times 10^{-3} = 41 - 3.095$
$= 37.90 \text{ kJ mol}^{-1} = 38 \text{ kJ mol}^{-1}$

Question130

200 mL of 0.2M HCl is mixed with 300 mL of 0.1M NaOH. The molar heat of neutralisation of this reaction is −57.1 kJ. The increase in temperature in °C of the system on mixing is $x \times 10^{-2}$. The value ofx is .... (Nearest integer)
[Given, specific heat of water $= 4.18 \text{Jg}^{-1}\text{K}^{-1}$Density of water $= 1.00 \text{gcm}^{-3}$]
(Assume no volume change on mixing)
[27 Aug 2021 Shift 1]

Answer: 82

Solution:

Millimoles of $HCl = 200 \times 0.2 = 40$
Millimoles of $NaOH = 300 \times 0.1 = 30$
Heat released $(q) = n \times$ molar heat
$= \frac{30}{1000} \times 57.1 \times 1000$
= 1713 J
Mass of solution = × 500 × 1= 500 g
We know that,
$\Delta T = \frac{q}{mc} = \frac{1713 \text{J}}{500 \text{g} \times 4.18 \text{J}/\text{g} - \text{K}}$
$= 0.8196 \text{K} = 81.96 \times 10^{-2} \text{K}$
x ≈ 82

Question131

The Born-Haber cycle for KCl is evaluated with the following data :

$\Delta_f H°$ for KCl $= -436.7\ kJ\ mol^{-1}$

$\Delta_{sub} H°$ for K $= 89.2\ kJ\ mol^{-1}$,

$\Delta_{ionisation} H°$ for K $= 419.0\ kJ\ mol^{-1}$;

$\Delta_{electron\ gain} H^{\cup}$ for Cl(g) $= -348.6\ kJ\ mol^{-1}$,

$\Delta_{bond} H°$ for $Cl_2 = 243.0\ kJ\ mol^{-1}$

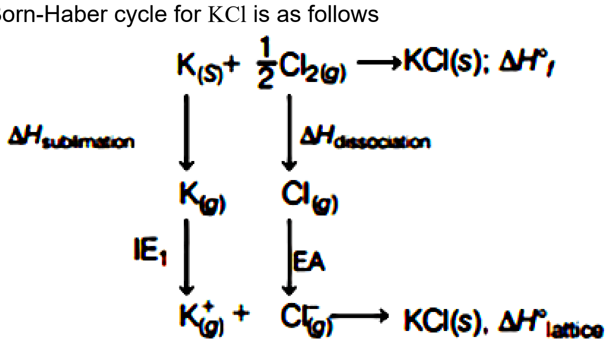
The magnitude of lattice enthalpy of KCl in kJ mol^{-1} is . (Nearest integer)

[26 Aug 2021 Shift 1]

Answer: 718

Solution:

Born-Haber cycle for KCl is as follows



$\therefore \Delta H_f° = \Delta H_{sublimation} + \frac{1}{2}\Delta H_{dissociation} + IE_1 + EA + \Delta H_{lattice}$

$-436.7\ kJ/mol = 89.2\ kJ/mol + \frac{1}{2} \times (243\ kJ/mol) + 419\ kJ/mol + (-348.6\ kJ/mol) + \Delta H°_{lattice}$

$\Rightarrow \Delta H°_{lattice} = -717.8\ kJ/mol$

$\therefore$ Magnitude of lattice enthalpy of KCl in kJ/mol is 718 (nearest).

Question132

The incorrect expression among the following is
[31 Aug 2021 Shift 2]

Options:

A. $\frac{\Delta G_{System}}{\Delta S_{Total}} = -T$ (at constant p)

B. $\ln k = \frac{\Delta H° - T\Delta S°}{RT}$

C. $k = e^{-\frac{\Delta G°}{RT}}$

D. For isothermal process, $W_{reversible} = -nRT \ln \frac{V_f}{V_i}$

Answer: B

Solution:

All the expression of thermodynamics are correct except
$\ln k = \frac{\Delta H° - T\Delta S°}{RT}$
As we know,
$\Delta G = \Delta H - T\Delta S°$ ...(i)
Also $\Delta G = -RT \ln K$ ...(ii)
∴ From (i) and (ii)
$-RT \ln K = \Delta H - T\Delta S°$
$\therefore \ln K = \frac{-\Delta H}{R} + \frac{\Delta S°}{R}$
The corect expression is
$\ln K = \frac{-\Delta H°}{RT} + \frac{\Delta S°}{R} = \frac{\Delta H° - T\Delta S°}{RT}$

Question133

Data given for the following reaction is as follows.
$FeO(s) + C_{(graphite)} \rightarrow Fe(s) + CO(g)$

Substance	$\Delta H°(kJ\ mol^{-1})$	$\Delta S°(J\ mol^{-1}K^{-1})$
$FeO(s)$	-266.3	57.49
$C_{(graphite)}$	0	5.74
$Fe(s)$	0	27.28
$CO(g)$	-110.5	197.6

The minimum temperature in K at which the reaction becomes spontaneous is......... (Integer answer)
[27 Aug 2021 Shift 2]

Answer: 964

Solution:

$FeO(s) + C_{(graphite)} \rightarrow Fe(s) + CO(g)$
$\Delta H°_{reaction} = \Delta H°_{f(product)} - \Delta H°_{f(reactants)}$
$= [\Delta H°_{f(Fe)} + \Delta H°_{f(CO)}] - [\Delta H°_{f(FeO)} - \Delta H°_{f(C)}]$
$= [0 + (-110.5)] - [-266.3 - 0]$
$= 156 kJ^{-1} mol^{-1}$
$\Delta S°_{reaction} = \Delta S°_{product} - \Delta S°_{reactant}$
$= [\Delta S°_{(Fe)} + \Delta S°_{(CO)}] - [\Delta S°_{(FeO)} - \Delta S°_{(O)}]$
$= [27.28 + 197.6] - [57.49 + 5.79]$
$= 161 JK^{-1} mol^{-1}$
According to Gibb's equation,
$\Delta G° = \Delta H° - T\Delta S°$
The reaction becomes spontaneous when $\Delta G°$ is atleast zero or negative.
$0 = \Delta H° - T\Delta S°$
$T\Delta S° = \Delta H°$

$\Rightarrow T = \frac{\Delta H^\circ}{\Delta S^\circ} = \frac{156\,kJ\,mol^{-1}}{161 JK^{-1} mol^{-1}}$

$= \frac{156000 mol^{-1}}{161 JK^{-1} mol^{-1}} = 964K$

The temperature at which reaction becomes spontaneous is 964 K.

Question134

For the reaction, $2NO_2(g) \rightleftharpoons N_2O_4(g)$, when $\Delta S = -176.0 \times JK^{-1}$ and $\Delta H = -57.8\,kJ\,mol^{-1}$, the magnitude of ΔG at 298K for the reaction is ....$kJ\,mol^{-1}$. (Nearest integer)
[1 Sep 2021 Shift 2]

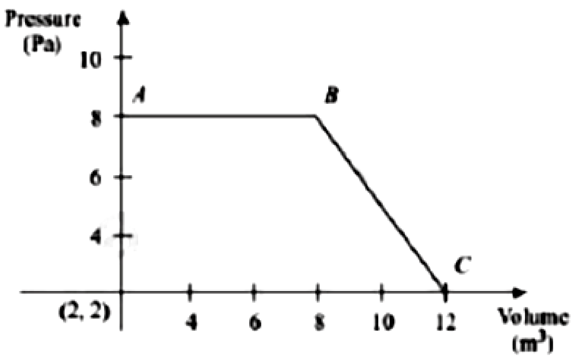
Answer: 5

Solution:

Given, $\Delta H = -57.8\,kJ\,mol^{-1}$
$\Delta S = -176 JK^{-1} mol^{-1}$
$T = 298K$
Using Gibb's free energy relation
$\Delta G = \Delta H - T\Delta S$
where, ΔG = change in Gibb's free energy
ΔH = change in enthalpy
T = temperature
ΔS = change in entropy
$\Delta G = 57.8\,kJ/mol - [298K \times (-176 Jk^{-1}\ mol^{-1})]$
$= 57.8\,kJ/mol - \left(298 \times \frac{-176}{1000} kJ\right)$ [$\therefore$1kJ = 1000 J]
= − 5.352 kJ/mol
| ΔG | = 5.352
Hence, answer is 5.

Question135

The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is ________.



[NV, Jan. 08,2020(I)]

Answer: 48

Solution:

Work done is given by the area under the trapezium.

$\therefore |w| = \frac{1}{2}(6+10) \times 6 = 48J$

Question136

At constant volume, 4 mol of an ideal gas when heated from 300K to 500K changes its internal energy by 5000J . The molar heat capacity at constant volume is
[NV, Jan. 08, 2020 (II)]

Answer: 6.25

Solution:

$$\begin{aligned} \Delta U &= nC_v \Delta T \\ 5000 &= 4 \times C_v(500-300) \\ C_v &= 6.25 J K^{-1} mol^{-1} \end{aligned}$$

Question137

If enthalpy of atomisation for $Br_2(\ell)$ is xkJ/mol and bond enthalpy for Br_2 is ykJ/mol, the relation between them:
[Jan. 09,2020 (I)]

Options:

A. is $x = y$

B. does not exist

C. is $x > y$

D. is $x < y$

Answer: C

Solution:

$\Delta H_{atomisation} = \Delta H_{vap} +$
Bond energy Hence $x > y$

Question138

For the reaction ;$A(l) \rightarrow 2B(g)$
$\Delta U = 2.1kcal$, $\Delta S = 20cal K^{-1}$ at 300K

Hence Δ G in kcal is.
[NV, Jan. 07, 2020 (I)]

Answer: -2.7

Solution:

$$\begin{aligned}
\Delta U &= 2.1\text{kcal} = 2.1 \times 10^3\text{cal} \\
\Delta n_g &= 2 \\
\Delta H &= \Delta U + \Delta n_g RT \\
&= 2.1 \times 10^3 + 2 \times 2 \times 300 \\
&= 2100 + 1200 \\
&= 3300\text{cal} \\
\Delta G &= \Delta H - T\Delta S \\
&= 3300 - 300 \times 20 \\
&= 3300 - 6000 \\
&= -2700\text{cal s} \\
&= -2.7\text{kcal}
\end{aligned}$$

Question139

The standard heat of formation ($\Delta_f H_{298}^{\circ}$) of ethane (in kJ/mol), if the heat of combustion of ethane, hydrogen and graphite are −1560, −393.5 and −286kJ /mol , respectively is ________.
[NV, Jan. 07,2020 (II)]

Answer: -192.5

Solution:

$C(s) + O_2(g) \rightarrow CO_2(g)$

$\Delta_C H^0[\text{Cgraphite}] = -286.0\text{kJ/mol} \;...\; (i)$

$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$

$\Delta_C H^0[H_2(g)] = -393.5\text{kJ/mol} \;...\; (ii)$

$C_2H_6(g) + \frac{7}{2}O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$

$\Delta_C H^0[C_2H_6(g)] = -1560\text{kJ/mol} \;...\; (iii)$

The reaction of formation of ethane is

$2C(s) + 3H_2(g) \rightarrow C_2H_6(g)$

$\therefore \Delta_f H^0$ of $C_2H_6(g) =$

$2 \times \Delta_C{}^0H[C_{\text{gaphite}}] + 3 \times \Delta_c H^0[H_2(g)] - \Delta_C H^0[C_2H_6(g)]$

$= 2 \times (-286.0) + 3(-393.5) - (-1560)$

$= -192.5\text{kJ/mol}$

Question140

The true statement amongst the following is :
[Jan. 09,2020 (II)]

Options:

A. Both ΔS and S are functions of temperature.

B. Both S and ΔS are not functions of temperature.

C. S is not a function of temperature but ΔS is a function of temperature.

D. S is a function of temperature but ΔS is not a function of temperature.

Answer: A

Solution:

A system at higher temperature has greater entropy (randomness). S and Δ S are related with T as:
$S_T = \int_0^T \frac{nC \cdot dT}{T}$ $\Delta S = \int \frac{dq}{T}$
Thus both S and ΔS are function of temperature.

Question141

Five moles of an ideal gas at 1 bar and 298K is expanded into vacuum to double the volume. The work done is:
[Sep. 04,2020 (II)]

Options:

A. $C_v(T_2 - T_1)$

B. $-RT(V_2 - V_1)$

C. $-RT \ln V_1 / V_1$

D. zero

Answer: D

Solution:

In expansion against vacuum
$P_{ext} = 0$
$W = -P_{ext} \Delta V = 0$
$W = 0$

Question142

Lattice enthalpy and enthalpy of solution of $NaCl$ are $788 kJ mol^{-1}$ and $4 kJ mol^{-1}$, respectively. The hydration enthalpy of $NaCl$ is :
[Sep. 05, 2020(II)]

Options:

A. $-780 kJ mol^{-1}$

B. $780 kJ mol^{-1}$

C. $-784 \mathrm{kJ} \mathrm{~mol}^{-1}$

D. $784 \mathrm{kJ} \mathrm{~mol}^{-1}$

Answer: C

Solution:

$\Delta_{\text{sol.}} H^{\circ}=\Delta_{\text{lattice}} H^{\circ}+\Delta_{\text{H yd .}} H^{\circ}$
$4=788+\Delta_{\text{H yd}} H^{\circ}$
$\Delta_{\text{H yd .}} H^{\circ}=-784 \mathrm{kJ} \mathrm{~mol}^{-1}$

Question143

For one mole of an ideal gas, which of these statements must be true?
(1) U and H each depends only on temperature (2) Compressibility factor z is not equal to 1
(3) $C_{P, m}-C_{V, m}=R$ (4) $d U=C_{y} d T$ for any process
[Sep. 04, 2020 (I)]

Options:

A. (1) and (3)

B. (2), (3) and (4)

C. (3) and (4)

D. (1), (3) and (4)

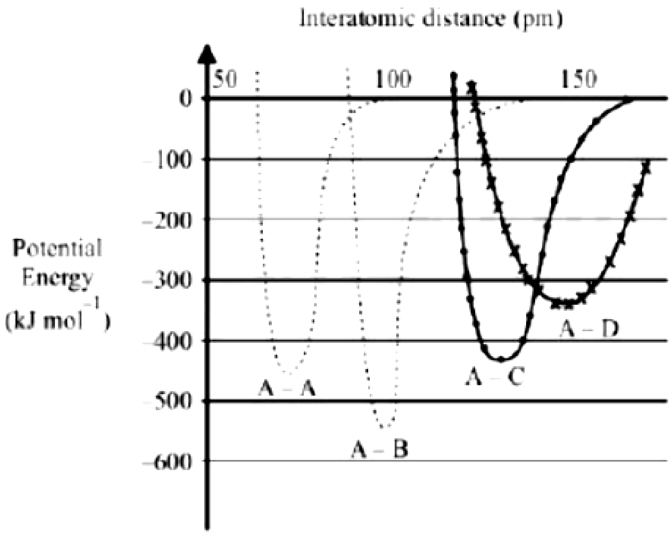
Answer: D

Solution:

(1) For ideal gas U and H are function of temperature.
(2) Compressibility factor for an ideal gas is 1 .
(3) $C_{P}-C_{V}=R$
(4) $\Delta U=$ CvdT for all processes.

Question144

The intermolecular potential energy for the molecules A, B, C and D given below suggests that :



[Sep. 04,2020 (1)]

Options:

A. A-D has the shortest bond length

B. A-A has the largest bond enthalpy

C. D is more electronegative than other atoms

D. A-B has the stiffest bond

Answer: D

Solution:

A-B bond has highest intermolecular potential energy among the given molecules. Hence, it is strongest bond and has maximum bond enthalpy.

Question145

The internal energy change (in J) when 90g of water undergoes complete evaporation at $100^{\circ}C$ is ______.
(Given : ΔH_{vap} for water at 373K = 41kJ / mol , R = 8.314J $K^{1}mol^{-1}$)
[NV, Sep. 02,2020(I)]

Answer: 189494

Solution:

$\Delta H = \Delta U + \Delta n_g RT$

$n = \frac{90}{18} = 5mol$

$H_2O(l) \rightleftharpoons H_2O(g)$

$\Delta n = 1$

$41000 = \Delta U + 1 \times 8.314 \times 373$

$\Rightarrow \Delta U = 37898.875J$

For 5 moles, $\Delta U = 37898.87 \times 5 = 189494J$

Question146

The heat of combustion of ethanol into carbon dioxids and water is −327 kcal at constant pressure. The heat evolved (in cal) at constant volume and 27°C (if all gases behave ideally) is ($R = 2 \text{cal mol}^{-1} K^{-1}$)_______.
[NV, Sep. 02, 2020 (II)]

Answer: -326400

Solution:

$C_2H_5OH(l) + O_2(g) \longrightarrow 2O_2(g) + 3H_2O(I)$
$\Delta H_C = -327K\ cal$
$\Delta H = \Delta U + \Delta n_g RT$
$\Rightarrow -327 \times 10^3 = \Delta U + (-1) \times 2 \times 300$
$\Rightarrow \Delta U = -327 \times 10^3 + 600$
$\therefore \Delta U = -326400 cal$

Question147

For a dimerization reaction, $2A(g) \rightarrow A_2(g)$,
at $298K$, $\Delta U^{\Theta} = -20 kJ\ mol^{-1}$, $\Delta S^{\Theta} = -30 J\ K^{-1} mol^{-1}$,
then the ΔG^{Θ} will be _______ J.
[NV, Sep. 05,2020 (II)]

Answer: -13538

Solution:

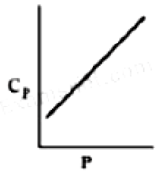
From $\Delta H^{\circ} = \Delta U^{\circ} + \Delta n_g RT$
$\Delta H^{\circ} = -20 \times 1000 - 1 \times 8.314 J/mol.K \times 298K$
$= -22477.57J$
$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ} = -22477.57 - (298 \times -30)$
$= -13538J$

Question148

For a diatomic ideal gas in a closed system, which of the following plots does not correctly describe the relation between various thermodynamic quantities?
[Jan. 12, 2019 (I)]

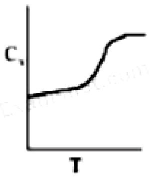
Options:

A.

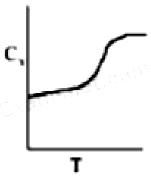


B.

C.



D.



Answer: A

Solution:

For ideal gas, C_p and C_v are dependent on temperaturc only.

$C_p = \frac{7}{2}R$ (Independent of P)

$C_v = \frac{5}{2}R$ (Independent of V)

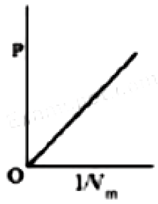
Thus, C_p will not change with pressure.

Question149

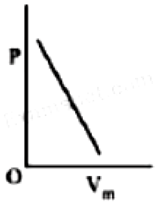
The combination of plots which does not represent isothermal expansion of an ideal gas is: [Jan. 12, 2019 (II)]

Options:

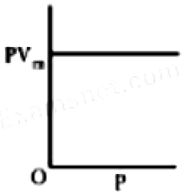
A.



B.



C.



D.

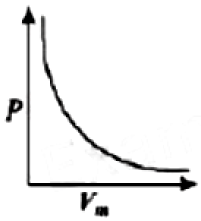
Answer: 0

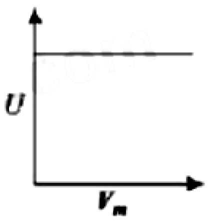
Solution:

Isothermal expansion

$PV_m = K$ (graph - C)

$P = \frac{K}{V_m}$(graph − A)





Therefore (B) and (D) are not correct representation.

Question150

An ideal gas undergoes isothermal compression from $5 m^3$ to $1 m^3$ against a constant external pressure of $4 N\ m^{-2}$. Heat released in this process is used to increase the temperature of 1 mole of Al. If molar heat capacity of Al is $24 J\ mol^{-1} K^{-1}$, the temperature of Al increases by: [Jan. 10, 2019 (II)]

Options:

A. $\frac{3}{2}K$

B. 2K

C. $\frac{2}{3}K$

D. 1K

Answer: C

Solution:

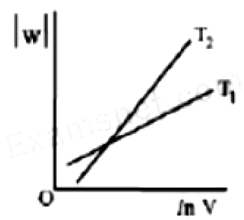
We know that,
$w = -P_{ext}(V_f - V_i)$
$w = -4N\,m^{-2}(1-5)m^3$
$w = 16N\,m \Rightarrow 16J$
For isothermal compression,
$\Delta U = q + w$
$\Rightarrow q = -w = -16J$ ($\because \Delta U = 0$ for isothermal process)
From calorimetry,
Heat given $= nC\,\Delta T$
So, $16 = \frac{1 \times 24J \times \Delta T}{mol\,K}$
$\therefore$ Change in temperature, $\Delta T = \frac{2}{3}K$

Question151

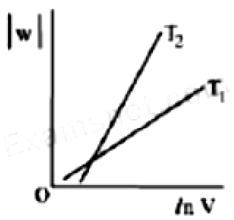
Consider the reversible isothermal expansion of an ideal gas in a closed system at two different temperatures, T_1 and $T_2 (T_1 < T_2)$. The correct graphical depiction of the dependence of work done (w) on the final volume (V) is:
[Jan. 9, 2019 (I)]

Options:

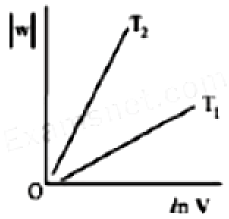
A.



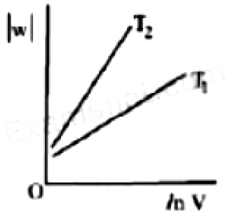
B.



C.



D.



Answer: B

Solution:

For reversible isothermal expansion,

$w = -nRT \ln \frac{V_2}{V_1}$

$\Rightarrow |w| = nRT \ln \frac{V_2}{V_1}$

$|w| = nRT (\ln V_2 - \ln V_1)$

$|w| = nRT \ln V_2 - nRT V_1$

$y = mx + c$

So, slope of curve 2 is more than curve 1 and intercept of curve 2 is more negative than curve 1 .

Question152

(i) C (graphite) +$O_2(g) \rightarrow CO_2(g)$; $\Delta_r H^0 = x kJ\ mol^{-1}$

(ii) C(graphite) + $\frac{1}{2}O_2(g) \rightarrow CO(g)$; $\Delta_r H^0 = y kJ\ mol^{-1}$

(iii) $CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g)$; $\Delta_r H^0 = z kJ\ mol^{-1}$

Based on the above thermochemical equations, find out which one of the following algebraic relationships is correct?
[Jan. 12, 2019 (II)]

Options:

A. $x = y + z$

B. $z = x + y$

C. $y = 2z - x$

D. $x = y - z$

Answer: A

Solution:

Equation (i) can be obtained by adding equations (ii) and equation (iii)

$C(\text{graphite}) + \frac{1}{2}O_2(g) \rightarrow CO(g); \Delta H_1^\circ = y\,kJ\,mol^{-1}$ ... (ii)

$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g); \Delta H_2^\circ = z\,kJ\,mol^{-1}$ . . . (iii)

$C\ (\text{graphite}) + O_2(g) \rightarrow CO_2(g); \Delta H_3^\circ = x\,kJ\,mol^{-1}$ ... (i)

$\because\ \Delta H_3^\circ = \Delta H_1^\circ + \Delta H_2^\circ$

$x = y + z$

Question153

Two blocks of the same metal having same mass and at temperature T_1, and T_2, respectively, are brought in contact with each other and allowed to attain thermal equilibrium at constant pressure. The change in entropy, ΔS, for this process is :
[Jan. 11, 2019 (I)]

Options:

A. $C_p \ln\left[\frac{(T_1+T_2)^2}{4T_1T_2}\right]$

B. $2C_p \ln\left[\frac{(T_1+T_2)^{\frac{1}{2}}}{T_1T_2}\right]$

C. $2C_p \ln\left(\frac{T_1+T_2}{4T_1T_2}\right)$

D. $2C_p \ln\left[\frac{T_1+T_2}{2T_1T_2}\right]$

Answer: A

Solution:

Final temperature $= \frac{T_1+T_2}{2}$, let $T_2 > T_1$

$\therefore dS = \frac{dq}{T} = \frac{C_p dT}{T}$

$\therefore \Delta S = C_p \ln\left(\frac{T_f}{T_i}\right)$

$\therefore \Delta S_{\text{Total}} = C_p \ln\left(\frac{T_1+T_2}{2T_1}\right) + C_p \ln\left(\frac{T_1+T_2}{2T_2}\right)$

$= C_p \ln\left[\frac{(T_1+T_2)^2}{4T_1T_2}\right]$

Question154

The process with negative entropy change is:
[Jan. 10, 2019 (II)]

Options:

A. Dissociation of $CaSO_4(s)$ to $CaO(s)$ and $SO_3(g)$

B. Sublimation of dry ice

C. Dissolution of iodine in water

D. Synthesis of ammonia from N_2 and H_2

Answer: D

Solution:

In the process of synthesis of ammonia from N_2 and H_2, number of moles decreases which implies that the change in entropy will be negative.
$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Question155

The entropy change associated with the conversion of 1 kg of ice at 273K to water vapours at 383K is:
(Specific heat of water liquid and water vapour are $4.2 kJ\ K^{-1} kg^{-1}$ and $2.0 kJ\ K^{-1} kg^{-1}$; heat of liquid fusion and vapourisation of water are $334 kJ\ kg^{-1}$ and $2491 kJ\ kg^{-1}$, respectively).
($\log 273 = 2.436$, $\log 373 = 2.572$, $\log 383 = 2.583$)
[Jan. 9, 2019 (II)]

Options:

A. $7.90 kJ\ kg^{-1} K^{-1}$

B. $2.64 kJ\ kg^{-1} K^{-1}$

C. $8.49 kJ\ kg^{-1} K^{-1}$

D. $9.26 kJ\ kg^{-1} K^{-1}$

Answer: D

Solution:

∴ Total entropy change
$\Delta S = 1.28 + 6.68 + 1.31 + 0.05$
$= 9.26 kJ\ kg^{-1} K^{-1}$

Question156

For the chemical reaction $X \rightleftharpoons Y$, the standard reaction Gibbs energy depends on temperature T (in K) as $\Delta_r G^{\circ}$(in $kJ\ mol^{-1}$) $= 120 - \frac{3}{8}T$
The major component of the reaction mixture at T is :
[Jan. 11, 2019 (I)]

Options:

A. Y if $T = 300K$

B. Y if $T = 280K$

C. X if $T = 350K$

D. X if $T = 315K$

Answer: D

Solution:

At 315K ; $\Delta G^{\circ} = 120 - \frac{3}{8}T$

$\Delta G^{\circ} = 120 - 118.125 =$ positive

Since ΔG° is positive then $K_{eq} < 1$.

So $\frac{[Y]}{[X]} < 1$.

$\therefore [X] > [Y]$

Question157

The reaction $MgO(s) + C(s) \rightarrow Mg(s) + CO(g)$, for which $\Delta H^{\circ} = +491.1 kJ\ mol^{-1}$ and $\Delta S^{\circ} = 198.0 J\ K^{-1} mol^{-1}$ is not feasible at 298K. Temperature above which reaction will be feasible is [Jan. 11, 2019 (II)]

Options:

A. 2040.5K

B. 1890.0K

C. 2480.3K

D. 2380.5K

Answer: C

Solution:

$MgO(s) + C(s) \rightarrow Mg(s) + CO(g)$

For a reaction to be spontaneous

$\Delta G < 0$

$\Delta H^{\circ} - T\ \Delta S^{\circ} < 0$

$\Rightarrow T > \frac{\Delta H^{\circ}}{\Delta S^{\circ}}$

$T > \frac{491.1 \times 1000}{198}$

$T > 2480.3K$

Question158

The standard reaction Gibbs energy for a chemical reaction at an absolute temperature T is given by $\Delta G^{\circ} = A - BT$ where A and B are non-zero constants. Which of the following is true about this reaction? [Jan. 11, 2019 (II)]

Options:

A. Exothermic if $B < 0$

B. Endothermic if $A > 0$

C. Endothermic if $A < 0$ and $B > 0$

D. Exothermic if $A > 0$ and $B < 0$

Answer: B

Solution:

$\Delta G^{\circ} = A - BT$
A and B are non-zero constants
$\therefore \Delta G^{\circ} = \Delta H^{\circ} - T \Delta S^{\circ} = A - BT$
$\therefore$ Reaction will be endothermic if $\Delta G^{0} > 0$.
Hence, $A > 0$ and $B < 0$.

Question159

A process has $\Delta H = 200 J\ mol^{-1}$ and $\Delta S = 40 J\ K^{-1} mol^{-1}$ Out of the values given below, choose the minimum temperature above which the process will be spontaneous: [Jan. 10, 2019 (I)]

Options:

A. 20K

B. 12K

C. 5K

D. 4K

Answer: C

Solution:

$\Delta H = 200 J\ mol^{-1}$
$\Delta S = 40 J\ K^{-1} mol^{-1}$
For spontaneous reaction. $\Delta G < 0$
$\Delta H - T \Delta S < 0; \Delta H < T \Delta S$
$\frac{\Delta H}{\Delta S} < T;\ \frac{200}{40} < T$
$5 < T$
So, minimum temperature is $5K$

Question160

An ideal gas is allowed to expand from 1L to 10L against a constant external pressure of 1 bar. The work done in kJ is: [April 12, 2019(I)]

Options:

A. -9.0

B. +10.0

C. −0.9

D. −2.0

Answer: C

Solution:

$$\begin{aligned} W &= -P\Delta V \\ &= -(1\text{ bar}) \times (9L) \\ &= -(10^5 Pa) \times (9 \times 10^{-3}) m^3 \\ &= -9 \times 10^2 N \cdot m \quad [\because 1Pa = 1N/m^2] \\ &= -900J = -0.9kJ \end{aligned}$$

Question161

Among the following, the set of parameters that represents path functions, is:
(A) $q + w$
(B) q
(C) w
(D) $H - TS$
[April 9, 2019 (I)]

Options:

A. (B) and (C)

B. (B), (C) and (D)

C. (A) and (D)

D. (A), (B) and (C)

Answer: A

Solution:

We know that heat and work are not state functions but $q + w = \Delta U$ is a state function. $H - TS$(i.e. G) is also a state function.

Question162

During compression of a spring the work done is 10kJ and 2kJ escaped to the surroundings as heat. The change in internal energy, "U (in kJ) is:
[April 9, 2019 (II)]

Options:

A. −12

B. −8

C. 8

D. 12

Answer: C

Solution:

$w = 10kJ$
$q = -2kJ$
$\Delta U = q + w = -2 + 10 = 8kJ$

Question163

Which one of the following equations does not correctly represent the first law of thermodynamics for the given processes involving an ideal gas ? (Assume non-expansion work is zero)
[April 8, 2019 (I)]

Options:

A. Cyclic process : $q = -w$

B. Adiabatic process : $\Delta U = -w$

C. Isochoric process: $\Delta U = q$

D. Isothermal process: $q = -w$

Answer: B

Solution:

From first law of thermodynamics, $\Delta U = q + w$
For adiabatic process, $q = 0$
$\therefore \Delta U = w$
For isothermal process, $\Delta U = 0 \Rightarrow q = -w$
For cyclic process, $\Delta U = 0 \Rightarrow q = -w$
For isochoric process, $w = 0 \Rightarrow \Delta U = q$

Question164

5 moles of an ideal gas at $100K$ are allowed to undergo reversible compression till its temperature becomes $200\ K$. If $C_V = 28J\ K^{-1}mol^{-1}$, calculate ΔU and ΔpV for this process. ($R = 8.0J\ K^{-1}mol^{-1}$)
[April 8, 2019 (II)]

Options:

A. $\Delta U = 14kJ$; $\Delta(pV) = 18kJ$

B. $\Delta U = 14kJ$; $\Delta(pV) = 0.8kJ$

C. $\Delta U = 14kJ$; $\Delta(pV) = 4kJ$

D. $\Delta U = 14kJ$; $\Delta(pV) = 8.0kJ$

Answer: C

Solution:

$\Delta U = nC_v \Delta T = 5 \times 28 \times 100 = 14kJ$
$\Delta(PV) = nR(T_2 - T_1) = 5 \times 8 \times 100 = 4kJ$

Question165

Enthalpy of sublimation of iodine is 24 cal g^{-1} at $200^{\circ}C$. If specific heat of $I_2(s)$ and I_2 (vap) are 0.055 and 0.031 cal $g^{1}K^{-1}$ respectively, then enthalpy of sublimation of iodine at $250^{\circ}C$ in cal g^{-1} is:
[April 12, 2019(I)]

Options:

A. 2.85

B. 5.7

C. 22.8

D. 11.4

Answer: C

Solution:

$I_2(s) \longrightarrow I_2(g)$
Heat of reaction depend upon temperature i.e., it varies with temperature, as given by Kirchoff's equation,
$\Delta H_{T_2} = \Delta H_{T_1} + \int_{T_1}^{T_2} \Delta C_p dT$
where $\Delta C_p = C_p$ of product $-C_p$ of reactant
$\therefore \quad \Delta C_p = 0.031 - 0.055 = -0.024 cal / g$
Now, $\Delta H_{T_2} - \Delta H_{T_1} = \Delta C_p(T_2 - T_1)$
$\Delta H_{(250)} - \Delta H_{(200)} = -0.024(523 - 473)$
$\Delta H_{(250)} = 24 - 50 \times 0.024 = 22.8 cal / g$

Question166

The difference between ΔH and ΔU ($\Delta H - \Delta U$), when the combustion of one mole of heptane (I) is carried out at a temperature T , is equal to:
[April 10, 2019 (II)]

Options:

A. $-4RT$

B. $-3RT$

C. $4RT$

D. $3RT$

Answer: A

Solution:

$C_7H_{16}(l) + 11O_2(g) \xrightarrow{\Delta} 7CO_2(g) + 8H_2O(l)$
$\Delta H - \Delta U = \Delta n_g RT$
$\Delta n_g =$ no. of moles of product in gaseous state - no. of moles of reactant in gaseous state.
$\because \Delta n_g = -4$
$\therefore \Delta H - \Delta U = -4RT$

Question167

For silver, $C_p(J\,K^{-1}mol^{-1}) = 23 + 0.01T$. If the temperature (T) of 3 moles of silver is raised from 300K to 1000K at 1 atm pressure, the value of ΔH will be close to: [April 8, 2019 (I)]

Answer: 62

Solution:

Given: $n = 3$
$T_1 = 300;\ T_2 = 1000$
$C_p = 23 + 0.01T$
The relation between ΔH and $\mathrm{mathcal}\,C_p$ is
$\Delta H = \int_{T_1}^{T_2} nC_p dT$ . . . (i)
After putting all variable values in eq. (i), we get
$\Delta H = n \int_{300}^{1000} (23 + 0.01T)dT$
$= 3\left[23T + \frac{0.01T^2}{2}\right]_{300}^{1000}$
$= 3\left[23\left(1000 - 300\right[+ \frac{0.01}{2}(1000^2 - 300^2)\right]$
$= 3[16100 + 4550]$
$= 3 \times 20650 = 61950J$
$= 61.95kJ$
$\approx 62kJ$

Question168

A process will be spontaneous at all temperatures if: [April 10, 2019 (I)]

Options:

A. $\Delta H < 0$ and $\Delta S < 0$

B. $\Delta H > 0$ and $\Delta S < 0$

C. $\Delta H < 0$ and $\Delta S > 0$

D. $\Delta H > 0$ and $\Delta S > 0$

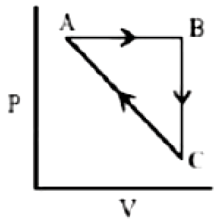
Answer: C

Solution:

A reaction is spontaneous if ΔG_{sys} is negative.
$\Delta G_{sys} = \Delta H_{sys} - T\ \Delta S_{sys}$
A reaction will be spontaneous at all temperatures if
ΔH_{sys} is negative and ΔS_{sys} is positive.

Question169

An ideal gas undergoes a cyclic process as shown in Figure.



$\Delta U_{BC} = -5\text{kJ mol}^{-1}$, $q_{48} = 2\text{kJ mol}^{-1}$
$W_{AB} = -5\text{kJ mol}^{-1}$, $W_{CA} = 3\text{kJ mol}^{-1}$
Heat absorbed by the system during process CA is:
[Online April 15,2018 (I)]

Options:

A. -5kJ mol^{-1}

B. $+5\text{kJ mol}^{-1}$

C. 18kJ mol^{-1}

D. -18kJ mol^{-1}

Answer: B

Solution:

$\Delta U_{AB} = q_{AB} + W_{AB} = 2 + (-5) = -3\text{kJ}/\text{mol}$
$\Delta U_{BC} = -5\text{kJ}/\text{mol}$
For cyclic process, $\Delta U = 0$
$\Delta U_{AB} + \Delta U_{BC} + \Delta U_{CA} = 0$
$\Delta U_{CA} = -\Delta U_{AB} - \Delta U_{BC}$
$\Delta U_{CA} = -(-3) - (-5) = 8\text{kJ}/\text{mol}$
$\Delta U_{CA} = q_{CA} + W_{CA}$
$8 = q_{CA} + 3$
$q_{CA} = +5\text{kJ}/\text{mol}$
Heat absorbed has positive sign.

Question170

The combustion of benzene (I) gives CO_2(g) and H_2O(l). Given that heat of combustion of benzene at constant volume is $-3263.9\text{kJ mol}^{-1}$ at 25°C; heat of combustion (in kJ mol^{-1}) of benzene at constant pressure will be:
$(R = 8.314\text{J K}^{-1}\text{mol}^{-1})$
[2018]

Options:

A. 4152.6

B. −452.46

C. 3260

D. −3267.6

Answer: D

Solution:

$C_6H_6(l) + \frac{15}{2}O_2(g) \longrightarrow {}_6CO_2(g) + 3H_2O(l)$

$\Delta n_g = 6 - \frac{15}{2} = -\frac{3}{2}$ $\Delta H = \Delta U + \Delta n_g RT$

$= -3263.9 + \left(-\frac{3}{2}\right) \times 8.314 \times 10^{-3} \times 298$

$= -3263.9 + (-3.71) = -3267.6 \text{kJ mol}^{-1}$

Question171

For which of the following reactions, ΔH is equal to ΔU ? [Online April 15, 2018 (I)]

Options:

A. $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

B. $2HI(g) \rightarrow H_2(g) + I_2(g)$

C. $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$

D. $2NO_2(g) \rightarrow N_2O_4(g)$

Answer: B

Solution:

$\Delta H = \Delta U + \Delta n_g RT$

$2HI(g) \rightarrow H_2(g) + I_2(g); \Delta n_g = (1+1) - 2 = 0$

$\therefore \Delta H = \Delta U + 0$

Question172

For which of the following processes, ΔS is negative? [Online April 16, 2018]

Options:

A. C (diamond) →C (graphite)

B. N_2(g, latm) → N_2(g, 5atm)

C. $N_2(g, 273K) \rightarrow N_2(g, 300K)$

D. $H_2(g) \rightarrow 2H(g)$

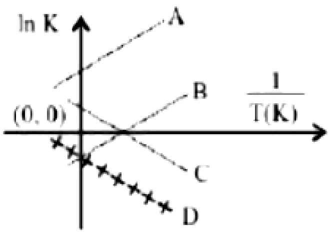
Answer: B

Solution:

When diamond is converted into graphite (it is heated to $1500^{\circ}C$) entropy is increased, $\Delta S > 0$
(b) When pressure increases then molecules of gas will come closer and intermolecular distance decreases, so cntropy will also decrease, $\Delta S < 0$
(c) When we increase the temperature of a gas then randomness is increased as the kinetic energy gained by molecules. So, $\Delta S > 0$
(d) H_2 molecule is converted into atoms, the no. of particles increases. Thus entropy will increase, $\Delta S > 0$

Question173

Which of the following lines correctly show the temperature dependence of equilibrium constant, K , for an exothermic reaction?



[2018]

Options:

A. A and B

B. B and C

C. C and D

D. A and D

Answer: A

Solution:

From thermodynamics relation.
$\Delta G^{\circ} = -RT \ell nK \;\; \Delta H^{\circ} - T \; \Delta S^{\circ} = RT \ell nK$
$-\frac{\Delta H^{\circ}}{RT} + \frac{T \; \Delta S^{\circ}}{RT} = \ln K$
or $\ln K = -\frac{\Delta H^{\circ}}{RT} + \frac{\Delta S^{\circ}}{R}$
For exothermic reaction, $\Delta H^{\circ} = -ve$
slope $= -\frac{\Delta H^{\circ}}{R} = +ve$
So from graph, lines should be A& B.

Question174

At 320K, a gas A_2 is 20% dissociated to A(g). The standard free energy change at 320K and 1atm in J mol^{-1} is approximately:
($R = 8.314 J K^{-1} mol^{-1}$; $\ln 2 = 0.693$; $\ln 3 = 1.098$)
[Online April 16, 2018]

Options:

A. 1844

B. 2068

C. 4281

D. 4763

Answer: D

Solution:

$\Delta G_{rxn}{}^{o} = \Delta_f G^{\circ}(\text{vapour}) - \Delta_f G^{\circ}(\text{liquid})$
$\Delta G_{rxn}{}^{o} = 103 - 100.7 = 2.3 kcal / mol = 2300 cal / mol$
$\Delta G_{rxn}{}^{0} = -RT \ln K_p (K_p = P_{vap})$
$2300 cal / mol = -2 cal / mol / K \times 500K \times \ln K_p$
$\ln K_p = -2.3 \log_{10} K_p = -1$
$K_p =$ Antilog $-1 = 0.1 atm$
∴ Vapour pressure of liquid ' S ' at 500K is approximately equal to 0.1atm.

Question175

$\Delta_f G^{\circ}$ at 500K for substance 'S' in liquid state and gaseous state are $+100.7 kcal\ mol^{-1}$ and $+103 kcal\ mol^{-1}$, respectively.
Vapour pressure of liquid 'S' at 500K is approximately equal to: ($R = 2 cal\ K^{-1} mol^{-1}$).
[Online April 15, 2018(II)]

Options:

A. 100atm

B. 1atm

C. 10atm

D. 0.1atm

Answer: C

Solution:

As we know that, at equilibrium,
$\Delta G^{\circ} = -2.303\ RT\ \log K_{p.....\ (1)}$
$S_{(l)} \rightleftharpoons S_{(g)}$
From the above reaction,
$\Delta G^{\circ} = \Delta G_{f_{(product)}} - \Delta G_{(reactant)}{}^{\circ}$
$\Rightarrow \Delta G^{\circ} = 103 - 100.7 = 2.3 kcal / mol = 2.3 \times 10^3 kcal / mol$
Given:-
$T = 500K$

$R = 2 \text{ cal/mol} - K$

From eq ^{n} (1), we have

$2.3 \times 10^3 = -2.303 \times 2 \times 500 \times \log K_p$

$\Rightarrow \log K_p = -1$

$\Rightarrow K_p = 10^{-1}$ atm

Now, from the above reaction,

$K_p = \dfrac{1}{P_{S_1}}$

$\Rightarrow P_{S_1} = \dfrac{1}{10^{-1}} = 10 \text{ atm}$

Hence the vapou pressure of liquid S is approximately equal to 10 atm .

Question176

Given
(i) $2Fe_2O_3(s) \rightarrow 4Fe(s) + 3O_2(g);$
$\Delta_r G^\circ = +1487.0 \text{kJ mol}^{-1}$
(ii) $2CO(g) + O_2(g) \rightarrow 2CO_2(g);$
$\Delta_r G^\circ = -514.4 \text{kJ mol}^{-1}$
Free energy change, $\Delta_r G^\circ$ for the reaction
$2Fe_2O_3(s) + 6CO(g) \rightarrow 4Fe(s) + 6CO_2(g)$ will be:
[Online April 15, 2018 (II)]

Options:

A. $-112.4 \text{kJ mol}^{-1}$

B. -56.2kJ mol^{-1}

C. $-208.0 \text{kJ mol}^{-1}$

D. $-168.2 \text{kJ mol}^{-1}$

Answer: B

Solution:

(i) $2Fe_2O_3(s) \rightarrow 4Fe(s) + 3O_2(g); \Delta_r G^0 = +1487.0 \text{ kJ mol}^{-1}$

(ii) $2CO(g) + O_2(g) \rightarrow 2CO_2(g); \Delta_r G^0 = -514.4 \text{ kJ mol}^{-1}$

Multiply above reaction with 3

(iii) $6CO(g) + 3O_2(g) \rightarrow 6CO_2(g); \Delta_r G^0 = 3 \times -514.4 = -1543.2 \text{ kJ mol}^{-1}$

When we add reaction (i) and reaction (iii), we get reaction (iv)

(iv) $2Fe_2O_3(s) + 6CO(g) \rightarrow 4Fe(s) + 6CO_2(g)$

Free energy change, $\Delta_r G^0$ for the reaction will be, $1487.0 - 1543.2 = -56.2 \text{ kJ mol}^{-1}$

Question177

ΔU is equal to
[2017]

Options:

A. Isochoric work

B. Isobaric work

C. Adiabatic work

D. Isothermal work

Answer: C

Solution:

From 1^{st} law of thermodynamics
$\Delta U = q + w$
For adiabatic process : $q = 0$
$\therefore \Delta U = w$

Question178

A gas undergoes change from state A to state B. In this process, the heat absorbed and work done by the gas is 5 J and 8J , respectively. Now gas is brought back to A by another process during which 3J of heat is evolved. In this reverse process of B to A: [Online April 9,2017]

Options:

A. 10J of the work will be done by the gas.

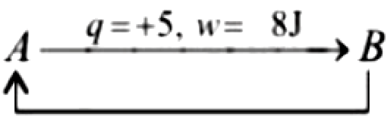
B. 6J of the work will be done by the gas.

C. 10J of the work will be done by the surrounding on gas.

D. 6J of the work will be done by the surrounding on gas.

Answer: D

Solution:



$\Delta U_{AB} = q + w = +5 + (-8) = -3$
$q = -3, \Delta U_{BA} = +3$
$\Delta U_{BA} = q + w$
$\Rightarrow 3 = -3 + w \Rightarrow w = +6$ (work done on the system).

Question179

Given

$\mathbf{C_{(graphite)} + O_2(g) \rightarrow CO_2(g); \Delta_r H^\circ = -393.5 kJ\ mol^{-1}}$
$\mathbf{H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l); \Delta_r H^\circ = -285.8 kJ\ mol^{-1}}$
$\mathbf{CO_2(g) + 2H_2O(l) \rightarrow CH_4(g) + 2O_2(g);}$
$\mathbf{\Delta_r H^\circ = +890.3 kJ\ mol^{-1}}$

Based on the above thermochemical equations, the value of $\Delta_r H^\circ$ at 298K for the reaction $C_{(gaphite)} + 2H_2(g) \rightarrow CH_4(g)$ will be
[2017]

Options:

A. $+74.8 \text{kJ mol}^{-1}$

B. $+144.0 \text{kJ mol}^{-1}$

C. -74.8kJ mol^{-1}

D. $-144.0 \text{kJ mol}^{-1}$

Answer: C

Solution:

$C_{(\text{ graphite})} + O_2(g) \longrightarrow CO_2(g);$
$\Delta H^\circ = -393.5 \text{kJ} / \text{mol}^{-1} \ldots$ (i)
$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(I)$
$\Delta_r H^\circ = -285.8 \text{kJ} / \text{mol}^{-1} \ldots$ (ii)
$CO_2(g) + 2H_2O(I) \longrightarrow CH_4(g) + 2O_2(g);$
$\Delta H_r^\circ = +890.3 \text{kJ} / \text{mol}^{-1} \ldots$ (iii)
$C_{(\text{graphite})} + 2H_2(g) \longrightarrow CH_4(g)$
$\Delta H = ? \ldots$ (iv)
$[$ Eq. (i) $+ Eq.(iii)] + [2 \times Eq.(ii)] = Eq(iv)$
$\therefore [\Delta H_1 + \Delta H_3] + [2 \times \Delta H_2] = \Delta H_4$
$[(-393.5) + (890.3)] + [2(-285.8)]$
$= -74.8 \text{kJ} / \text{mol}$

Question180

For a reaction, $A(g) \rightarrow A(l)$; $\Delta H = -3RT$. The correct statement for the reaction is:
[Online April 8,2017]

Options:

A. $\Delta H = \Delta U \neq O$

B. $\Delta H = \Delta U = O$

C. $|\Delta H| < |\Delta U|$

D. $|\Delta H| > |\Delta U|$

Answer: D

Solution:

$A(g) \longrightarrow A(I)$
$\Delta H = \Delta U + \Delta n_g RT$
Given, $\Delta H = -3RT$
Here
$\Delta n_g = n_p - n_r = 0 - 1 = -1$
$\Delta H = \Delta U + \Delta RT$
$\Rightarrow -3RT = \Delta U - RT$

$\Rightarrow -3RT + RT = \Delta U$
$\Rightarrow -2RT = \Delta U$
$|\Delta H| > |\Delta U|$

Question181

The enthalpy change on freezing of 1mol of water at $5^{\circ}C$ to ice at $-5^{\circ}C$is :
(Given $\Delta_{fis} H = 6kJ\ mol^{-1}$ at $0^{\circ}C$,
$C_p(H_2O, l) = 75.3J\ mol^{-1}K^{-1}$
$C_p(H_2O, s) = 36.8J\ mol^{-1}K^{-1}$)
[Online April 8, 2017]

Options:

A. $5.44kJ\ mol^{-1}$

B. $5.81kJ\ mol^{-1}$

C. $6.56kJ\ mol^{-1}$

D. $6.00kJ\ mol^{-1}$

Answer: B

Solution:

In order to calculate the enthalpy change for H_2O at $5^{\circ}C$ to ice at $-5^{\circ}C$, we need to calculate the enthalpy change of all the transformation involved in the process.
(a) Energy change of $1mol$, $H_2O(l)$, at $5^{\circ}C$
$\rightarrow 1mol, H_2O(I), 0^{\circ}C$
(b) Energy change of $1mol$, $H_2O(l)$, at $0^{\circ}C$
$\rightarrow 1mol, H_2O(s)(ice), 0^{\circ}C$
(c) Energy change of $1mol$, ice (s), at $0^{\circ}C$
$\rightarrow 1mol$, ice (s), $-5^{\circ}C$
Total ΔH
$= C_p[H_2O(l)]\Delta T + \Delta H\ \text{freezing} + C_p[H_2O(s)]\Delta T$
$= (75.3J\ mol^{-1}K^{-1})(0-5)K + (-6\times 10^3 J\ mol^{-1})$
$+(36.8J\ mol^{-1}K^{-1})(-5-0)K$
$\Delta H = -6.56kJ\ mol^{-1}$ (cxothermic process)
So, $\Delta H = 6.56kJ\ mol^{-1}$.

Question182

An ideal gas undergoes isothermal expansion at constant pressure. During the process: [Online April9, 2017]

Options:

A. enthalpy increases but entropy decreases.

B. enthalpy remains constant but entropy increases.

C. enthalpy decreases but entropy increases.

D. Both enthalpy and entropy remain constant.

Answer: B

Solution:

$\Delta H = nC_p \Delta T = 0$ means ΔH constant.

$\Delta S = nR \ln\left(\frac{V_f}{V_i}\right) \geq 0$ ΔS increases

----

Question183

If 100 mole of H_2O_2 decomposes at 1 bar and 300K, the work done (kJ) by one mole of $O_2(g)$ as it expands against 1 bar pressure is :
$2H_2O_2(l) \rightleftharpoons H_2O(l) + O_2(g)$
$(R = 83 J K^{-1} mol^{-1})$
[Online April 10, 2016]

Options:

A. 124.50

B. 249.00

C. 498.00

D. 62.25

Answer: A

Solution:

$2H_2O_2(1) \rightleftharpoons 2H_2O(1) + O_2(g)$
$w = -P_{ext}(\Delta V) = -n_{O_2}RT$
$\because 100 mole H_2O_2$ on decomposition give $50 mole O_2$.
$\therefore W = -(50)(8.3)(300) = -124500J = -124.5kJ$.

Question184

The heats of combustion of carbon and carbon monoxide are −393.5 and −283.5kJ mol^{-1}, respectively. The heat of formation (in kJ) of carbon monoxide per mole is :
[2016]

Options:

A. −676.5

B. −110.5

C. 110.5

D. 676.5

Answer: B

Solution:

Given

$C(s) + O_2(g) \rightarrow CO_2(g); \Delta H = -393.5 kJ\ mol^{-1}$ . . . (i)

$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g); \Delta H = -283.5 kJ\ mol^{-1}$ . . . (ii)

$\therefore$ Heat of formation of $CO = eq^n(i) - eq^n(ii)$

$= -393.5 - (-283.5)$

$= -110 kJ$

Question185

A reaction at 1 bar is non-spontaneous at low temperature but becomes spontaneous at high temperature. Identify the correct statement about the reaction among the following :
[Online April 9, 2016]

Options:

A. ΔH is negative while ΔS is positive

B. Both ΔH and ΔS are negative

C. ΔH is positive while ΔS is negative

D. Both ΔH and ΔS are positive.

Answer: D

Solution:

$\Delta G = \Delta H - T \Delta S$

At low temperature,

$T \Delta S < \Delta H$

$\Delta G > 0$ and the reaction is non spontaneous.

At high temperature,

$T \Delta S > \Delta H$

$\Delta G < 0$ and the reaction is spontaneous.

Question186

The heat of atomization of methane and ethane are 360kJ / mol and 620kJ / mol, respectively. The longest wavelength of light capable of breaking the $C-C$ bond is (Avogadro number $= 6.02 \times 10^{23}$, $h = 6.62 \times 10^{-34} J\ s$):
[Online April 10,2015]

Options:

A. $2.48 \times 10^4 nm$

B. $1.49 \times 10^3 nm$

C. $2.48 \times 10^3 nm$

D. $1.49 \times 10^4 nm$

Answer: B

Solution:

$In\ CH_4,\ 4 \times BE_{(C-H)} = 360kJ/mol$
$\therefore BE_{(C-H)} = 90kJ/mol$
In C_2H_6, $BE_{(C-C)} + 6 \times BE_{(C-H)} = 620kJ/mol$
$\therefore BE_{(C-C)} = 80kJ/mol$
$\therefore BE_{(C-C)} = \dfrac{80 \times 10^3}{6.023 \times 10^{23} J/mol}$
Now, $E = \dfrac{hc}{\lambda}$
$\therefore \lambda = \dfrac{6.626 \times 10^{-34} \times 3 \times 10^8 \times 6.023 \times 10^{23}}{80 \times 10^3}$
$\lambda = 1.49 \times 10^{-6} m \ (\because 1nm = 10^{-9} m)$
$\therefore \lambda = 1.49 \times 10^3 nm$

Question187

For complete combustion of ethanol,
$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$, the amount of heat produced as measured in bomb calorimeter, is $1364.47 kJ\ mol^{-1}$ at $25^\circ C$. Assuming ideality the enthalpy of combustion, $\Delta_c H$, for the reaction will be:
$(R = 8.314 kJ\ mol^{-1})$
[2014]

Options:

A. $-1366.95 kJ\ mol^{-1}$

B. $-1361.95 kJ\ mol^{-1}$

C. $-1460.95 kJ\ mol^{-1}$

D. $-1350.50 kJ\ mol^{-1}$

Answer: A

Solution:

$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$
Bomb calorimeter gives ΔU of the reaction
Given, $\Delta U = -1364.47 kJ\ mol^{-1}$
$\Delta n_g = -1$
$\Delta H = \Delta U + \Delta n_g RT$
$= -1364.47 - \dfrac{I \times 8.314 \times 298}{1000}$
$= -1366.95 kJ\ mol^{-1}$

Question188

The standard enthalpy of formation of NH_3 is $-46.0 kJ/mol$. If the enthalpy of formation of H_2 from its atoms is $436 kJ/mol$ and that of N_2 is $-712 kJ/mol$, the average bond enthalpy

of $N-H$ bond in NH_3 is:
[Online April 9, 2014]

Options:

A. $-1102 kJ/mol$

B. $-964 kJ/mol$

C. $+352 kJ/mol$

D. $+1056 kJ/mol$

Answer: C

Solution:

Given
$\frac{1}{2}N_2+\frac{3}{2}H_2 \rightleftharpoons NH_3$
$\Delta_f H=-46.0 kJ/mol$
$H+H \rightleftharpoons H_2; \Delta_f H=-436 kJ/mol$
$N+N \rightleftharpoons N_2; \Delta_f H=-712 kJ/mol$
$\Delta_f H(NH_3)=\frac{1}{2}\Delta H_{N-N}+\frac{3}{2}\Delta H_{H-H}-3\Delta H_{N-H}$
$-46=\frac{1}{2}(712)+\frac{3}{2}(436)-3\Delta H_{N-H}$
$-3\Delta H_{N-H}=-1056$
$\Delta H_{N-H}=352 kJ/mol$

Question189

The standard enthalpy of formation ($\Delta_f H_{298}^{\circ}$) for methane, CH_4 is $-74.9 kJ\ mol^{-1}$. In order to calculate the average energy given out in the formation of a $C-H$ bond from this it is necessary to know which one of the following?
[Online April 12, 2014]

Options:

A. The dissociation energy of the hydrogen molecule, H_2.

B. The first four ionisation energies of carbon.

C. The dissociation energy of H_2 and enthalpy of sublimation of carbon (graphite).

D. The first four ionisation energies of carbon and electron affinity of hydrogen.

Answer: C

Solution:

To calculate average enthalpy of $C-H$ bond in methane following informations are needed
(i) dissociation energy of H_2 i.e.
$\frac{1}{2}H_2(g) \longrightarrow H(g); \Delta H=x(\text{ suppose })$
(ii) Sublimation energy of C(graphite) to $C(g)$
$C_{(graphite)} \longrightarrow C(g); \Delta H=y(\text{ Suppose })$
Given

$C_{(graphite)} + 2H_2(g) \longrightarrow CH_4(g)$
$\Delta H\,g = -74.9\,kJ\,mol^{-1}$

Question190

The molar heat capacity (C_p) of CD_2O is 10 cals at 1000K. The change in entropy associated with cooling of 32g of CD_2O vapour from 1000K to 100K at constant pressure will be: (D = deuterium, atomic mass = 2u)
[Online April 11, 2014]

Options:

A. $23.03\ cal\ deg^{-1}$

B. $-23.03\ cal\ deg^{-1}$

C. $2.303\ cal\ deg^{-1}$

D. $-2.303\ cal\ deg^{-1}$

Answer: B

Solution:

(b) Given, $C_p = 10cal$ at $1000K$
$T_1 = 1000K$. $T_2 = 100K$
$m = 32g$
$\Delta S = ?$
at constant pressure

$$\Delta S = C_p \ln \frac{T_2}{T_1}$$
$$= 2.303 \times C_p \log \frac{T_2}{T_1}$$
$$= 2.303 \times 10 \log \frac{100}{1000}$$
$$= -23.03 cal\ deg^{-1}$$

Question191

The (S°) of the following substances are:
$CH_4(g) 186.2 J\,K^{-1} mol^{-1}$ $O_2(g) 205.2 J\,K^{-1} mol^{-1}$
$CO_2(g) 213.6 J\,K^{-1} mol^{-1}$
$H_2O(l) 69.9 J\,K^{-1} mol^{-1}$
The entropy change (ΔS°) for the reaction
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ is:
[Online April 12, 2014]

Options:

A. $-312.5 J\,K^{-1} mol^{-1}$

B. $-242.8 J\,K^{-1} mol^{-1}$

C. $-108.1 \mathrm{J\,K^{-1} mol^{-1}}$

D. $-37.6 \mathrm{J\,K^{-1} mol^{-1}}$

Answer: B

Solution:

$$\begin{aligned}\Delta S^\circ &= S^\circ CO_2 + 2 \times S^\circ H_2O - \left(S^\circ CH_4 + 2 \times S_{O_2}^\circ\right) \\ &= (213.6 + 2 \times 69.9) - (186.2 + 2 \times 205.2) \\ &= -242.8 \mathrm{J\,K^{-1} mol^{-1}}\end{aligned}$$

Question192

A piston filled with 0.04 mol of an ideal gas expands reversibly from 50.0mL to 375mL at a constant temperature of 37.0°C. As it does so, it absorbs 208J of heat. The values of q and w for the process will be:
$(R = 8.314 \mathrm{J / mol\,K})(\ln 7.5 = 2.01)$
[2013]

Options:

A. $q = +208\mathrm{J}$, $w = -208\mathrm{J}$

B. $q = -208\mathrm{J}$, $w = -208\mathrm{J}$

C. $q = -208\mathrm{J}$, $w = +208\mathrm{J}$

D. $q = +208\mathrm{J}$, $w = +208\mathrm{J}$

Answer: A

Solution:

Process is isothermal reversible expansion, hence
$\Delta U = 0$, therefore $q = -w$
Since $q = +208\mathrm{J}$, $w = -208\mathrm{J}$

Question193

Which of the following statements\/relationships is not correct in thermodynamic changes? [Online April 23,2013]

Options:

A. $\Delta U = 0$ (isothermal reversible expansion of a gas)

B. $w = -nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

C. $w = nRT \ln \frac{V_2}{V_1}$ (isothermal reversible expansion of an ideal gas)

D. For a system of constant volume heat involved directly changes to internal energy.

Answer: C

Solution:

For isothermal reversible expansion.

$w = -nRT \ln \frac{V_2}{V_1}$

Question194

Given:

(I) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$

$\Delta H_{298K}^{\circ} = -285.9 kJ\ mol^{-1}$

(II) $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g);$

$\Delta H_{298K}^{\circ} = -241.8 kJ\ mol^{-1}$

The molar enthalpy of vapourisation of water will be:
[Online April 9, 2013]

Options:

A. $241.8 kJ\ mol^{-1}$

B. $22.0 kJ\ mol^{-1}$

C. $44.1 kJ\ mol^{-1}$

D. $527.7 kJ\ mol^{-1}$

Answer: C

Solution:

Given

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$

$\Delta H^{\circ} = -285.9 kJ\ mol^{-1}$ . . . (i)

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(g)$

$\Delta H^{\circ} = -241.8 kJ\ mol^{-1}$ . . . (ii)

We have to calculate

$H_2O(l) \longrightarrow H_2O(g); \Delta H^{\circ} = ?$

On substracting eqn. (ii) from eqn. (i) we get

$H_2O(l) \longrightarrow H_2O(g); \Delta H^{\circ} = -241.8 - (-285.9)$

$= 44.1 kJ\ mol^{-1}$

Question195

Reaction	Energy Change (in kJ)
$Li(s) \rightarrow Li(g)$	161
$Li(g) \rightarrow Li^{+}(g)$	520
$\frac{1}{2}F_2(g) \rightarrow F(g)$	77
$F(g) + e^{-} \rightarrow F^{-}(g)$	(Electron gain enthalpy)
$Li^{+}(g) + F(g) \rightarrow LiF(s)$	-1047
$Li(s) + \frac{1}{2}F_2(g) \rightarrow LiF(s)$	-617

Based on data provided, the value of electron gain enthalpy of fluorine would be : [Online April 22,2013]

Options:

A. $-300 \mathrm{kJ\,mol}^{-1}$

B. $-350 \mathrm{kJ\,mol}^{-1}$

C. $-328 \mathrm{kJ\,mol}^{-1}$

D. $-228 \mathrm{kJ\,mol}^{-1}$

Answer: C

Solution:

Applying Hess's Law

$\Delta_f H^\circ = \Delta_{sub} H + \frac{1}{2}\Delta_{diss} H + \text{I.E.} + \text{E.A.} + \Delta_{lattice} H$

$-617 = 161 + 520 + 77 + \text{E.A.} + (-1047)$

E.A. $= -617 + 289 = -328 \mathrm{kJ\,mol}^{-1}$

$\therefore$ electron affinity of fluorine $= -328 \mathrm{kJ\,mol}^{-1}$

Question196

In which of the following exothermic reactions, the heat liberated per mole is the highest? [Online April 25, 2013]

Options:

A. $CaO + H_2O \rightarrow Ca(OH)_2$

B. $SrO + H_2O \rightarrow Sr(OH)_2$

C. $BaO + H_2O \rightarrow Ba(OH)_2$

D. $MgO + H_2O \rightarrow Mg(OH)_2$

Answer: A

Question197

Given that:
(i) $\Delta_f H^\circ$ of N_2O is 82kJ mol^{-1}
(ii) Bond energies of $N \equiv N$, $N = N$, $O = O$ and $N = O$ are 946, 418, 498 and 607kJ mol^{-1} respectively,
The resonance energy of N_2O is:
[Online April 25, 2013]

Options:

A. $-88kJ$

B. $-66kJ$

C. $-62kJ$

D. $-44kJ$

Answer: A

Solution:

$N_2(g) + \frac{1}{2}O_2 \rightarrow N_2O(g)$

$N \equiv N(g) + \frac{1}{2}(O = O) \rightarrow \overset{+}{N} = N = O(g)$

$\Delta_f H^o = [$
Energy required for breaking of bonds]
$-[$ Energy released for forming of bonds]

$= \left(\Delta H_{N \equiv N} + \frac{1}{2}\Delta H_{O=O} - (\Delta H_{N=N} + \Delta H_{N=O}\right)$

$= \left(946 + \frac{1}{2} \times 498\right) - (418 + 607) = 170kJ\ mol^{-1}$

Resonance energy = observed $\Delta_f H^\circ$ − calculated $\Delta_f H^\circ$

$82 - 170 = -88kJ\ mol^{-1}$

Question198

The difference between the reaction enthalpy change (ΔH) and reaction internal energy change ($\Delta_f U$) for the reaction:
$2C_6H_6(I) + 15O_2(g) \longrightarrow 12CO_2(g) + 6H_2O(I)$
at 300K is ($R = 8.314 J\ mol^{-1} K^{-1}$)
[Online May 12, 2012]

Options:

A. $0 J\ mol^{-1}$

B. $2490 J\ mol^{-1}$

C. $-2490 J\ mol^{-1}$

D. $-7482 J\ mol^{-1}$

Answer: D

Solution:

$\Delta H = \Delta U + \Delta n_g RT$
For the reaction $\Delta n_g = 12 - 15 = -3$
$\Delta H - \Delta U = -3 \times 8.314 \times 300$
$= -7482.6 J\ mol^{-1}$

Question199

The enthalpy of neutralisation of NH_4OH with HCl is $-51.46 kJ\ mol^{-1}$ and the enthalpy of neutralisation of $NaOH$ with HCl is $-55.90 kJ\ mol^{-1}$. The enthalpy of ionisation of NH_4OH is
[Online May 19, 2012]

Options:

A. $-107.36 kJ\ mol^{-1}$

B. $-4.44 kJ\ mol^{-1}$

C. $+107.36 kJ\ mol^{-1}$

D. $+4.44 kJ\ mol^{-1}$

Answer: D

Solution:

$\underset{\text{Strong acid}}{HCl} \longrightarrow \underset{\text{(Complete ionisation)}}{H^+ + Cl^-}$ ...(i)
$\underset{\text{Weak base}}{NH_4OH} \rightleftharpoons NH_4^+ + OH^-$ . . . (ii)
$\Delta H = x kJ\ mol^{-1}$
$H^+ + OH^- \longrightarrow H_2O$ . . . (iii)
$\Delta H = -55.90 kJ\ mol^{-1}$

(from neutralisation of strong acid and strong base)
From equation (i), (ii) and (iii)
$NH_4OH + HCl \longrightarrow NH_4^+ + Cl^- + H_2O$
$\Delta H = -51.46 kJ\ mol^{-1}$
$\therefore\ x + (-55.90) = -51.46$
$x = -51.46 + 55.90$
$= 4.44 kJ\ mol^{-1}$
∴ Enthalpy of ionisation of
$NH_4OH = 4.44 kJ\ mol^{-1}$

Question200

The entropy of a sample of a certain substance increases by $0.836 J\ K^{-1}$ on adding reversibly $0.3344 J$ of heat at constant temperature. The temperature of the sample is: [Online May 7, 2012]

Options:

A. 2.5K

B. 0.3K

C. 0.016K

D. 0.4K

Answer: D

Solution:

$\Delta S = \frac{\Delta H}{T}; T = \frac{\Delta H}{\Delta S} = \frac{0.3344}{0.836} = 0.4 K$

Question201

One mole of an ideal gas is expanded isothermally and reversibly to half of its initial pressure. ΔS for the process in $J\ K^{-1} mol^{-1}$ is $[\ln 2 = 0.693$ and $R = 8.314, J/(mol\ K)]$ [Online May 26, 2012]

Options:

A. 6.76

B. 5.76

C. 10.76

D. 8.03

Answer: B

Solution:

For isothermal process
$(\Delta T = 0)$

$$\Delta S = R \ln \frac{P_1}{P_2} = 8.314 \ln 2$$
$$= 8.314 \times 0.693 = 5.76$$

Question202

The incorrect expression among the following is:
[2012]

Options:

A. $\frac{\Delta G_{system}}{\Delta S_{total}} = -T$

B. In isothermal process, $w_{reversible} = -nRT \ln \frac{V_f}{V_i}$

C. $\ln K = \frac{\Delta H^\circ - T \Delta S^\circ}{RT}$

D. $K = e^{-\Delta G^\prime / RT}$

Answer: C

Solution:

$$\Delta G^\circ = \Delta H^\circ - T \Delta S^\circ$$
$$-RT \ln K = \Delta H^\circ - T \Delta S^\circ$$
$$\ln K = -\frac{\Delta H^\circ - T \Delta S^\circ}{RT}$$

Question203

The value of enthalpy change (ΔH) for the reaction $C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$ at $27^\circ C$ is $-1366.5 kJ\ mol^{-1}$. The value of internal energy change for the above reaction at this temperature will be :
[2011 RS]

Options:

A. $-1369.0 kJ$

B. $-1364.0 kJ$

C. $-1361.5 kJ$

D. $-1371.5 kJ$

Answer: B

Solution:

$$C_2H_5OH(l) + 3O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$$
$$\Delta n_g = 2 - 3 = -1$$
$$\Delta U = \Delta H - \Delta n_g RT$$
$$= -1366.5 - [(-1) \times 8.314 \times 300]$$

$$= -1366.5 + \left[(1) \times \frac{8.314}{10^3} \times 300 \right]$$
$$= -1366.5 + 0.8314 \times 3 = -1364 \text{kJ}$$

Question204

Consider the reaction:
$4NO_2(g) + O_2(g) \rightarrow 2N_2O_5(g)$ $\Delta H = -111\text{kJ}$ If $N_2O_5(s)$ is formed instead of $N_2O_5(g)$ in the above reaction, the $\Delta_r H$ value will be :
(given, ΔH of sublimation for N_2O_5 is 54kJ mol^{-1})
[2011 RS]

Options:

A. $+54\text{kJ}$

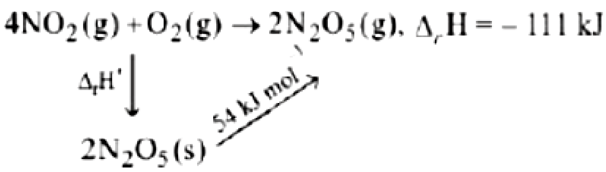
B. $+219\text{kJ}$

C. -219J

D. -165kJ

Answer: D

Solution:



$$\Delta_r H' = \Delta_r H - 54$$
$$= -111 - 54$$
$$= -165 \text{ kJ}$$

Question205

The standard enthalpy of formation of NH_3 is -46.0kJ mol^{-1}. If the enthalpy of formation of H_2 from its atoms is -436kJ mol^{-1} and that of N_2 is -712kJ mol^{-1}, the average bond enthalpy of $N-H$ bond in NH_3 is
[2010]

Options:

A. -964kJ mol^{-1}

B. $+352\text{kJ mol}^{-1}$

C. $+1056\text{kJ mol}^{-1}$

D. -1102kJ mol^{-1}

Answer: B

Solution:

(b) $N_2 + 3H_2 \longrightarrow 2NH_3 \quad \Delta H = 2 \times -46.0 kJ\ mol^{-1}$

Let x be the bond enthalpy of $N-H$ bond then [Note: Enthalpy of formation or bond formation enthalpy is given which is negative but the given reaction involves bond breaking hence values should be taken as positive.] $\Delta H = \Sigma$ Bond energies of reactants $-\Sigma$ Bond energies of products

$2 \times -46 \quad = 712 + 3 \times (436) - 6x - 92 = 2020 - 6x$

$6x \quad = 2020 + 92$

$6x \quad = 2112$

$x \quad = +352 kJ / mol$

Question206

For a particular reversible reaction at temperature T, ΔH and ΔS were found to be both +ve. If T_e is the temperature at equilibrium, the reaction would be spontancous when
[2010]

Options:

A. $T_c > T$

B. $T > T_c$

C. T_e is 5 times T

D. $T = T_e$

Answer: B

Solution:

At equilibrium $\Delta G = 0$

Hence, $\Delta G = \Delta H - T \Delta S = 0$

$\therefore \Delta H = T_e \Delta S$ or $T_e = \dfrac{\Delta H}{\Delta S}$

For a spontaneous reaction

ΔG must be negative which is possible only if $\Delta H < T \Delta S$

or $T > \dfrac{\Delta H}{\Delta S}; T_e < T$

Question207

On the basis of the following thermochemical data :
$(\Delta_f G^\circ H^+_{(aq)} = 0)$

$H_2O(l) \rightarrow H^+(aq) + OH^-(aq); \Delta H = 57.32 kJ$

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l); \Delta H = -286.20 kJ$

The value of enthalpy of formation of OH^- ion at 25°C is:
[2009]

Options:

A. $-228.88 kJ$

B. +228.88kJ

C. −343.52kJ

D. −22.88kJ

Answer: A

Solution:

$H_2O(l) \longrightarrow H^+(aq.) + OH^-(aq.)$
$\Delta H = 57.32kJ$ . . . (i)
$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$
$\Delta H = -286.20kJ$ . . . (ii)
By adding equation (i) & (ii) we get,
$H_2(g) + \frac{1}{2}(O_2) \longrightarrow H^+(aq) + OH^-(aq)$
$\Delta H = 57.32 - 286.2 = -228.88kJ$
Here $\Delta_f H^\circ$ of $H^+(aq.) = 0$
$\therefore \Delta_f H^\circ$ of $OH^- = -228.88kJ$

Question208

Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below:

$$\frac{1}{2}Cl_2(g) \xrightarrow{\frac{1}{2}\Delta_{diss}H^\circ} Cl(g) \xrightarrow{\Delta_{eg}H^\circ} Cl^-(g) \xrightarrow{\Delta_{hyd}H^\circ} Cl^-(aq)$$

The energy envolved in the conversion of
$\frac{1}{2}Cl_2(g)$toCl^{-1}(aq)
(using the data,
$\Delta_{diss} H_{Cl_2}^\circ = 240kJ\,mol^{-1}$, $\Delta_{eg} H_{Cl}^o = -349kJ\,mol^{-1}$
$\Delta_{hyd} H_{Cl}^\circ = -381kJ\,mol^{-1}$), will be
[2008]

Options:

A. $+152kJ\,mol^{-1}$

B. $-610kJ\,mol^{-1}$

C. $-850kJ\,mol^{-1}$

D. $+120kJ\,mol^{-1}$

Answer: B

Solution:

The energy involved in the conversion of
$\frac{1}{2}Cl_2(g)$ to $Cl^-(aq)$ is given by
$\Delta H = \frac{1}{2}\Delta_{diss}H_{Cl_2}{}^O + \Delta_{eg}H_{Cl}{}^\circ + \Delta_{hyd}H_{Cl}{}^\circ$
Substituting various values from given data, we get

$$\Delta H = \left(\frac{1}{2} \times 240\right) + (-349) + (-381)$$
$$= (120 - 349 - 381) = -610 \text{kJ mol}^{-1}$$

Question209

Standard entropy of X_2, Y_2 and XY_3 are 60,40 and 50J $K^{-1}mol^{-1}$, respectively. For the reaction,
$\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3$, $\Delta H = -30kJ$, to be at equilibrium, the temperature will be
[2008]

Options:

A. 1250K

B. 500K

C. 750K

D. 1000K

Answer: C

Solution:

For a reaction to be at equilibrium $\Delta G^{\circ} = 0$. Since $\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$ so at equilibrium $\Delta H^{\circ} - T\Delta S^{\circ} = 0$
or $\Delta H^{\circ} = T\Delta S^{\circ}$
For the reaction
$\frac{1}{2}X_2 + \frac{3}{2}Y_2 \rightarrow XY_3; \Delta H^{\circ} = -30kJ$
Calculating ΔS° for the above reaction, we get
$$\Delta S^{\circ} = 50 - \left[\frac{1}{2} \times 60 + \frac{3}{2} \times 40\right]$$
$$= 50 - (30 + 60) = -40 J K^{-1}$$
At equilibrium, $T\Delta S^{\circ} = \Delta H^{\circ}$ $[\because \Delta G^{\circ} = 0]$
$\therefore T \times (-40 J K^{-1}) = -30 \times 1000 J$ $[\because 1kJ = 1000J]$
or $T = \frac{-30 \times 1000}{-40} = 750K$

Question210

Assuming that water vapour is an ideal gas, the internal energy change (ΔU) when 1mol of water is vapourised at 1 bar pressure and $100^{\circ}C$, (given : molar enthalpy of vapourisation of water at 1 bar and 373K = 41kJ mol^{-1} and R = 8.3J $mol^{-1}K^{-1}$) will be
[2007]

Options:

A. 41.00kJ mol^{-1}

B. 4.100kJ mol^{-1}

C. 3.7904kJ mol^{-1}

D. 37.904kJ mol^{-1}

Answer: D

Solution:

Given $\Delta H = 41 kJ\ mol^{-1} = 41000 J\ mol^{-1}$
$T = 100^{\circ}C = 273 + 100 = 373K$
$H_2O(I) \xrightarrow{\Delta} H_2O(g)$
$\Delta n_g = 1 - 0 = 1$
$\Delta U = \Delta H - \Delta nRT = 41000 - (1 \times 8.314 \times 373)$
$= 37898.88 J\ mol^{-1} \sim eq 37.9 kJ\ mol^{-1}$

Question211

In conversion of lime-stone to lime, $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$ the values of ΔH° and ΔS° are $+179.1 kJ\ mol^{-1}$ and $160.2 J / K$ respectively at $298K$ and 1 bar. Assuming that ΔH° and ΔS° do not change with temperature, temperature above which conversion of limestone to lime will be spontaneous is [2007]

Options:

A. 1118K

B. 1008K

C. 1200K

D. 845K .

Answer: A

Solution:

$\Delta G^{\circ} = \Delta H^{\circ} - T \Delta S^{\circ}$
For a spontaneous reaction $\Delta G^{\circ} < 0$
or $\Delta H^{\circ} - T \Delta S^{\circ} < 0 \Rightarrow T > \dfrac{\Delta H^{\circ}}{\Delta S^{\circ}}$
$\Rightarrow T > \dfrac{179.3 \times 10^3}{160.2}$
$> 1117.9K \approx 1118K$

Question212

Identify the correct statement regarding a spontaneous process: [2007]

Options:

A. Lowering of energy in the process is the only criterion for spontaneity.

B. For a spontaneous process in an isolated system, the change in entropy is positive.

C. Endothermic processes are never spontaneous.

D. Exothermic processes are always spontaneous.

Answer: B

Solution:

Spontaneity of reaction depends on tendency to acquire minimum energy state and maximum randomness. For a spontaneous process in an isolated system the change in entropy is positive.

Question213

An ideal gas is allowed to expand both reversibly and irreversibly in an isolated system. If T_i is the initial temperature and T_f is the final temperature, which of the following statements is correct?
[2006]

Options:

A. $(T_f)_{rev} = (T_f)_{irrev}$

B. $T_f = T_i$ for both reversible and irreversible processes

C. $(T_f)_{imcv} > (T_f)_{rev}$

D. $T_f > T_i$ for reversible process but $T_f = T_i$ for irreversible process

Answer: C

Solution:

According to first law of thermodynamics $\Delta Q = \Delta U + \Delta W$
An isolated system is adiabatic. This means $\Delta Q = 0$. The first law in this case yields
$0 = \Delta U + \Delta W \Rightarrow \Delta W = -\Delta U$ . . . (i)
For expansion, ΔW is positive and hence ΔU is negative.
This means T_f is less than T_i in both the cases.
For the same expansion of volume, the work done in irreversible process is greater than that in reversible one because the system has to work against friction etc. Thus
$\Delta W_{irreversible} > \Delta W_{reversible}$
$\Rightarrow -\Delta U_{irreversible} > -\Delta U_{reversible}$ [from equation (i)]
$\Rightarrow \Delta U_{irreversible} < \Delta U_{reversible}$
$\Rightarrow \Delta T_{irreversible} < \Delta T_{reversible}$
$\Rightarrow T_{f\ irreversible} > T_{f\ reversible}$

Question214

The standard enthalpy of formation $(\Delta_f H^{\circ})$ at 298K for methane, $CH_4(g)$ is $-74.8 kJ\ mol^{-1}$. The additional information required to determine the average energy for $C-H$ bond formation would be
[2006]

Options:

A. the first four ionization energies of carbon and electron gain enthalpy of hydrogen

B. the dissociation energy of hydrogen molecule, H_2

C. the dissociation energy of H_2 and enthalpy of sublimation of carbon

D. latent heat of vapourization of methane

Answer: C

Solution:

The standard enthalpy of formation of CH_4 is given by the equation:
$C(s) + 2H_2(g) \longrightarrow CH_4(g)$
Hence, dissociation energy of hydrogen and enthalpy of sublimation of carbon is required.

Question215

The enthalpy changes for the following processes are listed below:
$Cl_2(g) = 2Cl(g),\ \ 242.3 kJ\ mol^{-1}$
$I_2(g) = 2I(g),\ \ 151.0 kJ\ mol^{-1}$
$ICl(g) = I(g) + Cl(g),\ \ 211.3 kJ\ mol^{-1}$
$I_2(s) = I_2(g),\ \ 62.76 kJ\ mol^{-1}$
Given that the standard states for iodine and chlorine are $I_2(s)$ and $Cl_2(g)$, the standard enthalpy of formation for $ICl(g)$ is :
[2006]

Options:

A. $+16.8 kJ\ mol^{-1}$

B. $+244.8 kJ\ mol^{-1}$

C. $-14.6 kJ\ mol^{-1}$

D. $-16.8 kJ\ mol^{-1}$

Answer: A

Solution:

$I_2(s) + Cl_2(g) \longrightarrow 2ICl(g)$
$\Delta H = [\Delta H_{12(s) \rightarrow 12(g)} + \Delta H_{1-1} + \Delta H_{Cl-Cl}] - 2[\Delta H_{1-Cl}]$
$= 62.76 + 151.0 + 242.3 - 2 \times 211.3 = 33.46$
$\Delta H_f^o(ICl) = \dfrac{33.46}{2} = 16.73 kJ/mol$

Question216

$(\Delta H - \Delta U)$ for the formation of carbon monoxide (CO) from its elements at 298K is $(R = 8.314 J\ K^{-1} mol^{-1})$
[2006]

Options:

A. $-2477.57 J\ mol^{-1}$

B. $2477.57 \mathrm{J\ mol}^{-1}$

C. $-1238.78 \mathrm{J\ mol}^{-1}$

D. $1238.78 \mathrm{J\ mol}^{-1}$

Answer: D

Solution:

For the reaction,
$C(s)+\frac{1}{2}O_2(g) \longrightarrow CO(g)$
$\Delta H=\Delta U+\Delta n_g RT \quad \text{or} \quad \Delta H-\Delta U=\Delta n_g RT$
$\Delta n_g=1-\frac{1}{2}=\frac{1}{2};$
$\Delta H-\Delta U=\frac{1}{2}\times 8.314\times 298$
$=1238.78 \mathrm{J\ mol}^{-1}$

Question217

Consider the reaction : $N_2+3H_2 \rightarrow 2NH_3$ carried out at constant temperature and pressure. If ΔH and ΔU are the enthalpy and internal energy changes for the reaction, which of the following expressions is true? [2005]

Options:

A. $\Delta H>\Delta U$

B. $\Delta H<\Delta U$

C. $\Delta H=\Delta U$

D. $\Delta H=0$

Answer: B

Solution:

$\Delta H=\Delta U+\Delta n_g RT$ for
$N_2+3H_2 \longrightarrow 2NH_3$
$\Delta n_g=2-4=-2$
$\therefore \Delta H>\Delta U-2RT$
or $\Delta U=\Delta H+2RT \quad \therefore \Delta U>\Delta H$

Question218

If the bond dissociation energies of XY, X_2 and Y_2 (all diatomic molecules) are in the ratio of 1 : 1 : 0.5 and ΔH_f for the formation of XY is $-200 \mathrm{kJ\ mole}^{-1}$. The bond dissociation energy of X_2 will be [2005]

Options:

A. $400 kJ\ mol^{-1}$

B. $300 kJ\ mol^{-1}$

C. $200 kJ\ mol^{-1}$

D. $100 kJ\ mol^{-1}$

E. None of above

Answer: E

Solution:

(N) $\frac{1}{2}X_2 + \frac{1}{2}Y_2 \longrightarrow XY, \Delta H = (-200)J$

Let x be the bond dissociation energy of X_2. Then

$\Delta H_f = -200 = \frac{1}{2}\Delta H_{x-x} + \frac{1}{2}\Delta H_{y-y} - \Delta H_{x-y}$

$-400 = x + 0.5x - 2x = -0.5x$

or $x = \frac{400}{0.5} = 800 kJ\ mol^{-1}$

(In the question paper, this option was not mentioned. So the answer has been marked 'N')

Question219

An ideal gas expands in volume from 1×10^{-3} to $1 \times 10^{-2} m^3$ at 300K against a constant pressure of $1 \times 10^5 N\ m^{-2}$. The work done is [2004]

Options:

A. 270kJ

B. −900kJ

C. −900J

D. 900kJ

Answer: C

Solution:

$w = -P\Delta V = -10^5[(1 \times 10^{-2}) - (1 \times 10^{-3})]$
$= -900J$

Question220

The enthalpies of combustion of carbon and carbon monoxide are −393.5 and $-283 kJ\ mol^{-1}$ respectively. The enthalpy of formation of carbon monoxide per mole is [2004]

Options:

A. −676.5kJ

B. 676.5kJ

C. 110.5kJ

D. −110.5kJ

Answer: D

Solution:

(i) $C + O_2 \rightleftharpoons CO_2, \Delta H = -393.5 kJ\ mol^{-1}$
(ii) $CO + \frac{1}{2}O_2 \rightleftharpoons CO_2, \Delta H = -283.0 kJ\ mol^{-1}$
Operating (i) - (ii), we have
$C + \frac{1}{2}O_2 \rightarrow CO\ \Delta H = -110.5 kJ\ mol^{-1}$

Question221

The internal energy change when a system goes from state A to B is 40kJ/mole. If the system goes from A to B by a reversible path and returns to state A by an irreversible path what would be the net change in internal energy?
[2003]

Options:

A. >40kJ

B. <40kJ

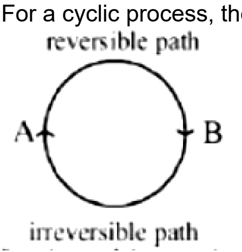
C. Zero

D. 40kJ

Answer: C

Solution:

For a cyclic process, the net change in the internal energy is zero because the change in internal energy does not depend on the path.



Question222

If at 298K the bond energies of $C-H$, $C-C$, $C=C$ and $H-H$ bonds are respectively 414, 347, 615 and 435kJ mol^{-1}, the value of enthalpy change for the reaction
$H_2C=CH_2(g) + H_2(g) \rightarrow H_3C-CH_3(g)$ at 298K will be
[2003]

Options:

A. −250kJ

B. +125kJ

C. −125kJ

D. +250kJ

Answer: C

Solution:

$CH_2 = CH_2(g) + H_2(g) \rightarrow CH_3 - CH_3$
Enthalpy change = Bond energy of reactants − Bond energy of products.
$\Delta H \;\; = 1(C = C) + 4(C - H) + 1(H - H) - 1(C - C) - 6(C - H)$
$= 1(C = C) + 1(H - H) - 1(C - C) - 2(C - H)$
$= 615 + 435 - 347 - 2 \times 414 = 1050 - 1175$
$= -125kJ$.

Question223

The enthalpy change for a reaction does not depend upon [2003]

Options:

A. use of different reactants for the same product

B. the nature of intermediate reaction steps

C. the differences in initial or final temperatures of involved substances

D. the physical states of reactants and products

Answer: B

Solution:

Enthalpy change for a reaction does not depend upon the nature of intermediate reaction steps.

Question224

In an irreversible process taking place at constant T and P and in which only pressure-volume work is being done, the change in Gibbs free energy (d G) and change in entropy (dS), satisfy the criteria [2003]

Options:

A. $(d\,S)_{V,\,E} > 0, (d\,G)_{T,\,P} < 0$

B. $(d\,S)_{V,\,E} = 0, (d\,G)_{T,\,P} = 0$

C. $(d\,S)_{V,\,E} = 0, (d\,G)_{T,\,P} > 0$

D. $(d\,S)_{V,\,E} < 0, (d\,G)_{T,\,P}{}^{T} < 0$

Answer: A

Solution:

For spontaneous reaction, $dS > 0$ and dG should be negative i.e. <0.

Question225

The correct relationship between free energy change in a reaction and the corresponding equilibrium constant K_c is
[2003]

Options:

A. $-\Delta G = RT / nK_c$

B. $\Delta G^\circ = RT / nK_c$

C. $-\Delta G^\circ = RT \ln K_c$

D. $\Delta G = RT \ln K_c$

Answer: C

Solution:

$\Delta G^\circ = -RT \ln K_c$ or $-\Delta G^0 = RT \ln K_c$

Question226

A heat engine absorbs heat Q_1 at temperature T_1 and heat Q_2 at temperature T_2. Work done by the engine is $J(Q_1 + Q_2)$. This data
[2002]

Options:

A. violates 1^{st} law of thermodynamics

B. violates 1^{st} law of themodynamics if Q_1 is $-$ve

C. violates 1^{st} law of thermodynamics of Q_2 is $-$ve

D. does not violate 1^{st} law of themodynamics.

Answer: A

Solution:

According to first law of thermodynamics energy can neither be created nor destroyed although it can be converted from one form to another.
Note: Carnot cycle is based upon this principle but during the conversion of heat into work some mechanical energy is always converted to other form of energy hence this data violates 1 st law of thermodynamics.

Question227

The heat required to raise the temperature of body by $1K$ is called
[2002]

Options:

A. specific heat

B. thermal capacity

C. water equivalent

D. none of these.

Answer: B

Solution:

The heat required to raise the temperature of body by $1K$ is called thermal capacity or heat capacity.

Question228

For the reactions,
$2C + O_2 \rightarrow 2CO_2; \quad \Delta H = -393J$
$2Zn + O_2 \rightarrow 2ZnO; \quad \Delta H = -412J$
[2002]

Options:

A. carbon can oxidise Zn

B. oxidation of carbon is not feasible

C. oxidation of Zn is not feasible

D. Zn can oxidise carbon.

Answer: D

Solution:

ΔH negative shows that the reaction is spontancous. Higher negative value for Zn shows that the reaction is more feasible.

Question229

If an endothermic reaction is non-spontaneous at freezing point of water and becomes feasible at its boiling point, then
[2002]

Options:

A. ΔH is $-$ve, ΔS is $+$ve

B. ΔH and ΔS both are +ve

C. ΔH and ΔS both are −ve

D. ΔH is +ve, ΔS is −ve

Answer: B

Solution:

$\Delta G = \Delta H - T\ \Delta S$
For an endothermic reaction,
$\Delta H = +$ ve and at low temperature $\Delta S = +$ ve
Hence $\Delta G = (+)\ \Delta H - T\ (+)\ \Delta S$
and if $T\ \Delta S < \Delta H$ (at low temp)
$\Delta G = +ve$ (non spontancous)
But at high temperature, reaction becomes spontaneous i.e. $\Delta G = -ve$
because at higher temperature $T\ \Delta S > \Delta H$
